

TRANSMISSION/TRANSAXLE

05

SECTION

OUTLINE 05-00
CLUTCH..... 05-10

MANUAL TRANSMISSION
[R15M-D] 05-11A
MANUAL TRANSMISSION
[S15M-D, S15MX-D] 05-11B

05-00 OUTLINE

TRANSMISSION/TRANSAXLE
ABBREVIATIONS 05-00-1

TRANSMISSION/TRANSAXLE
FEATURES 05-00-1
TRANSMISSION/TRANSAXLE
SPECIFICATIONS 05-00-1

TRANSMISSION/TRANSAXLE ABBREVIATIONS

dcf050000000t01

MT	Manual Transmission
1GR	First Gear
2GR	Second Gear
5GR	Fifth Gear

TRANSMISSION/TRANSAXLE FEATURES

dcf050000000t02

CLUTCH	
Reduced noise and vibration	• Dual-mass flywheel adopted. (S15M-D, S15MX-D)
Improved durability	• Clutch cover with wear assurance function adopted.
Improved startability	• One-way valve that fits inside a clutch master cylinder adopted.
MT [R15M-D]	
Improved operability	• A double cone synchronizer mechanism has been adopted for 2nd gear.
Improved reliability	• A double engagement prevention mechanism (interlock mechanism) has been adopted.
Mis-shift prevention	• A stopper pin-type reverse lock-out mechanism has been adopted.
MT [S15M-D, S15MX-D]	
Improved operability	• A linked, triple cone synchronizer mechanism has been adopted for 1st and 2nd gears. • A linked, double cone synchronizer mechanism has been adopted for 3rd gear.
Improved reliability	• A double engagement prevention mechanism (interlock mechanism) has been adopted.
Mis-shift prevention	• A cam-type reverse lock-out mechanism has been adopted.

TRANSMISSION/TRANSAXLE SPECIFICATIONS

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Clutch

Item		Specifications	
Manual transmission type		R15M-D	S15M-D, S15MX-D
Clutch control		Hydraulic	
Clutch cover	Spring type	Diaphragm	
	Set load (N {kgf, lbf})	5,290 {539, 1,189}	10,100 {1,030, 2,271}
Clutch disc	Outer diameter (mm {in})	240 {9.45}	250 {9.84}
	Inner diameter (mm {in})	160 {6.30}	155 {6.10}
Clutch pedal	Type	Suspended	
	Pedal ratio	6.0	5.2
	Full stroke (mm {in})	152 {5.984}	
Clutch master cylinder inner diameter (mm {in})		15.87 {0.6248}	

05-00-1

OUTLINE

Item		Specifications	
Manual transmission type		R15M-D	S15M-D, S15MX-D
Clutch release cylinder inner diameter	(mm {in})	19.05 {0.7500}	22.23 {0.8752}
Clutch fluid type		SAE J1703, FMVSS 116 DOT-3	

Manual Transmission [R15M-D]

Item			Specifications
Transmission type			R15M-D
Transmission control			Floor-shift
Shift assist			Synchromesh
Gear ratio	1GR		4.250
	2GR		2.356
	3GR		1.428
	4GR		1.000
	5GR		0.827
	Reverse		3.656
Oil	Grade		API service GL-4 or GL-5
	Viscosity	All season	SAE 75W-90
		Above 10°C {50°F}	SAE 80W-90
	Capacity (approx. quantity)	(L {US qt, Imp qt})	2.8 {3.0, 2.5}

Manual Transmission [S15M-D, S15MX-D]

Item			Specifications
Transmission type			S15M-D, S15MX-D
Transmission control			Floor-shift
Shift assist			Synchromesh
Gear ratio	1GR		3.905
	2GR		2.248
	3GR		1.491
	4GR		1.000
	5GR		0.800
	Reverse		3.391
Oil	Type		Mercon® Multi-purpose AFT XT-2-QDX
	Capacity (approx. quantity)	(L {US qt, Imp qt})	3.55 {3.75, 3.12}

TRANSMISSION/TRANSAXLE

05
SECTION

OUTLINE 05-00
ON-BOARD
DIAGNOSTIC[5R55S] 05-02

AUTOMATIC TRANSMISSION
[5R55S] 05-13
AUTOMATIC TRANSMISSION
SHIFT MECHANISM 05-14

05-00 OUTLINE

TRANSMISSION/TRANSAXLE
FEATURES [5R55S] 05-00-1

TRANSMISSION/TRANSAXLE
SPECIFICATIONS [5R55S] 05-00-1

TRANSMISSION/TRANSAXLE FEATURES [5R55S]

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Improved marketability	- Five-speed 5R55S automatic transmission has been adopted. - A water-cooling type and an air-cooling type AT oil cooler has been adopted
Superior shift quality	- Adaptive learn strategy system has been adopted - Engine-transaxle total control system has been adopted

TRANSMISSION/TRANSAXLE SPECIFICATIONS [5R55S]

Id0500001006f1

Item		Specification
Transmission type		5R55S
Transmission control		Floor-shift
Gear ratio	1GR	3.22
	2GR	2.29
	3GR	1.55
	4GR	1.00
	5GR	0.71
	Reverse	3.07
ATF	Type	Mercon® V
	Capacity (approx. quantity) (L {US qt, Imp qt})	9.93 {10.49, 8.74}
Hydraulic system (Number of drive/driven gear plates)	Forward clutch	5/5
	Coast clutch	2/2
	Direct clutch	5/5

05

05-02 ON-BOARD DIAGNOSTIC [5R55S]

ON-BOARD DIAGNOSTIC (OBD)	
SYSTEM OUTLINE [5R55S]	05-02-1
ON-BOARD DIAGNOSTIC (OBD)	
SYSTEM BLOCK DIAGRAM	
[5R55S]	05-02-1
MALFUNCTION DETECTION	
FUNCTION [5R55S]	05-02-2
MEMORY FUNCTION [5R55S]	05-02-3

MALFUNCTION INDICATION	
FUNCTION [5R55S]	05-02-3
FAIL-SAFE FUNCTION [5R55S]	05-02-3
PARAMETER IDENTIFICATION (PID)	
DATA MONITORING FUNCTION	
[5R55S]	05-02-4
SIMULATION FUNCTION [5R55S]	05-02-4
DLC-2 OUTLINE [5R55S]	05-02-5

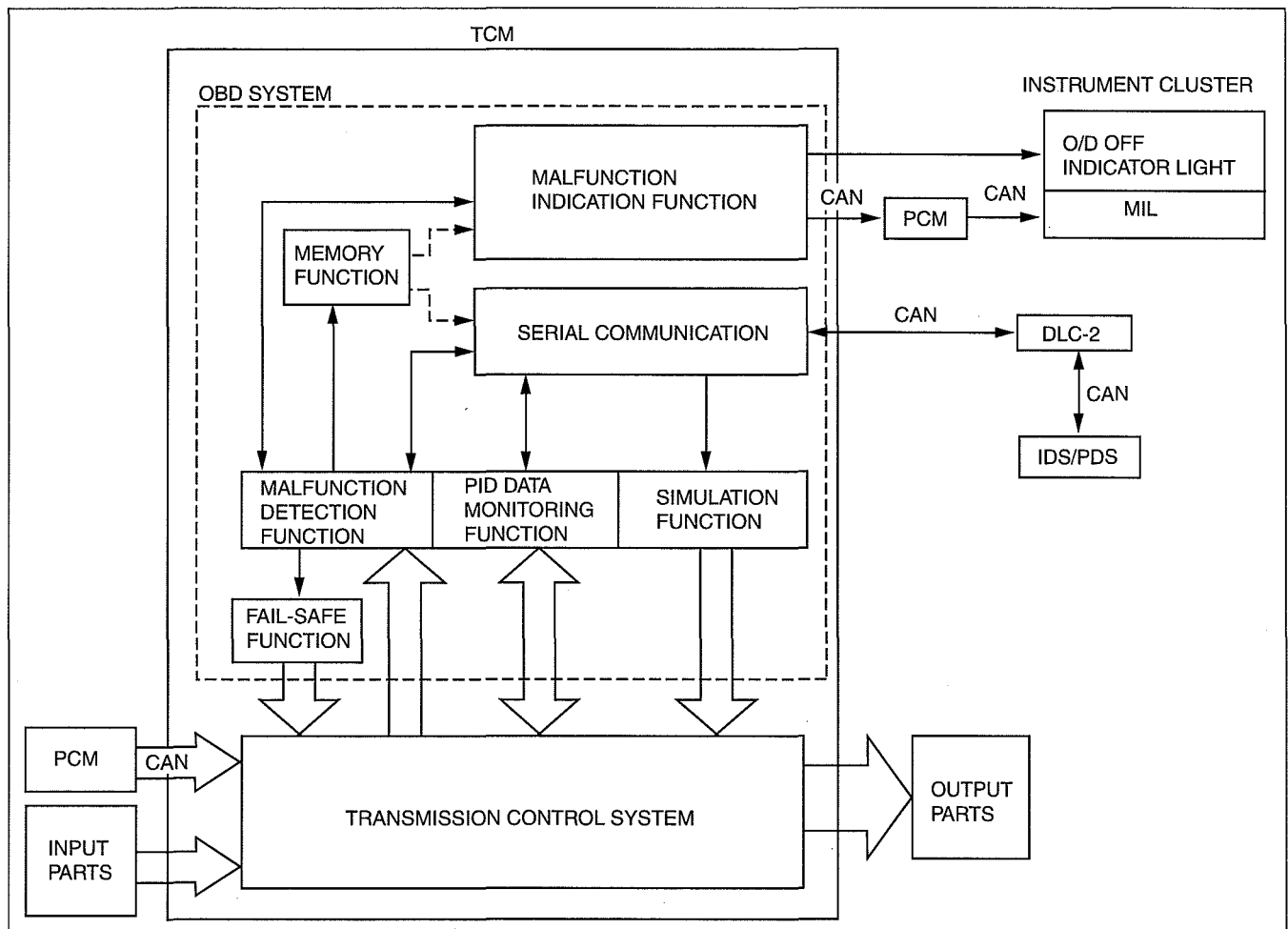
ON-BOARD DIAGNOSTIC (OBD) SYSTEM OUTLINE [5R55S]

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- The OBD system has the following functions:
 - Malfunction detection function: Detects malfunctions of the input/output devices and system components of the AT.
 - Fail-safe function: Fixes the output device function and input value of the sensors/switches to ensure minimum vehicle driveability when a malfunction is detected.
 - Memory function: Stores the DTC when a malfunction is detected.
 - PID data monitored function: Monitors the input/output signal and calculated value of the TCM, and sends the monitoring data to the IDS/PDS.
 - Simulation function: Allows override operation of simulation items for input/output system parts preset in the TCM.

ON-BOARD DIAGNOSTIC (OBD) SYSTEM BLOCK DIAGRAM [5R55S]

id0502c1100200



arnfn00000333

ON-BOARD DIAGNOSTIC [5R55S]

MALFUNCTION DETECTION FUNCTION [5R55S]

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Malfunction Detection Function

- In the malfunction detection function, the TCM detects malfunctions in the automatic transmission while driving.
- When vehicle driving conditions correspond with a preset malfunction detection condition, the TCM determines that the automatic transmission has a malfunction and stores the corresponding DTC.
- When a malfunction is detected, stored DTCs can be retrieved using the IDS/PDS connected to the DLC-2.

DTC Table

DTC No.	Condition
P0657	Solenoid valve power supply circuit open
P0658	Solenoid valve power supply circuit voltage low
P0705	Digital transmission range (TR) sensor circuit malfunction (invalid bit pattern)
P0708	Digital transmission range (TR) sensor circuit malfunction (TR3A circuit open)
P0711	Transmission fluid temperature (TFT) sensor malfunction (ATF temperature no change)
P0712	Transmission fluid temperature (TFT) sensor circuit malfunction (short to ground)
P0713	Transmission fluid temperature (TFT) sensor circuit malfunction (open circuit)
P0715	Turbine shaft speed (TSS) sensor malfunction
P0717	Turbine shaft speed (TSS) sensor signal is not input
P0718	Turbine shaft speed (TSS) sensor signal noise
P0720	Output shaft speed (OSS) sensor malfunction
P0721	Output shaft speed (OSS) sensor signal noise
P0722	Output shaft speed (OSS) sensor signal is not input
P0731	Gear 1 incorrect
P0732	Gear 2 incorrect
P0733	Gear 3 incorrect
P0734	Gear 4 incorrect
P0735	Gear 5 incorrect
P0740	Torque converter clutch (TCC) control solenoid circuit malfunction (open circuit)
P0741	Torque converter clutch (TCC) control solenoid circuit malfunction (stuck off)
P0742	Torque converter clutch (TCC) control solenoid circuit malfunction (stuck on)
P0743	Torque converter clutch (TCC) control solenoid circuit malfunction (open or short circuit)
P0744	Torque converter clutch (TCC) control solenoid circuit malfunction (short to power)
P0745	Pressure control solenoid A malfunction (low pressure)
P0748	Pressure control solenoid A circuit malfunction (non-operation)
P0750	Shift solenoid A circuit malfunction (open or short circuit)
P0753	Shift solenoid A circuit malfunction (open or short circuit)
P0755	Shift solenoid B circuit malfunction (open or short circuit)
P0757	Shift solenoid B malfunction (stuck on)
P0758	Shift solenoid B circuit malfunction (open or short circuit)
P0760	Shift solenoid C circuit malfunction (open or short circuit)
P0762	Shift solenoid C malfunction (stuck on)
P0763	Shift solenoid C circuit malfunction (open or short circuit)
P0765	Shift solenoid D circuit malfunction (open or short circuit)
P0768	Shift solenoid D circuit malfunction (open or short circuit)
P0775	Pressure control solenoid B malfunction (low pressure)
P0778	Pressure control solenoid B circuit malfunction (non-operation)
P0791	Intermediate shaft speed (ISS) sensor malfunction
P0794	Intermediate shaft speed (ISS) sensor signal noise
P0795	Pressure control solenoid C malfunction (low pressure)
P0798	Pressure control solenoid C circuit malfunction (non-operation)
P0960	Pressure control solenoid A circuit malfunction (open circuit)
P0962	Pressure control solenoid A circuit malfunction (short to ground)
P0963	Pressure control solenoid A circuit malfunction (short to power supply)
P0964	Pressure control solenoid B circuit malfunction (open circuit)
P0966	Pressure control solenoid B circuit malfunction (short to ground)
P0967	Pressure control solenoid B circuit malfunction (short to power supply)

ON-BOARD DIAGNOSTIC [5R55S]

DTC No.	Condition
P0968	Pressure control solenoid C circuit malfunction (open circuit)
P0970	Pressure control solenoid C circuit malfunction (short to ground)
P0971	Pressure control solenoid C circuit malfunction (short to power supply)
P1635	Tire/axle ratio out of acceptable range
P1700	Transmission indeterminate malfunction
P1702	Digital transmission range (TR) sensor circuit intermittent
P1704	Digital transmission range (TR) sensor circuit not indicating P or N position during on-board diagnostic
P1705	Digital transmission range (TR) sensor circuit not indicating P or N position during on-board diagnostic
P1711	ATF temperature out of on-board diagnostic range
P1780	O/D OFF switch circuit malfunction (open or short circuit)
P1783	ATF overheating
U0100	Communication error to PCM
U0294	Communication error to PCM

MEMORY FUNCTION [5R55S]

id0502c1100400

- The memory function stores malfunction information detected in the malfunction detection function. Once malfunction information is stored, the memory will not be cleared even when the engine switch is turned off (LOCK position) or the malfunction is repaired.
- The stored memory (malfunction information) can be cleared using the IDS/PDS, or by disconnecting the negative battery cable.

MALFUNCTION INDICATION FUNCTION [5R55S]

id0502c1100500

- The malfunction indication function illuminates the MIL or flashing the O/D OFF indicator light when the malfunction detection function determines that there is a malfunction.

FAIL-SAFE FUNCTION [5R55S]

id0502c1100600

- In the fail-safe function, minimum vehicle driveability is obtained by changing the signals that are determined to be irregular by the malfunction detection function to the preset values, and limiting TCM control.

ON-BOARD DIAGNOSTIC [5R55S]

PARAMETER IDENTIFICATION (PID) DATA MONITORING FUNCTION [5R55S]

id0502c1100700

- The PID mode allows access to certain data values, analog and digital input and output, calculations and system state information.

PID/DATA MONITOR AND RECORD function table

Monitor item	Definition	Unit/ Condition	TCM terminal
BPP	Brake switch	on/off	N/A
FIRM_ST	Firm shaft control (FSC) status requested	0/1	N/A
GEAR	Calculated gear range in TCM	1/2/3/4/5	N/A
GEAR_OSC	PID used to command gear changes during Simulation Function	1/2/3/4/5	N/A
ISS	ISS sensor	rpm	AM, AS
OSS	OSS sensor	rpm	AD, AG
PCA	Pressure control solenoid A	pis	I, CK
PCB	Pressure control solenoid B	pis	J, CK
PCC	Pressure control solenoid C	pis	M, CK
RPM	Engine speed	rpm	N/A
SSA	Shift solenoid A	on/off	B, CK
SSB	Shift solenoid B	on/off	C, CK
SSC	Shift solenoid C	on/off	F, CK
SSD	Shift solenoid D	on/off	D, CK
TCC	TCC control solenoid	%	A, CK
TCCRAT	Torque converter speed ratio	ratio	N/A
TFT	ATF temperature	°C, °F	AY, BH
TR_D	Digital TR sensor signal		AX, BU, BY, CA
TR_V	Digital TR sensor signal voltage	V	AX, BU, BY, CA
TR1	TR1 switch	0/1	BY
TR2	TR2 switch	0/1	BU
TR3A	TR3 switch	0/1	AX
TR4	TR4 switch	0/1	CA
TSS	TSS sensor	rpm	AH, AP
VSS	Vehicle speed	mph	AD, AG

SIMULATION FUNCTION [5R55S]

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- By using the IDS/PDS, simulation items for input/output parts preset in the TCM can be optionally selected and operated regardless of TCM control conditions.

Simulation item table

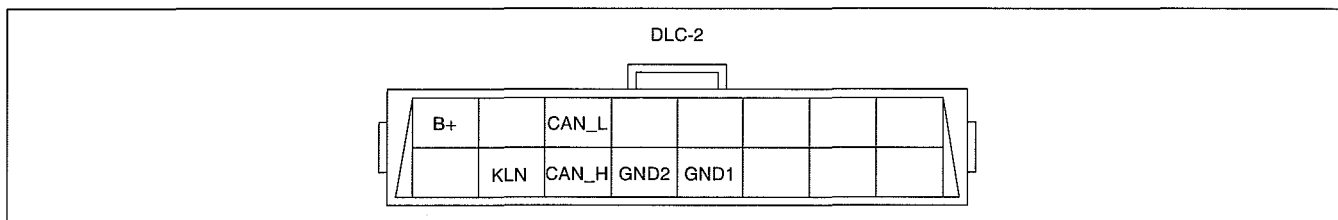
Simulation item	Applicable component	Unit	TCM terminal
PCA	Pressure control solenoid A	pis	I, CK
PCB	Pressure control solenoid B	pis	J, CK
PCC	Pressure control solenoid C	pis	M, CK
SSA	Shift solenoid A	on/off	B, CK
SSB	Shift solenoid B	on/off	C, CK
SSC	Shift solenoid C	on/off	F, CK
SSD	Shift solenoid D	on/off	D, CK
TCC	TCC control solenoid	%	A, CK

ON-BOARD DIAGNOSTIC [5R55S]

DLC-2 OUTLINE [5R55S]

id0502c1100900

- A connector (DLC-2) conforming to International Organization for Standardization (ISO) standards has been adopted.
- The shape and terminal arrangement as stipulated by the ISO 15031-3 (SAE J1962) international standard has been adopted for this connector. The connector has a 16-pin construction that includes the B+, CAN_H, CAN_L, GND1, GND2 and KLN terminals.



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Terminal	Function
B+	Battery power supply terminal
CAN_L	Serial communication terminal (Lo)
CAN_H	Serial communication terminal (Hi)
GND1	Body ground terminal
GND2	Serial communication ground terminal
KLN	Serial communication terminal (Malfunction diagnosis use)



(1)



CLUTCH

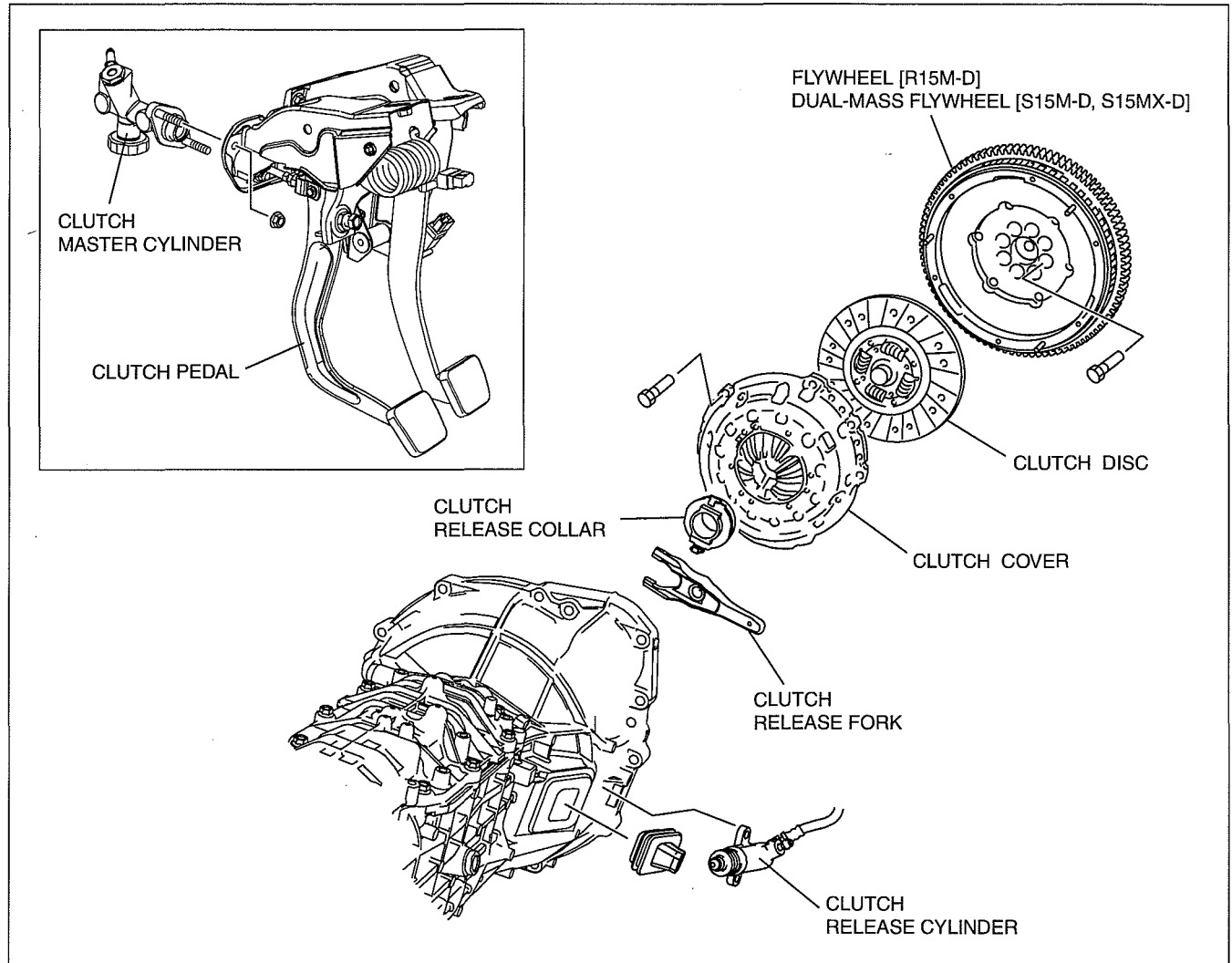
05-10 CLUTCH

CLUTCH STRUCTURAL VIEW.....	05-10-1
CLUTCH MASTER CYLINDER CONSTRUCTION	05-10-2
CLUTCH RELEASE CYLINDER construction	05-10-2

CLUTCH COVER OUTLINE [S15M-D, S15MX-D]	05-10-2
DUAL-MASS FLYWHEEL FUNCTION [S15M-D, S15MX-D]	05-10-3

CLUTCH STRUCTURAL VIEW

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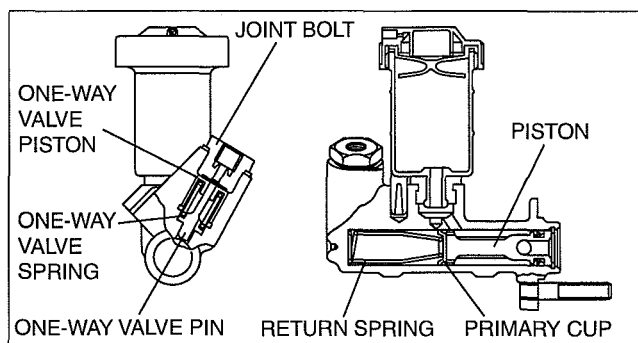
CLUTCH

CLUTCH MASTER CYLINDER CONSTRUCTION

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- A clutch master cylinder with a built-in one-way valve has been adopted preventing sudden clutch engagement and ensuring smooth startability.

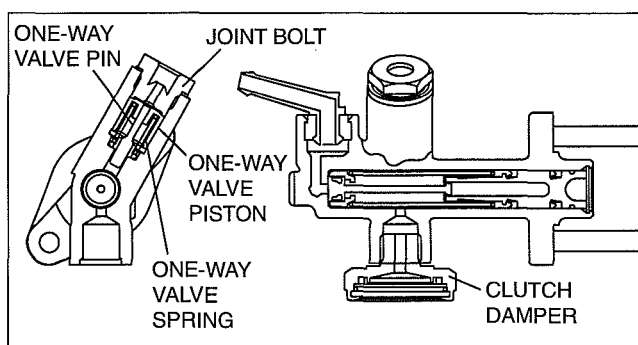
Vehicles without clutch damper



DCG510ZTB002

Vehicles with clutch damper

- The clutch damper controls hydraulic pulsation during clutch operation, reducing harsh vibration transmitted to the clutch pedal and operation noise.

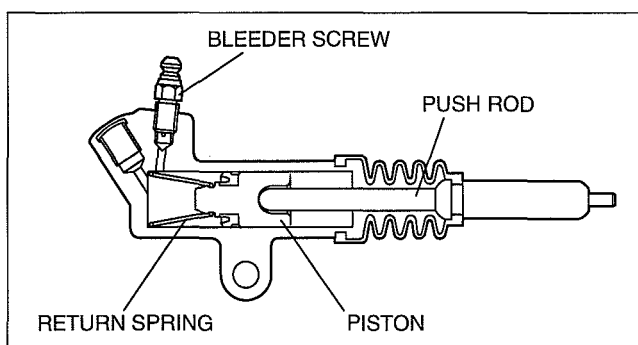


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CLUTCH RELEASE CYLINDER construction

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- The clutch release cylinder consists of a return spring, piston, push rod and a bleeder screw for bleeding air.
- Due to spring pressure maintaining play between the push rod end and the release fork at zero, an automatic adjusting, maintenance-free design has been achieved.



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CLUTCH COVER OUTLINE [S15M-D, S15MX-D]

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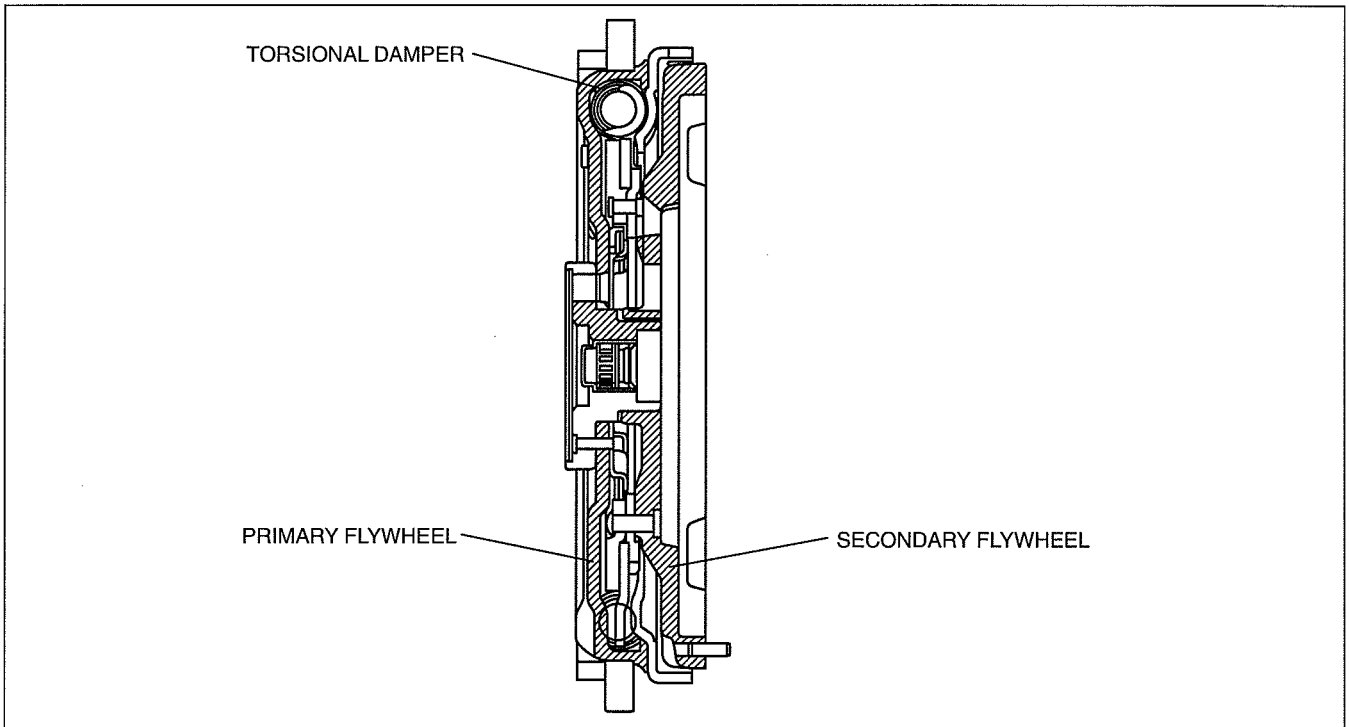
- The wear assurance function of the clutch cover maintains the set load of the clutch cover unchanged even as the clutch disc becomes worn, thereby extending disc life.

CLUTCH

DUAL-MASS FLYWHEEL FUNCTION [S15M-D, S15MX-D]

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- The dual-mass flywheel has separated primary and secondary flywheel, and a torsional damper that is located inside the flywheel. Due to this structure, the transmission main drive gear shaft rotation can be stabilized, decreasing noise inside the transmission (gear teeth rattling) created by engine speed fluctuation, and thereby greatly reducing vehicle booming noise also.



DCG510ZTB005



05-11B MANUAL TRANSMISSION [S15M-D, S15MX-D]

MANUAL TRANSMISSION OUTLINE [S15M-D, S15MX-D]	05-11B-1	DOUBLE cone SYNCHRONIZER MECHANISM OUTLINE [S15M-D, S15MX-D]	05-11B-7
MANUAL TRANSMISSION CROSS-SECTIONAL VIEW [S15M-D, S15MX-D]	05-11B-2	DOUBLE CONE SYNCHRONIZER MECHANISM CONSTRUCTION/ OPERATION [S15M-D, S15MX-D]	05-11B-7
MANUAL TRANSMISSION POWER FLOW [S15M-D, S15MX-D]	05-11B-3	SHIFT INTERLOCK MECHANISM FUNCTION [S15M-D, S15MX-D]	05-11B-7
SHIFT MECHANISM OUTLINE [S15M-D, S15MX-D]	05-11B-4	SHIFT INTERLOCK MECHANISM OPERATION [S15M-D, S15MX-D]	05-11B-7
TRIPLE CONE SYNCHRONIZER MECHANISM STRUCTURE [S15M-D, S15MX-D]	05-11B-5	REVERSE LOCKOUT MECHANISM FUNCTION [S15M-D, S15MX-D]	05-11B-8
TRIPLE CONE SYNCHRONIZER MECHANISM OPERATION [S15M-D, S15MX-D]	05-11B-5	REVERSE LOCKOUT MECHANISM CONSTRUCTION/OPERATION [S15M-D, S15MX-D]	05-11B-8

MANUAL TRANSMISSION OUTLINE [S15M-D, S15MX-D]

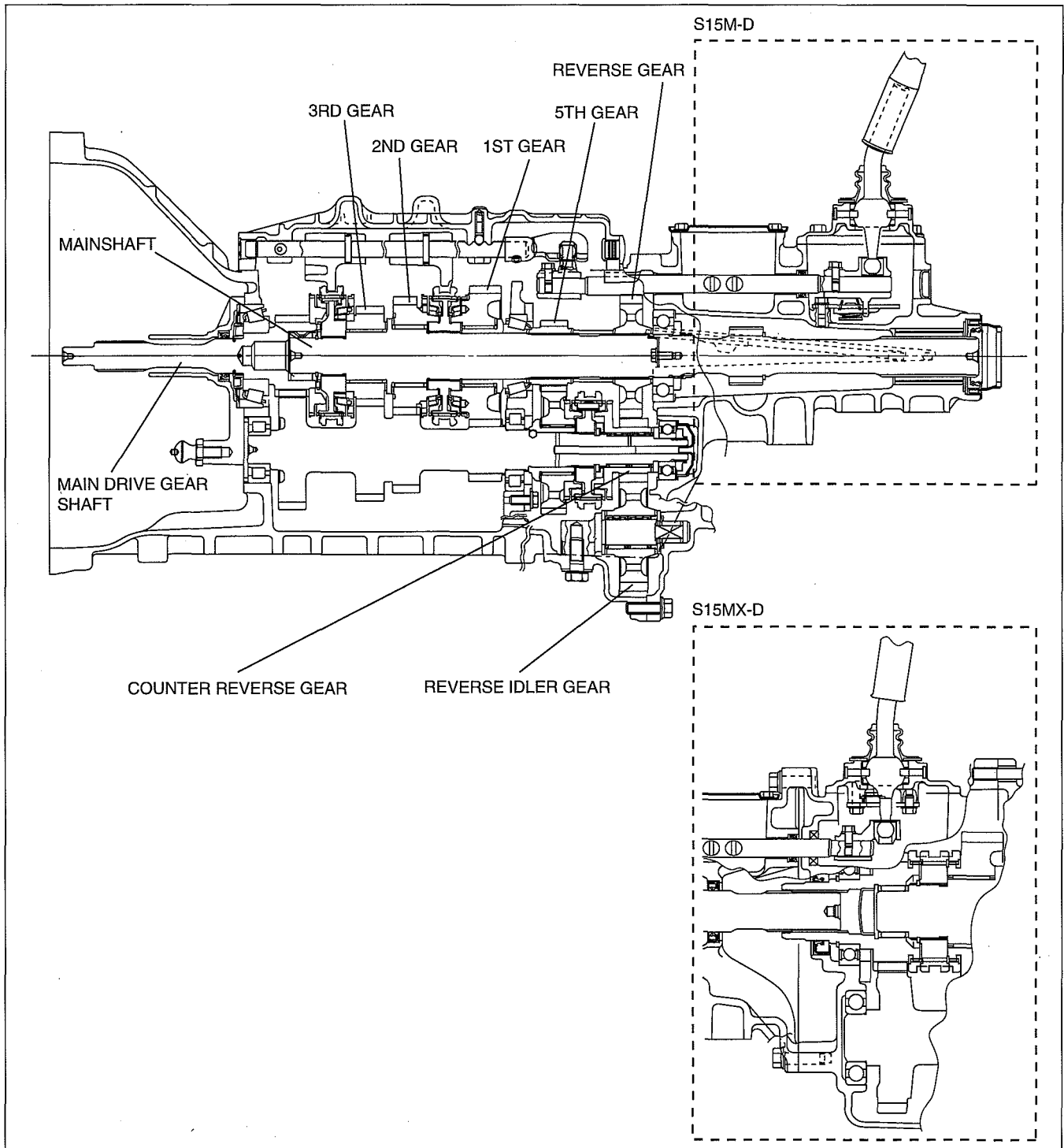
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- A linked, triple cone synchronizer mechanism has been adopted for 1st and 2nd gears.
- A linked, double cone synchronizer mechanism has been adopted for 3rd gear.
- A cam-type reverse lockout mechanism has been adopted.

MANUAL TRANSMISSION [S15M-D, S15MX-D]

MANUAL TRANSMISSION CROSS-SECTIONAL VIEW [S15M-D, S15MX-D]

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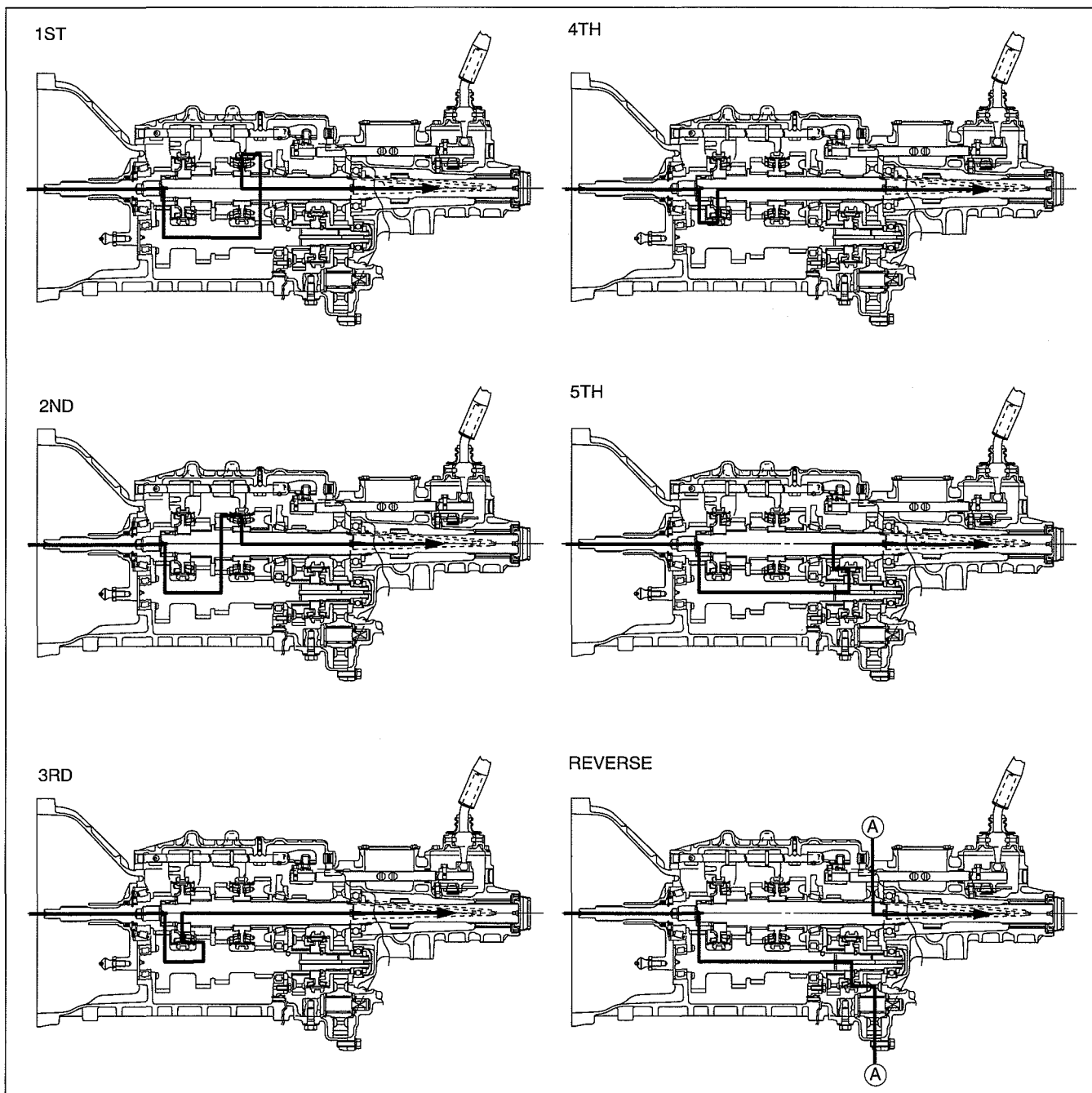


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MANUAL TRANSMISSION [S15M-D, S15MX-D]

MANUAL TRANSMISSION POWER FLOW [S15M-D, S15MX-D]

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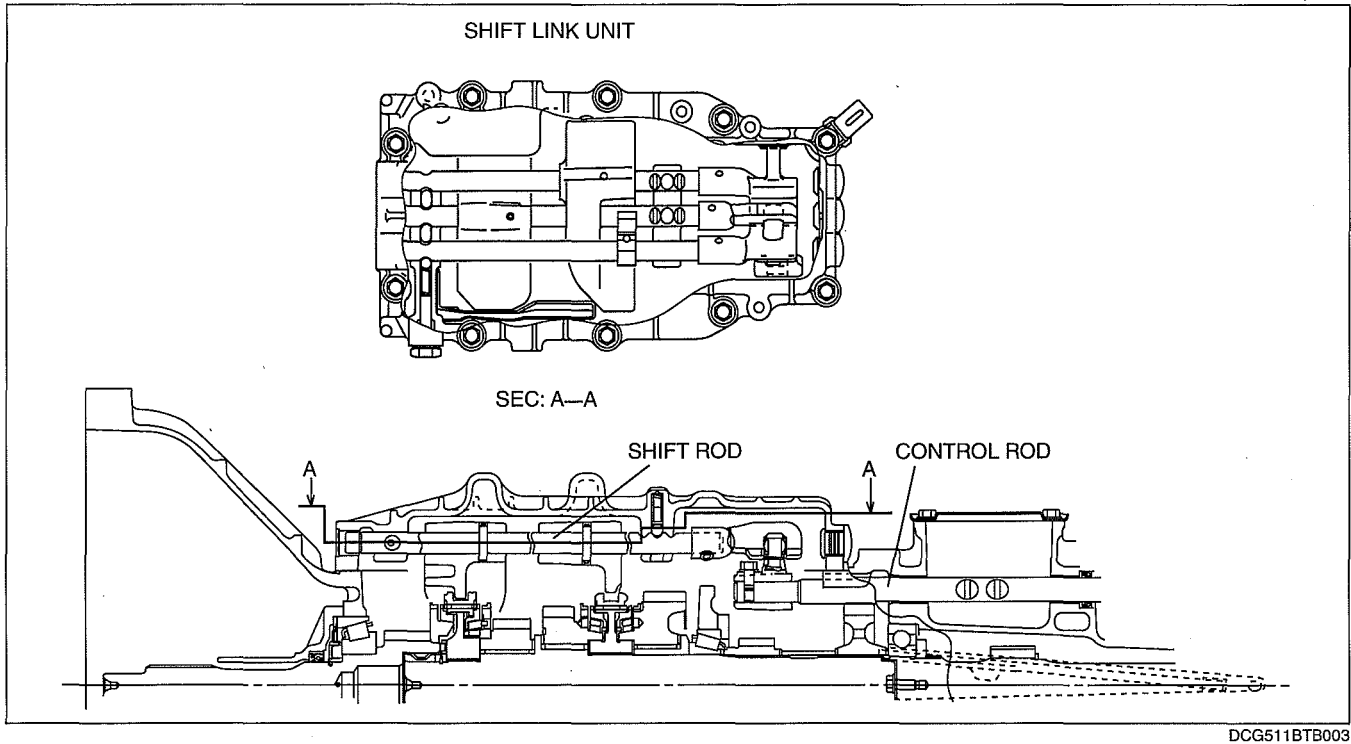
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MANUAL TRANSMISSION [S15M-D, S15MX-D]

SHIFT MECHANISM OUTLINE [S15M-D, S15MX-D]

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- To realize assured shift feel, the shift link mechanism has been integrated.



MANUAL TRANSMISSION [S15M-D, S15MX-D]

TRIPLE CONE SYNCHRONIZER MECHANISM STRUCTURE [S15M-D, S15MX-D]

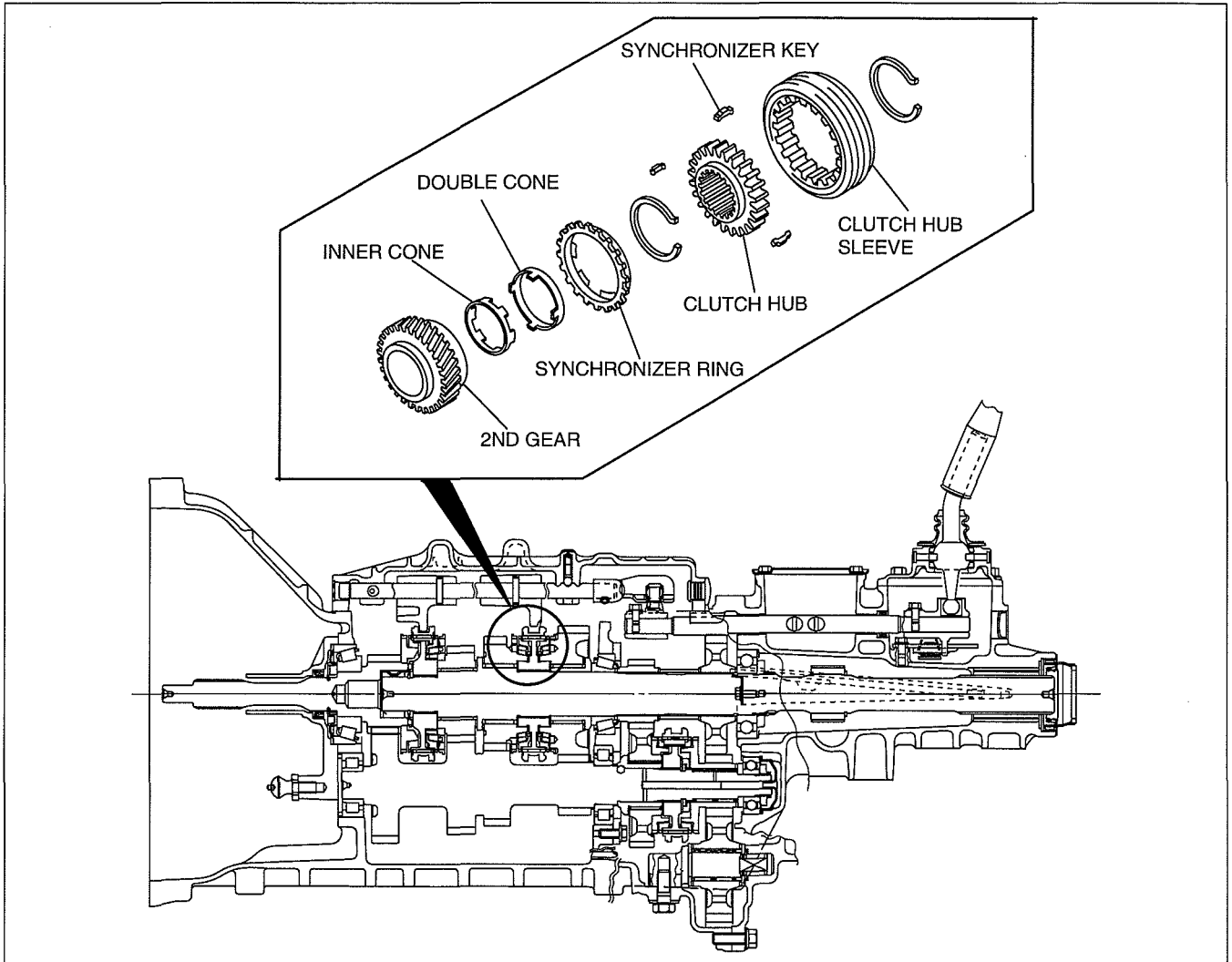
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Features

- A triple cone synchronizer mechanism is used for the 1st and 2nd gears.
- The triple cone synchronizer mechanism is a compact device capable of heavy duty meshing.
- The synchro mechanism reduces meshing time and improves operation.
- The triple cone synchro mechanism includes a synchronizer ring, a double cone, and an inner cone.

Structure

Structural view



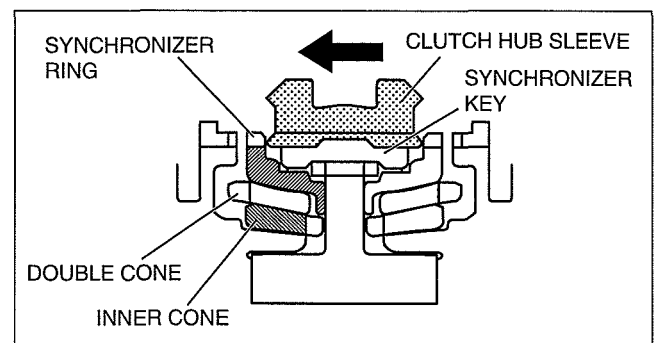
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TRIPLE CONE SYNCHRONIZER MECHANISM OPERATION [S15M-D, S15MX-D]

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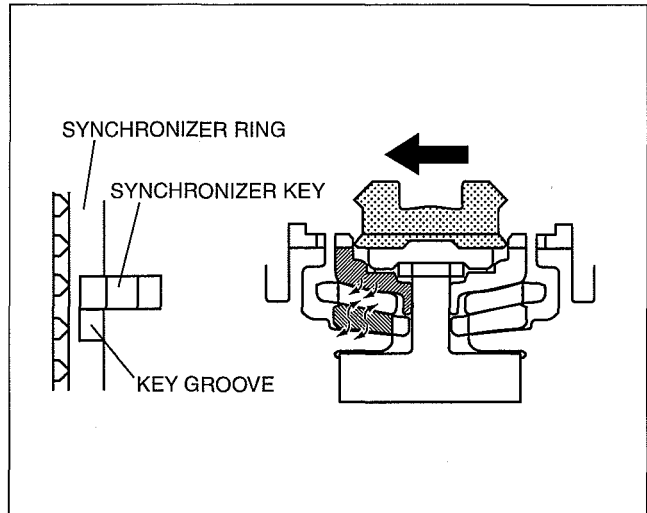
1. When the hub sleeve moves to the left (in the direction of the arrow), the synchronizer key presses against the synchronizer ring.



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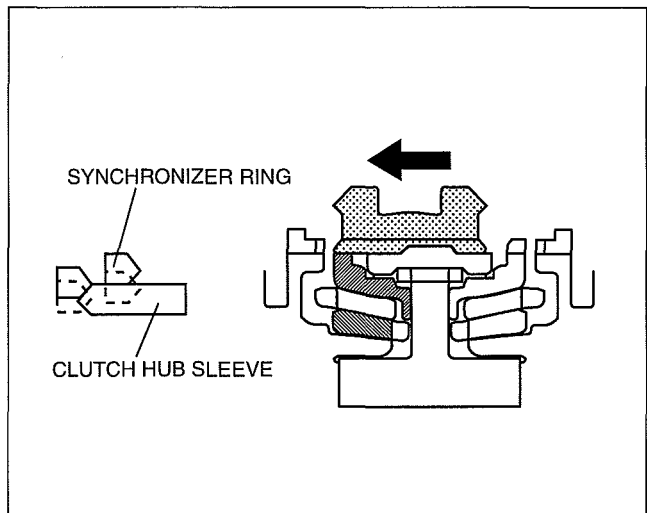
MANUAL TRANSMISSION [S15M-D, S15MX-D]

2. As the hub sleeve continues moving to the left, the key causes friction between the synchronizer ring, double cone, and inner cone and 2nd gear. The synchronizer ring turns only the distance that the key groove gap allows, aligning the teeth of the hub sleeve and the synchronizer ring. As the hub sleeve continues moving, the friction between the cones becomes greater, and the difference between the rotational speeds of the synchronizer ring, inner cone, and double cone (unified with the gear) gradually disappears.



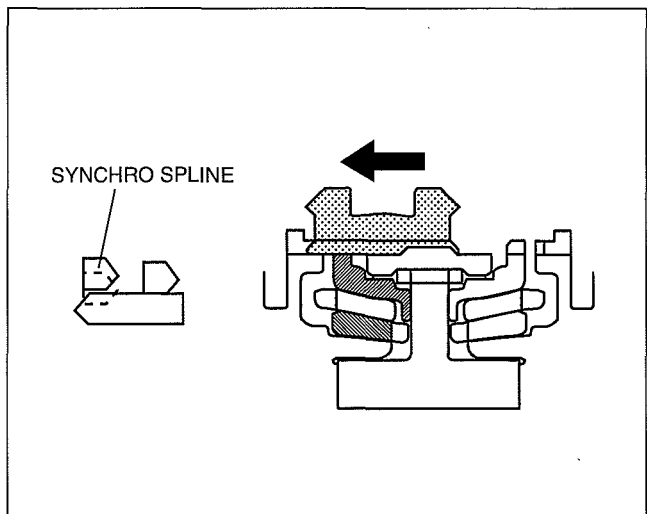
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3. The hub sleeve then moves up onto the synchronizer key and engages the synchronizer ring.



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4. The hub sleeve then engages the synchro spline of the gear to complete shifting.



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MANUAL TRANSMISSION [S15M-D, S15MX-D]

DOUBLE cone SYNCHRONIZER MECHANISM OUTLINE [S15M-D, S15MX-D]

dcf051100000t44

- A linked, double cone synchronizer mechanism has been adopted for 3rd gear.
- The double cone synchronizer mechanism consists of a synchronizer ring, double cone, and inner cone the same as a triple cone synchronizer mechanism.

DOUBLE CONE SYNCHRONIZER MECHANISM CONSTRUCTION/OPERATION [S15M-D, S15MX-D]

dcf051100000t45

- The basic construction of the double cone synchronizer mechanism is the same as a triple cone synchronizer mechanism except that the inner side of the inner cone is not used as a friction surface.
- The basic construction of the double cone synchronizer mechanism is the same as a triple cone synchronizer mechanism except that friction force is not generated because there is no contact surface between the inner cone and each gear.
- For the double cone synchronizer mechanism operation, refer to the triple cone synchronizer mechanism. (See 05-11B-5 TRIPLE CONE SYNCHRONIZER MECHANISM OPERATION [S15M-D, S15MX-D].)

SHIFT INTERLOCK MECHANISM FUNCTION [S15M-D, S15MX-D]

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- This provides reliable double-engagement prevention.

SHIFT INTERLOCK MECHANISM OPERATION [S15M-D, S15MX-D]

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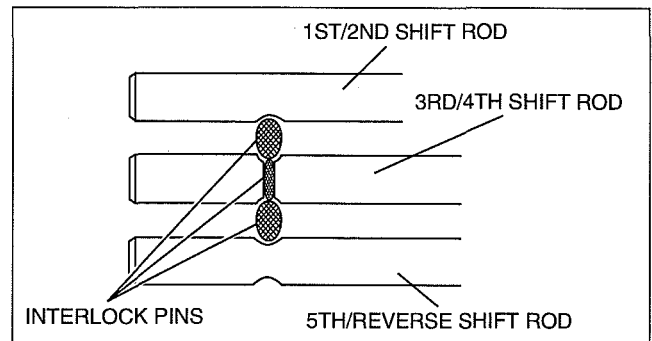
Structure

- During shifting, the shift rods, except for the one in operation, are locked in the neutral position by the interlock pins.

Operation

Neutral

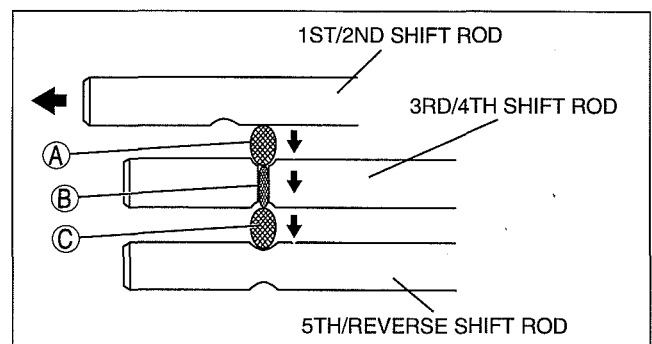
- Because no shift rod is operated, the interlock pins are seated in the grooves.



DCG511BTB010

1st/2nd shifting

- Movement of the 1st/2nd shift rod forces interlock pin A out of the 1st/2nd shift rod groove, and locks the 3rd/4th shift rod. Pin B, forced by pin A, pushes out pin C to lock the 5th/Reverse shift rod.

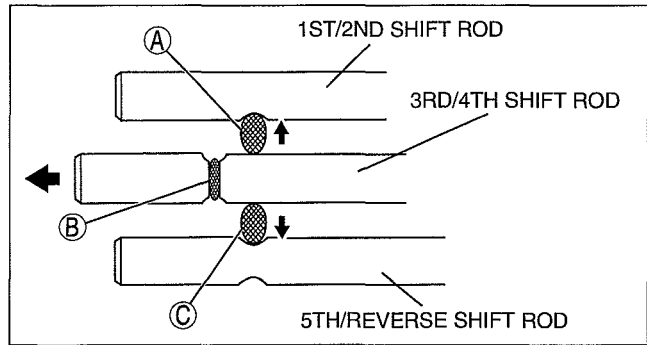


DCG511BTB011

MANUAL TRANSMISSION [S15M-D, S15MX-D]

3rd/4th shifting

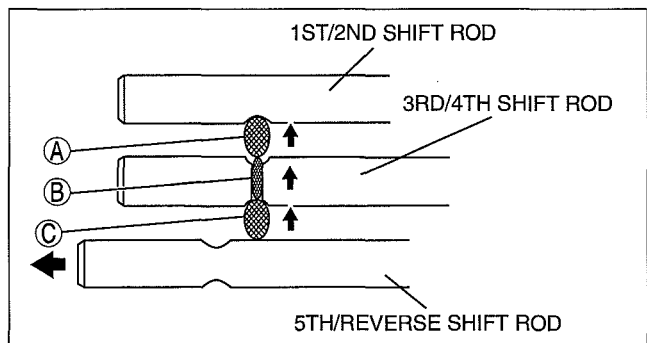
- Movement of the 3rd/4th shift rod forces out pins A and C, and locks the 1st/2nd and 5th/Reverse shift rods. Pin B does not affect the other pins or shift rods during 3rd/4th shifting.



DCG511BTB012

5th/Reverse shifting

- When performing 5th/Reverse shifting, the interlock pins function the same way as in 1st/2nd shifting, except the pin movement order is in reverse, and the 3rd/4th and 1st/2nd shift rods are locked.



DCG511BTB013

REVERSE LOCKOUT MECHANISM FUNCTION [S15M-D, S15MX-D]

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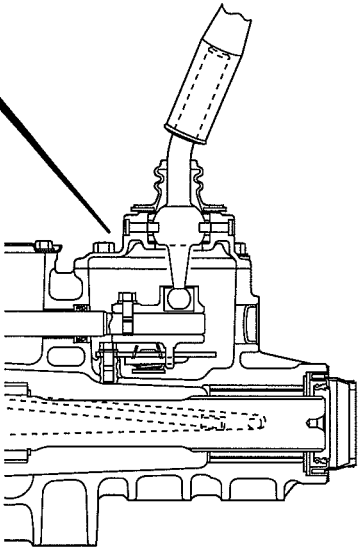
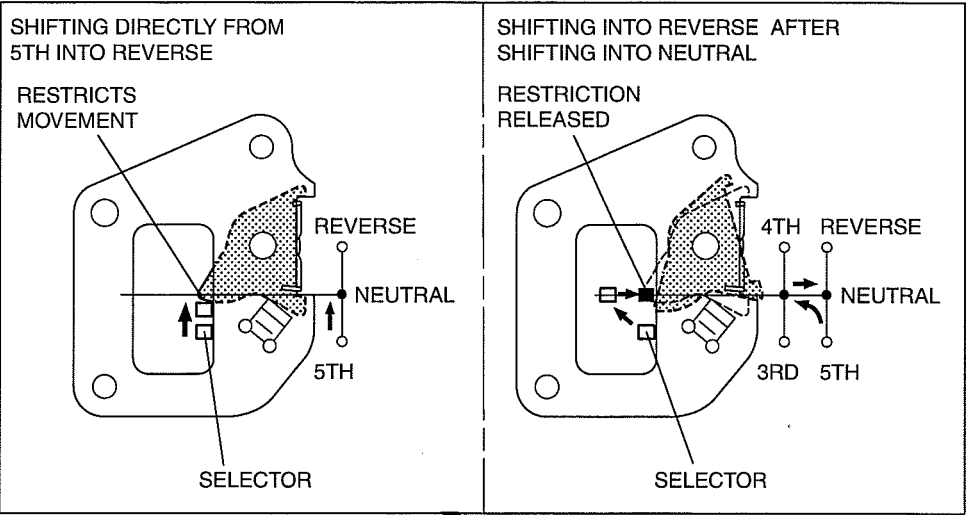
- The reverse lockout mechanism prevents the driver from accidentally shifting into reverse gear when downshifting from 5th to 4th gear.

REVERSE LOCKOUT MECHANISM CONSTRUCTION/OPERATION [S15M-D, S15MX-D]

dcf051117570t10

- A cam-type reverse lockout mechanism is adopted to ensure reliability.
- A cam, which is installed to the shift guide plate in the shift control case, restricts the selector movement to prevent the driver from miss shifting.
- When shifting into reverse, by shifting the selector back into the neutral position once and then shifting to the 5th/reverse direction, the selector pushes the cam outward to release the shifting restriction, and shifting into reverse is made possible.

S15M-D

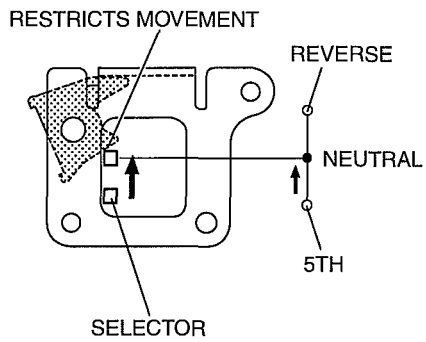


DCG511BTB009

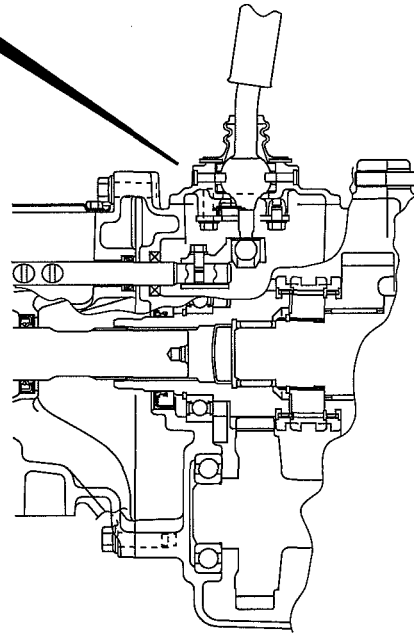
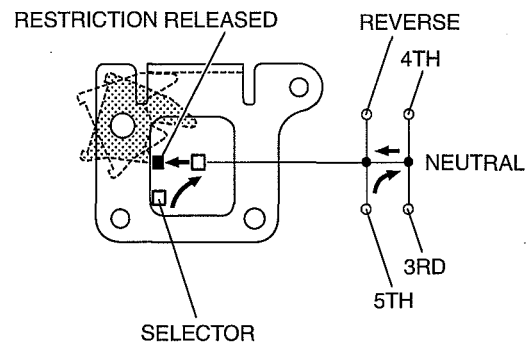
MANUAL TRANSMISSION [S15M-D, S15MX-D]

S15MX-D

SHIFTING DIRECTLY FROM 5TH INTO REVERSE



SHIFTING INTO REVERSE AFTER SHIFTING INTO NEUTRAL

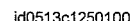


DCG511BTB014

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AUTOMATIC TRANSMISSION OUTLINE [5R55S]

- This transmission has the following features:
 - Five forward speeds
 - Electronic shift, pressure and torque converter clutch controls
 - Three compound planetary gearsets
 - Three bands (overdrive, intermediate, low/reverse)
 - Three multi-plate clutches (coast, direct, forward)
 - Two one-way clutches (Overdrive, low/reverse)
- All hydraulic functions are directed by electronic solenoids. The solenoids control:
 - Static engagement feel.
 - Shift feel.
 - Shift scheduling.
 - Modulated torque converter clutch (TCC) applications.
 - Engine braking utilizing the coast clutch and band.
 - Manual 1st and 2nd timing.
 - Reverse inhibit timing.

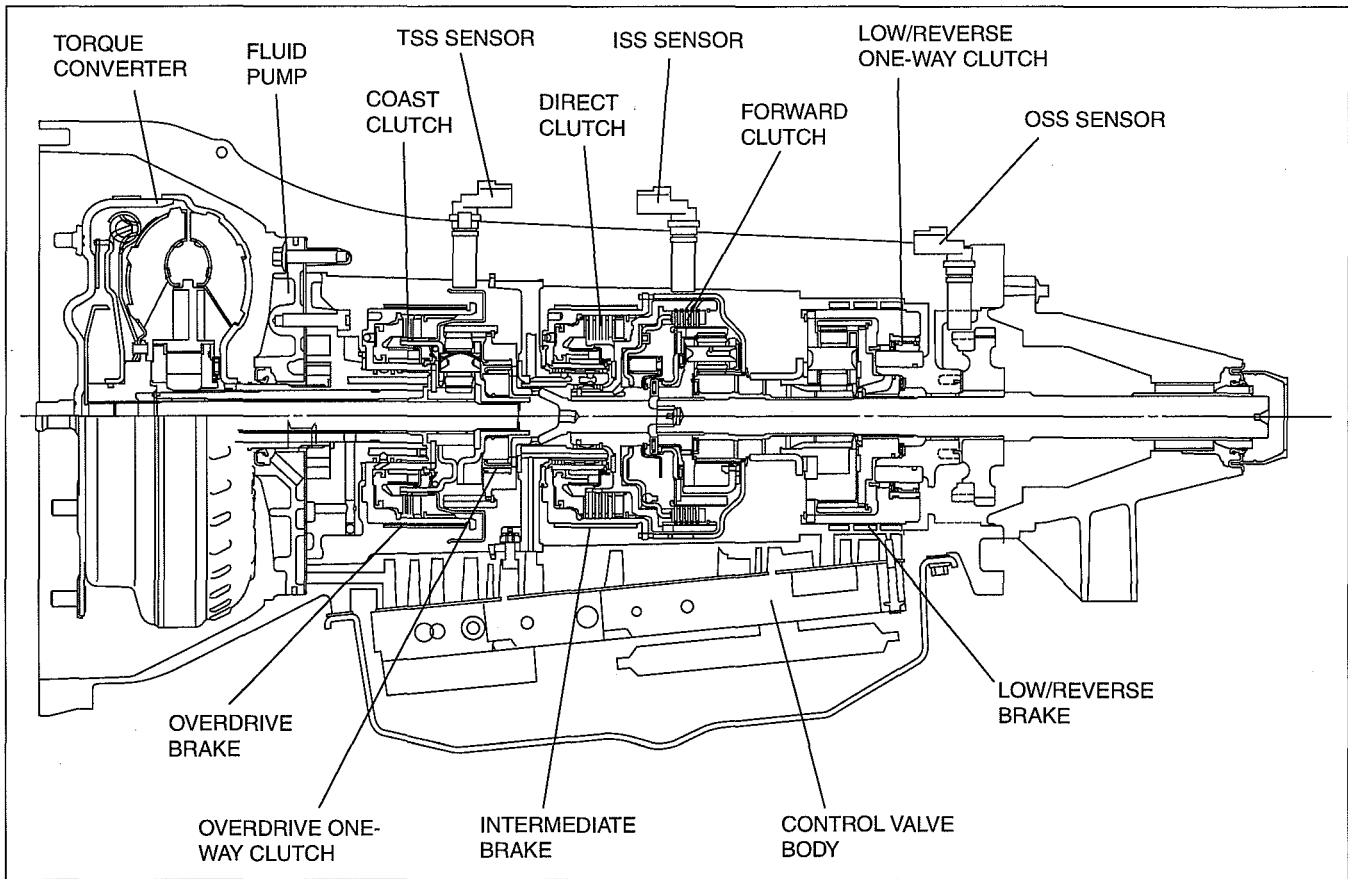


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AUTOMATIC TRANSMISSION [5R55S]

AUTOMATIC TRANSMISSION CROSS-SECTIONAL VIEW [5R55S]

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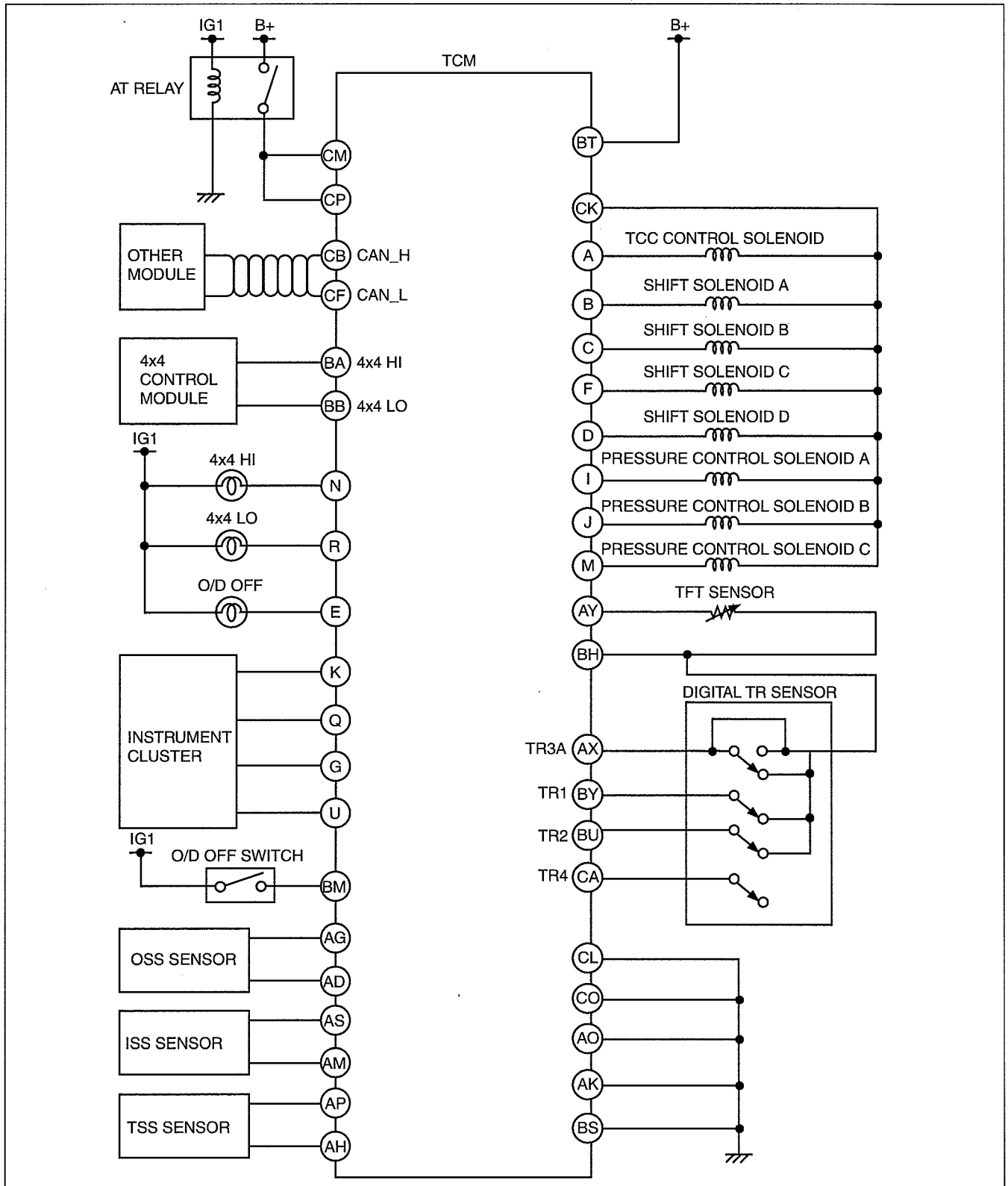


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AUTOMATIC TRANSMISSION [5R55S]

AUTOMATIC TRANSMISSION CONTROL SYSTEM WIRING DIAGRAM [5R55S]

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AUTOMATIC TRANSMISSION [5R55S]

EC-AT OPERATION CHART [5R55S]

id0513c1251000

Solenoid Operation Chart

Selector position		Gear position	Shift solenoid A	Shift solenoid B	Shift solenoid C	Shift solenoid D	Pressure control solenoid A	Pressure control solenoid B	Pressure control solenoid C
P	—	—	On	Off	Off	On	Low	High/Low	Low
R	—	Reverse	On	Off	Off	On	Low/High	Low	High
N	—	—	On	Off	Off	On	Low	High/Low	Low
D	Normal	1GR	On	Off	Off	On	High	High/Low	Low
		2GR	On	Off	On	On	Low/High	High	Low
		3GR	On	On	Off	On	High	Low/High	Low
		4GR	Off	Off	Off	On	High	Low/High	High
		5GR	Off	Off	On	On	High	High/Low	High
	O/D OFF	1GR	On	Off	Off	On	High	High/Low	Low
		2GR	On	Off	On	On	Low/High	High	Low
		3GR	On	On	Off	On	High	Low/High	Low
		4GR	Off	Off	Off	Off	Low/High	High	High
3	—	3GR	On	On	Off	Off	High	Low	High/Low
2	—	2GR	On	Off	On	Off	High	Low	High/Low
1	—	1GR	On	Off	Off	Off	High	Low	High/Low

Band and Clutch Application Chart A

Selector position		Gear position	Overdrive band	Intermediate band	Low/reverse band	Forward clutch	Direct clutch	Coast clutch
P	—	—	—	—	—	—	—	—
R	—	Reverse	—	—	A	—	A	Ac
N	—	—	—	—	—	—	—	—
D	Normal	1GR	—	—	—	A	—	—
		2GR	A	—	—	A	—	—
		3GR	—	A	—	A	—	—
		4GR	—	—	—	A	A	—
		5GR	A	—	—	A	A	—
	O/D OFF	1GR	—	—	—	A	—	—
		2GR	A	—	—	A	—	—
		3GR	—	—	—	A	—	—
		4GR	—	—	—	A	A	Ac
3	—	3GR	—	Ac	—	A	—	Ac
2	—	2GR	A	—	Ac	A	—	—
1	—	1GR	—	—	Ac	A	—	Ac

A : Applied

Ac : Applied to carry coast torque.

— : N/A

AUTOMATIC TRANSMISSION [5R55S]

Band and Clutch Application Chart B

Selector position		Gear position	Direct one-way clutch		Low/reverse one-way clutch		Engine braking
			Drive	Coast	Drive	Coast	
P	—	—	—	—	—	—	—
R	—	Reverse	H	NE	NE	NE	No
N	—	—	—	—	—	—	—
D	Normal	1GR	H	NE	H	NE	No
		2GR	OR	OR	H	NE	No
		3GR	H	NE	OR	OR	No
		4GR	H	NE	OR	OR	No
		5GR	OR	OR	OR	OR	Yes
	O/D OFF	1GR	H	NE	H	NE	Yes
		2GR	OR	OR	H	NE	Yes
		3GR	H	NE	OR	OR	Yes
		4GR	H	NE	OR	OR	Yes
3	—	3GR	H	NE	OR	OR	Yes
2	—	2GR	OR	OR	H	NE	Yes
1	—	1GR	H	NE	H	NE	Yes

H : Hold

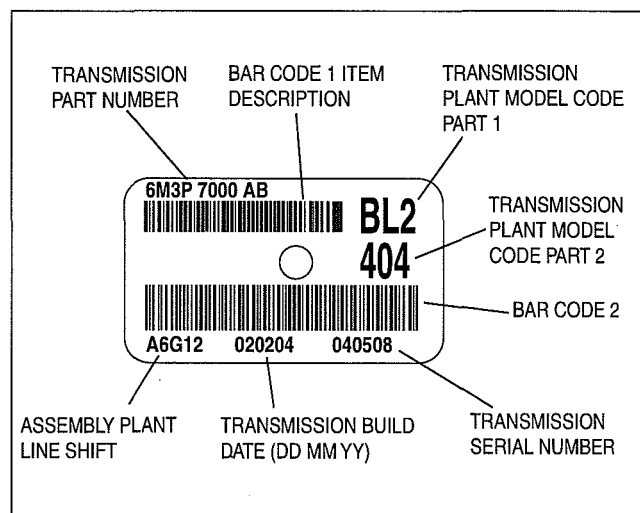
NE: No effect

OR: Overrunning

— : N/A

IDENTIFICATION TAG OUTLINE [5R55S]

- All vehicles are equipped with a vehicle certification label, located on the driver side door lock post. Refer to the code in the space marked TR. For model, service ID level or build date information, refer to the transmission service ID tag located on the transmission case.

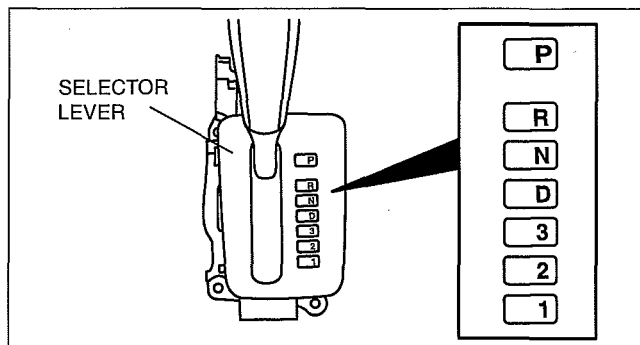


AUTOMATIC TRANSMISSION [5R55S]

RANGE SELECTION [5R55S]

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- The transmission has 7 range positions: P, R, N, D, 3, 2 and 1.



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Park (P Position)

- In the P position:
 - There is no power flow through the transmission.
 - The parking pawl locks the output shaft to the case.
 - The engine may be started.
 - The ignition key may be removed.

Reverse (R Position)

- In the R position:
 - The vehicle may be operated in a rearward direction, at a reduced gear ratio.

Neutral (N Position)

- In the N position:
 - There is no power flow through the transmission.
 - The output shaft is not held and is free to turn.
 - The engine may be started.

Drive—Overdrive Enabled (D range)

- D range (overdrive enabled) is the normal position for most forward driving.
- The D range provides:
 - Automatic shifts 1—5 and 5—1.
 - Apply and release of the torque converter clutch.
 - Maximum fuel economy during normal operation.
 - Engine braking in 5GR

Drive—Overdrive Canceled (D range, O/D OFF mode)

- The D range provides:
 - Automatic shifts 1—4 and 4—1.
 - Apply and release of the torque converter clutch.
 - Maximum fuel economy during normal operation.
 - Engine braking in 4GR

Manual 3 (3 range)

- The 3 range provides:
 - 3GR start and hold.
 - Torque converter clutch apply and release.
 - Improved traction on slippery roads.
 - Engine braking.

Manual 2 (2 range)

- If this position is selected at normal road speeds, the transmission will downshift into the next lower gear and continue downshifting until the vehicle reaches 2GR.
- The 2 range provides:
 - 2GR start and hold.
 - Torque converter clutch apply and release.
 - Improved traction on slippery roads.
 - Engine braking.

Manual 1 (1 range)

- If this position is selected at normal road speeds, the transmission will downshift into the next lower gear and continue downshifting until the vehicle reaches 1GR.
- This position provides:
 - 1GR operation only.
 - Engine braking for descending steep grades.

AUTOMATIC TRANSMISSION [5R55S]

SHIFT PATTERNS [5R55S]

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Upshifts

- Transmission upshifting is controlled by the TCM. The TCM receives inputs from various engine or vehicle sensors and driver demands to control shift scheduling, shift feel and torque converter clutch (TCC) operation.
- The TCM has an adaptive learn strategy to electronically control the transmission which will automatically adjust the shift feel.
- Even if the battery is disconnected or a new battery installed, transmission operating parameters are kept. The TCM must relearn these parameters.

Downshifts

- Under certain conditions the transmission will downshift automatically to a lower gear range (without moving the range selector lever).
- There are 3 categories of automatic downshifts: coastdown, torque demand and forced or kickdown shifts.

Coastdown

- The coastdown downshift occurs when the vehicle is coasting down to a stop.

Torque Demand

- The torque demand downshift occurs (automatically) during part throttle acceleration when the demand for torque is greater than the engine can provide at that gear ratio.
- If applied, the transmission will disengage the TCC to provide added acceleration.

Kickdown

- For maximum acceleration, the driver can force a downshift by pressing the accelerator pedal to the floor.
- A forced downshift into a lower gear is possible below calibrated speeds.
- Specifications for downshift speeds are subject to variations due to tire size and engine and transmission calibration requirements.

AUTOMATIC TRANSMISSION [5R55S]

POWERFLOW OPERATION [5R55S]

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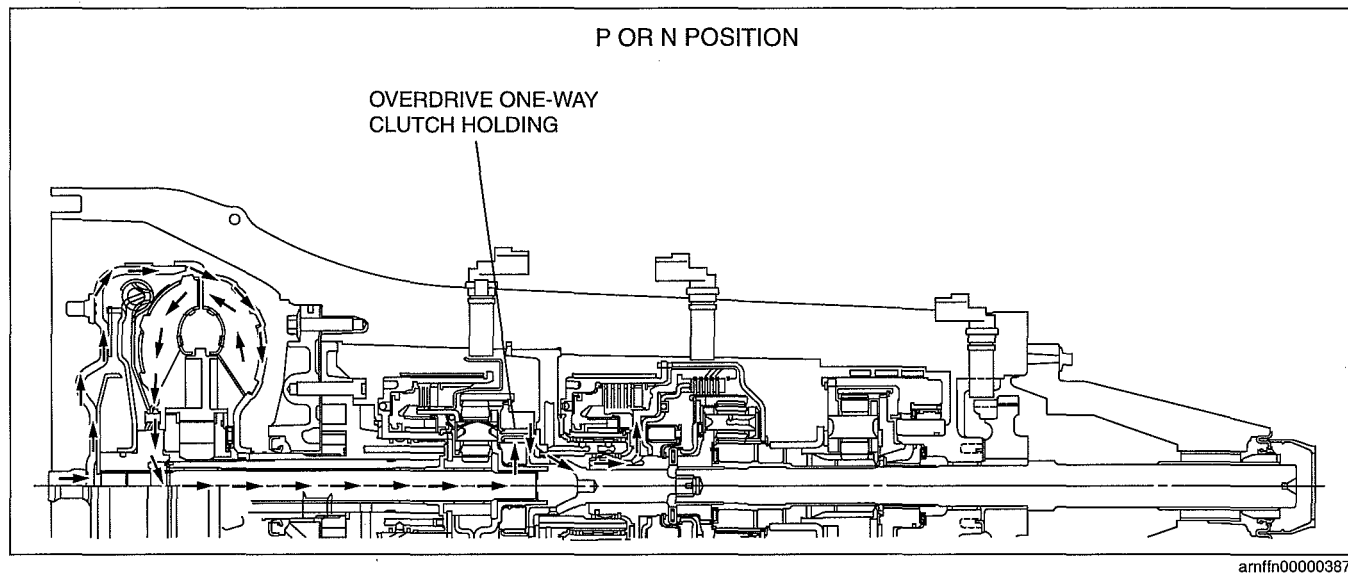
P or N Position

Apply component operation

- Overdrive one-way clutch holding
- Output shaft stationary with the parking pawl holding the parking gear (transmission shifter in P position)

Planetary gearset operation

- Overdrive gearset indirect
- Forward/Reverse gearset driving member:
 - None
- Forward/Reverse gearset driven member:
 - None
- Forward/Reverse gearset held member:
 - None
- Forward/Reverse gearset member also rotating:
 - None



R Position

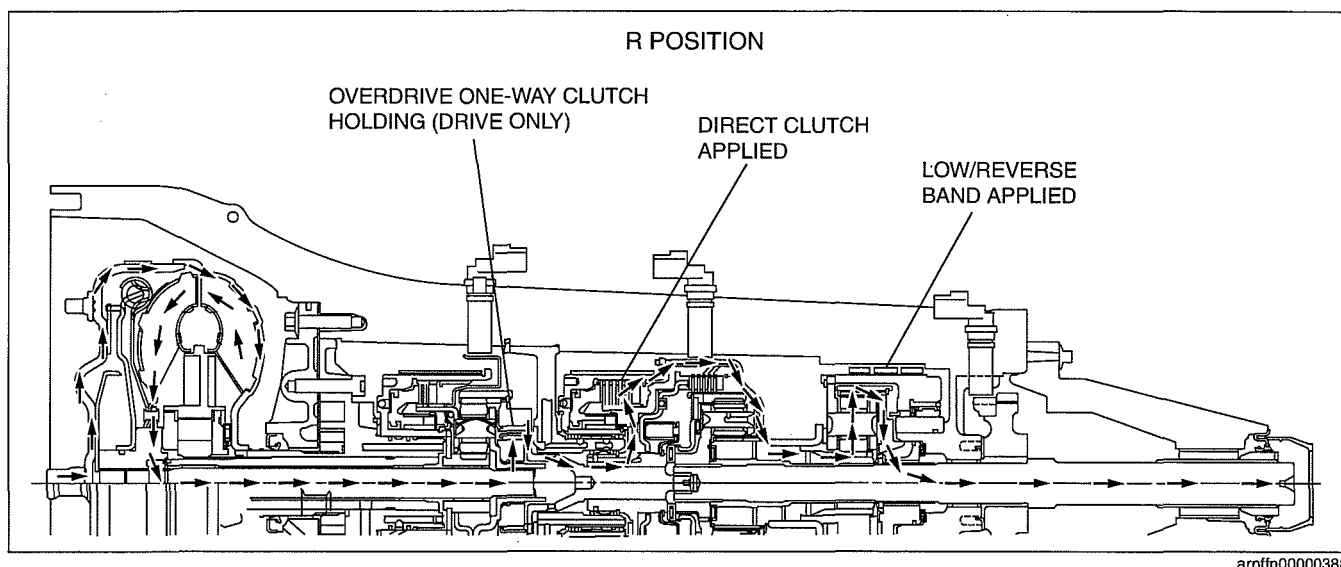
Apply component operation

- Overdrive one-way clutch holding (drive)
- Direct clutch applied
- Low/reverse band applied

Planetary gearset operation

- Overdrive gearset indirect
- Forward/Reverse gearset driving member:
 - Forward/reverse sun gear
- Forward/Reverse gearset driven member:
 - Forward planetary carrier component and reverse ring gear
- Forward/Reverse gearset held member:
 - Reverse planetary carrier component
- Forward/Reverse gearset member also rotating:
 - Forward ring gear

AUTOMATIC TRANSMISSION [5R55S]



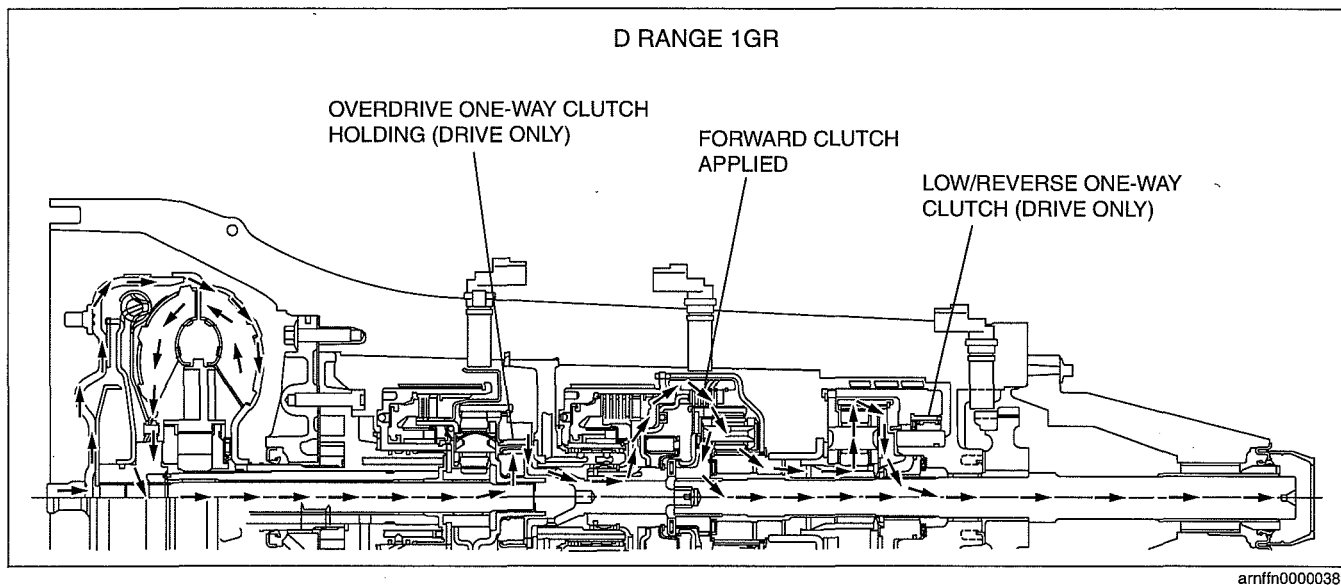
D Range 1GR

Apply component operation

- Overdrive one-way clutch holding (drive)
- Forward clutch applied
- Low/reverse one-way clutch holding (drive)

Planetary gearset operation

- Overdrive gearset indirect
- Forward/Reverse gearset driving member:
 - Forward ring gear
- Forward/Reverse gearset driven member:
 - Forward planetary carrier component and reverse ring gear
- Forward/Reverse gearset held member:
 - Reverse planetary carrier component
- Forward/Reverse gearset member also rotating:
 - Forward/reverse sun gear



AUTOMATIC TRANSMISSION [5R55S]

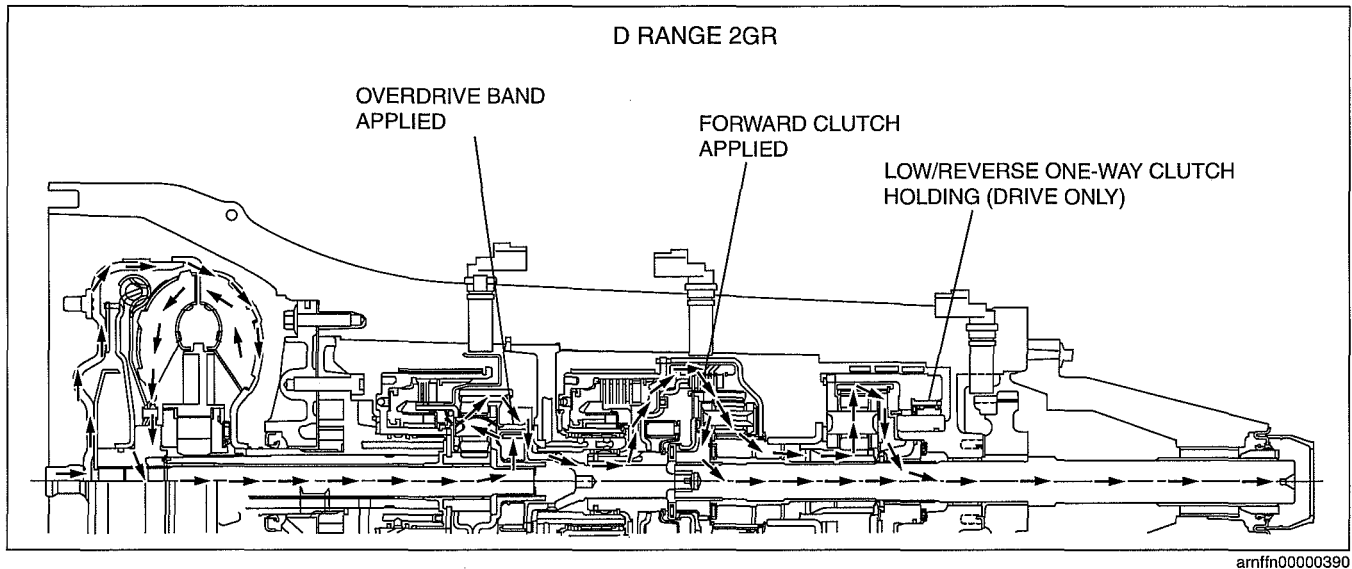
D Range 2GR

Apply component operation

- Overdrive band applied
- Forward clutch applied
- Low/reverse one-way clutch holding (drive)

Planetary gearset operation

- Overdrive gearset in overdrive
- Forward/Reverse gearset driving member:
 - Forward ring gear
- Forward/Reverse gearset driven member:
 - Forward planetary carrier component and reverse ring gear
- Forward/Reverse gearset held member:
 - Reverse planetary carrier component
- Forward/Reverse gearset member also rotating:
 - Forward/reverse sun gear



D Range 3GR

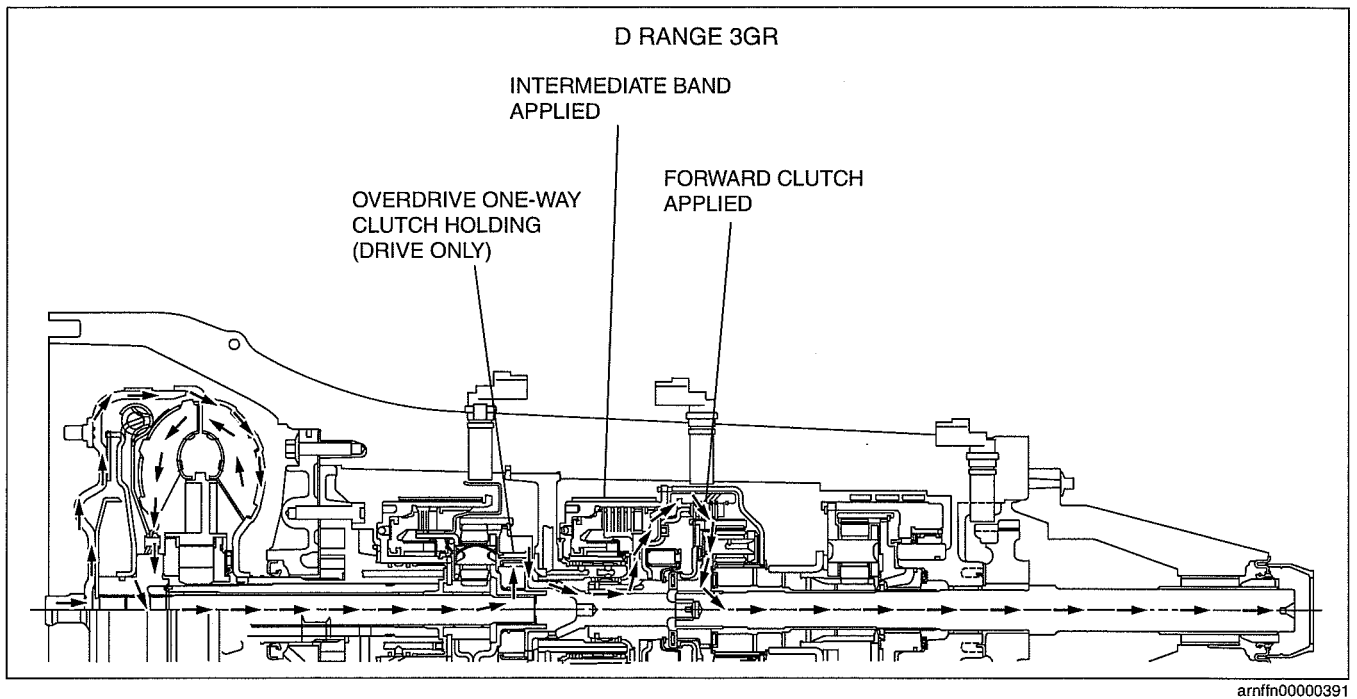
Apply component operation

- Overdrive one-way clutch holding (drive)
- Forward clutch applied
- Intermediate band applied

Planetary gearset operation

- Overdrive gearset indirect
- Forward/Reverse gearset driving member:
 - Forward ring gear
- Forward/Reverse gearset driven member:
 - Forward planetary carrier component and reverse ring gear
- Forward/Reverse gearset held member:
 - Forward/reverse sun gear
- Forward/Reverse gearset member also rotating:
 - Reverse planetary carrier component

AUTOMATIC TRANSMISSION [5R55S]



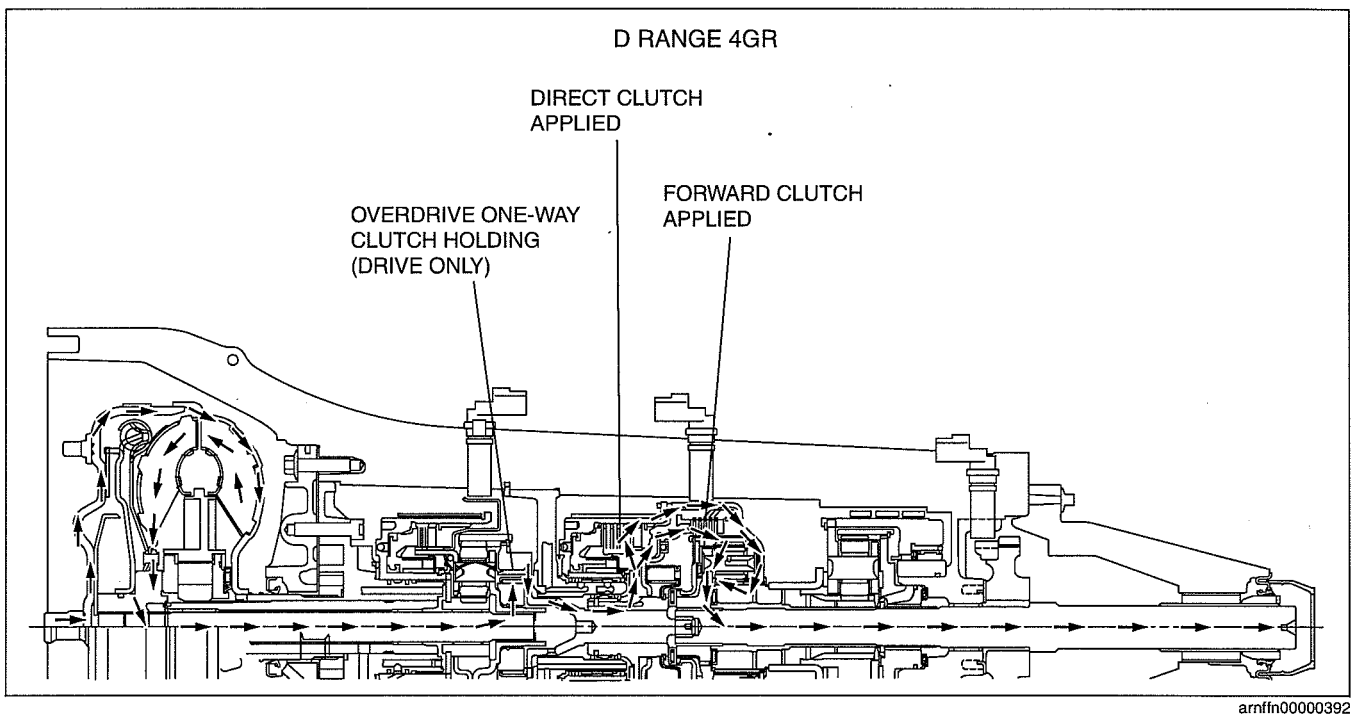
D Range 4GR

Apply component operation

- Overdrive one-way clutch holding (drive)
- Forward clutch applied
- Direct clutch applied

Planetary gearset operation

- Overdrive gearset indirect
- Forward/Reverse gearset driving member:
 - Forward ring gear
 - Forward/reverse sun gear
- Forward/Reverse gearset driven member:
 - Forward planetary carrier component and reverse ring gear
- Forward/Reverse gearset held member:
 - None
- Forward/Reverse gearset member also rotating:
 - Reverse planetary carrier component



AUTOMATIC TRANSMISSION [5R55S]

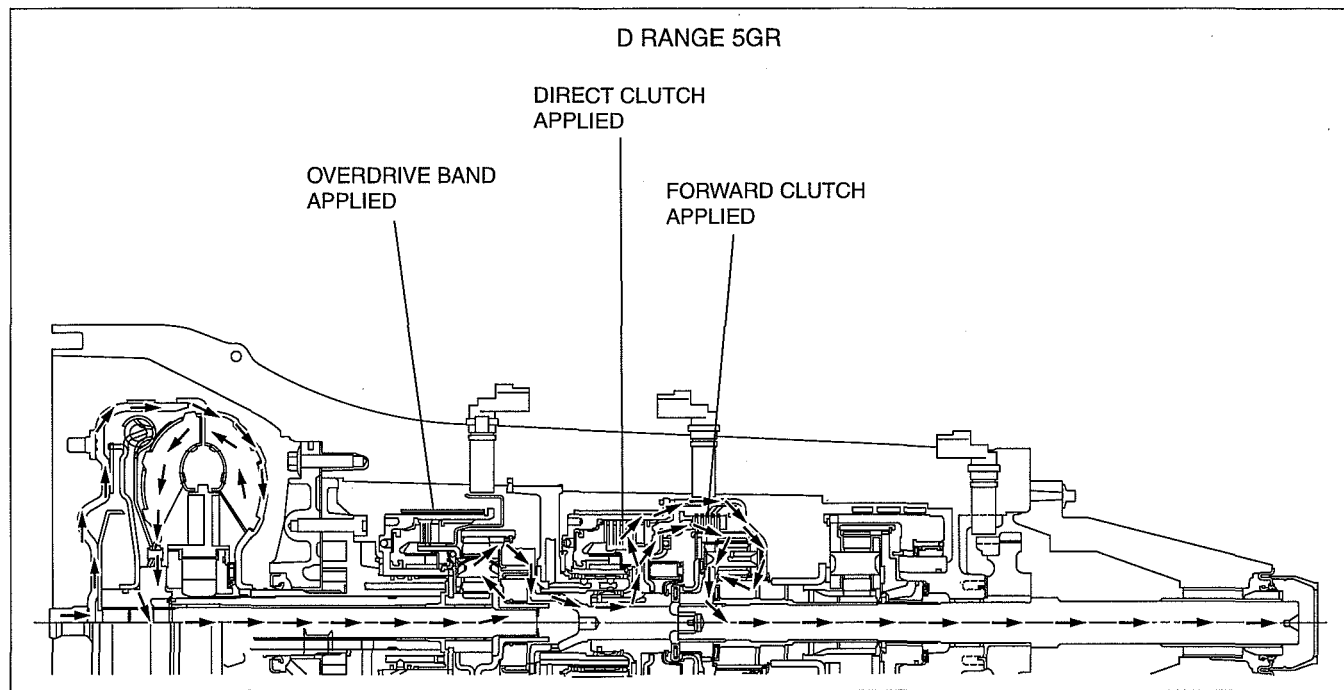
D Range 5GR

Apply component operation

- Overdrive band applied
- Forward clutch applied
- Direct clutch applied

Planetary gearset operation

- Overdrive gearset in overdrive
- Forward/Reverse gearset driving member:
 - Forward ring gear
 - Forward/reverse sun gear
- Forward/Reverse gearset driven member:
 - Forward planetary carrier component and reverse ring gear
- Forward/Reverse gearset held member:
 - None
- Forward/Reverse gearset member also rotating:
 - Reverse planetary carrier component



D Range 4GR, O/D OFF Mode (Manual 4GR)

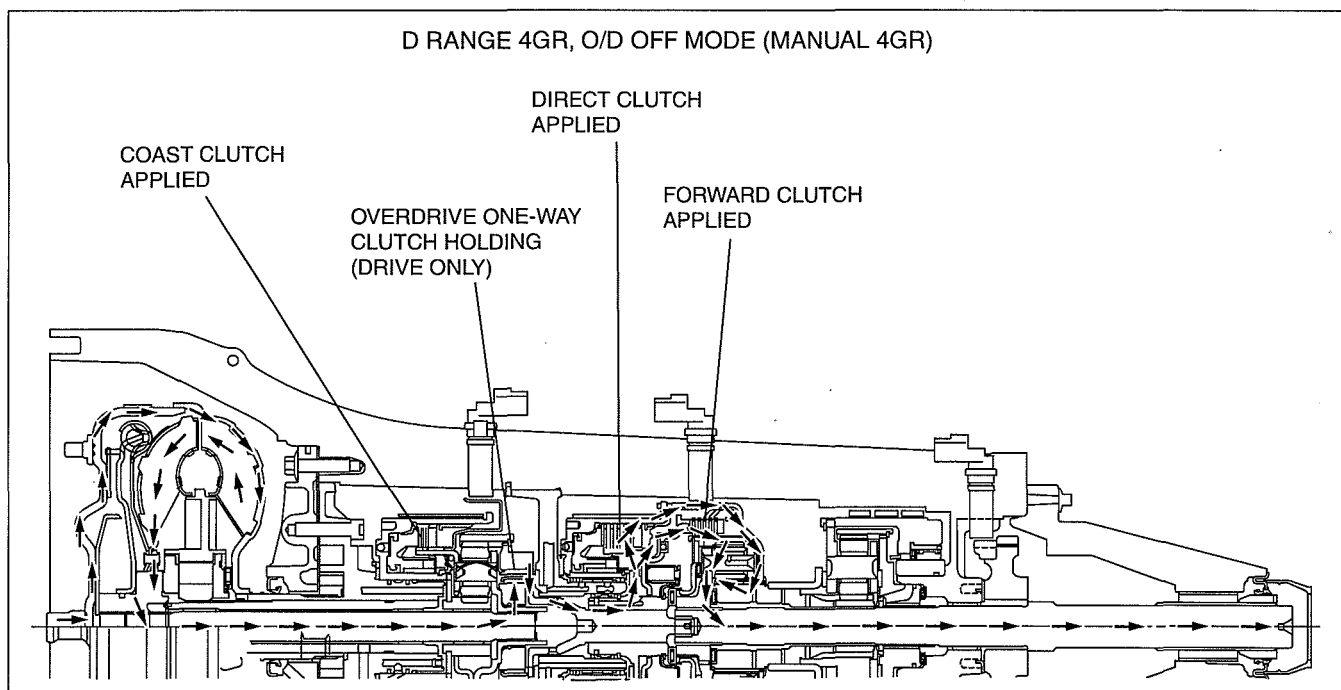
Apply component operation

- Coast clutch applied
- Overdrive one-way clutch holding (drive)
- Forward clutch applied
- Direct clutch applied

Planetary gearset operation

- Overdrive gearset indirect
- Forward/Reverse gearset driving member:
 - Forward ring gear
 - Forward/reverse sun gear
- Forward/Reverse gearset driven member:
 - Forward planetary carrier component and reverse ring gear
- Forward/Reverse gearset held member:
 - None
- Forward/Reverse gearset member also rotating:
 - Reverse planetary carrier component

AUTOMATIC TRANSMISSION [5R55S]



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3 Range (Manual 3GR)

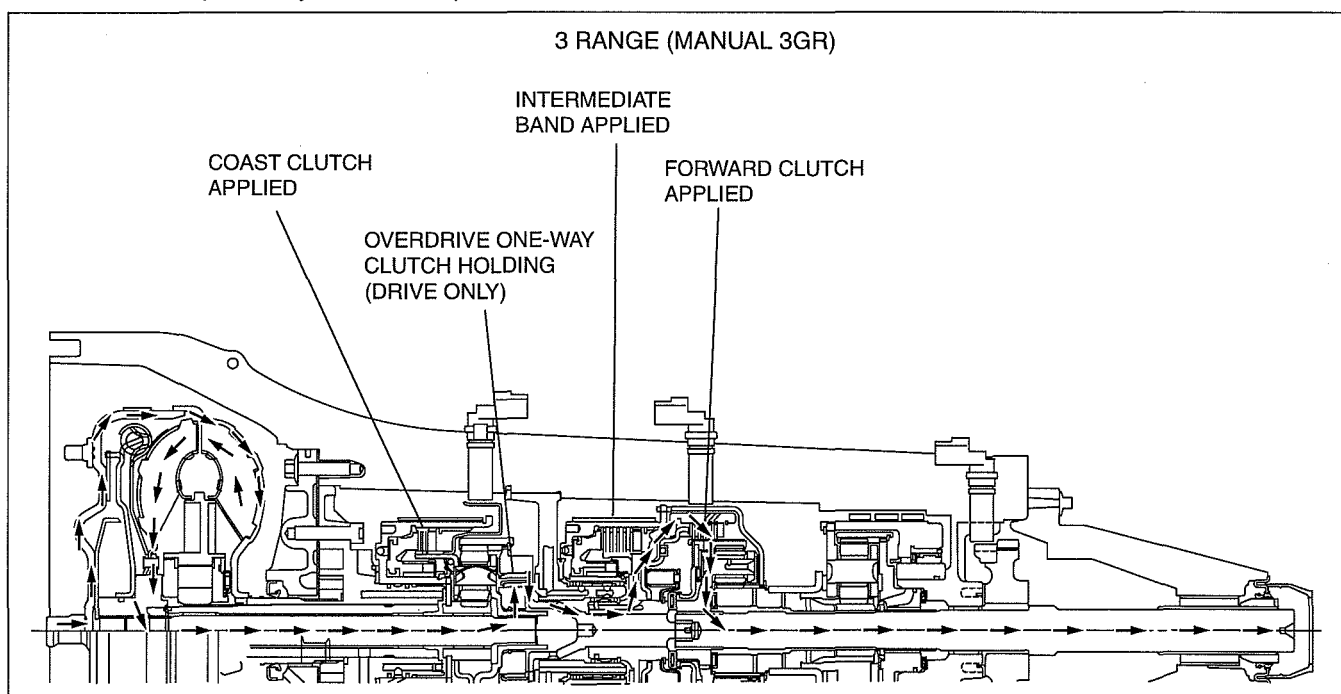
Apply component operation

- Coast clutch applied
- Overdrive one-way clutch holding (drive)
- Forward clutch applied
- Intermediate band applied

Planetary gearset operation

- Overdrive gearset indirect
- Forward/Reverse gearset driving member:
 - Forward ring gear
- Forward/Reverse gearset driven member:
 - Forward planetary carrier component and reverse ring gear
- Forward/Reverse gearset held member:
 - Forward/reverse sun gear
- Forward/Reverse gearset member also rotating:
 - Reverse planetary carrier component

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AUTOMATIC TRANSMISSION [5R55S]

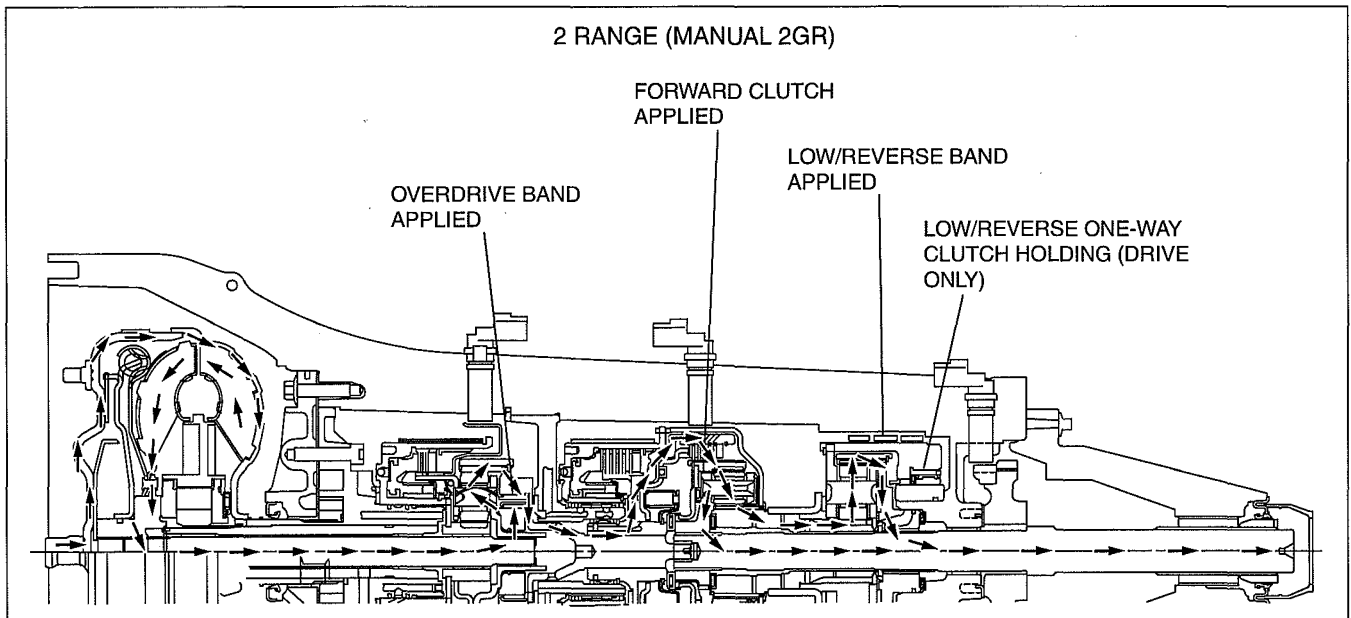
2 Range (Manual 2GR)

Apply component operation

- Overdrive band applied
- Forward clutch applied
- Low/reverse one-way clutch holding (drive)
- Low/reverse band applied

Planetary gearset operation

- Overdrive gearset in overdrive
- Forward/Reverse gearset driving member:
 - Forward ring gear
- Forward/Reverse gearset driven member:
 - Forward planetary carrier component and reverse ring gear
- Forward/Reverse gearset held member:
 - Reverse planetary carrier component
- Forward/Reverse gearset member also rotating:
 - Forward/reverse sun gear



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AUTOMATIC TRANSMISSION [5R55S]

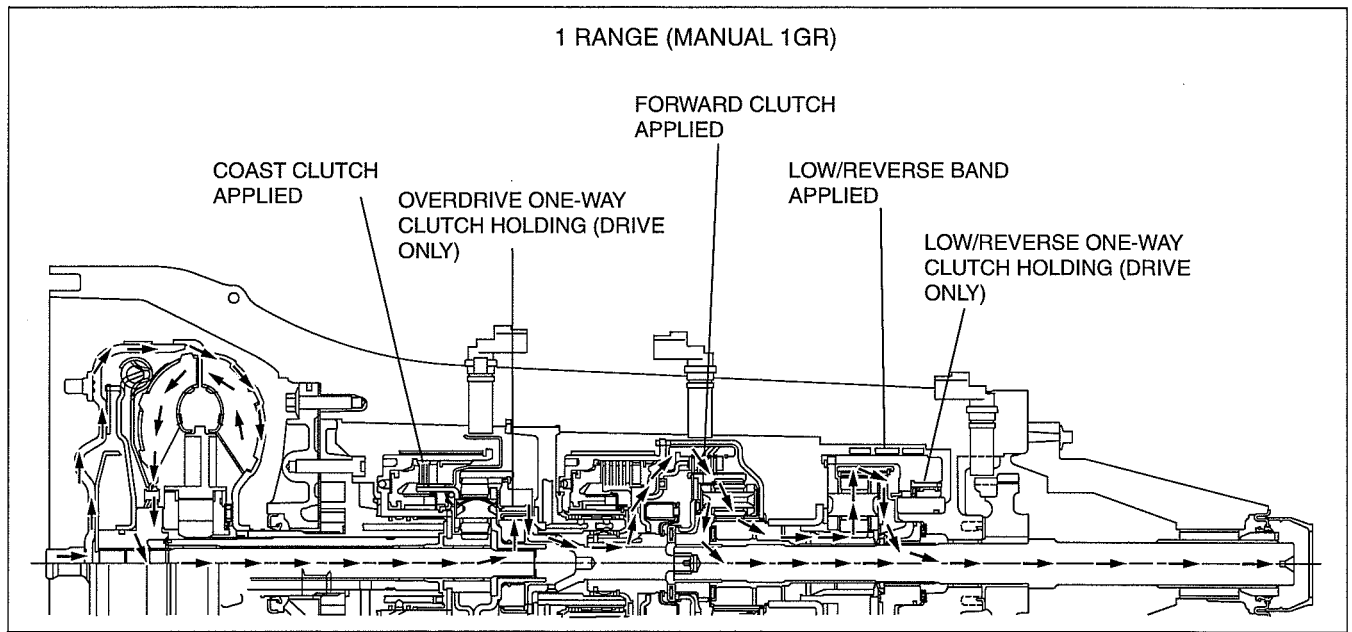
1 Range (Manual 1GR)

Apply component operation

- Coast clutch applied
- Overdrive one-way clutch holding (drive)
- Forward clutch applied
- Low/reverse one-way clutch holding (drive)
- Low/reverse band applied

Planetary gearset operation

- Overdrive gearset in overdrive
- Forward/Reverse gearset driving member:
 - Forward ring gear
- Forward/Reverse gearset driven member:
 - Forward planetary carrier component and reverse ring gear
- Forward/Reverse gearset held member:
 - Reverse planetary carrier component
- Forward/Reverse gearset member also rotating:
 - Forward/reverse sun gear

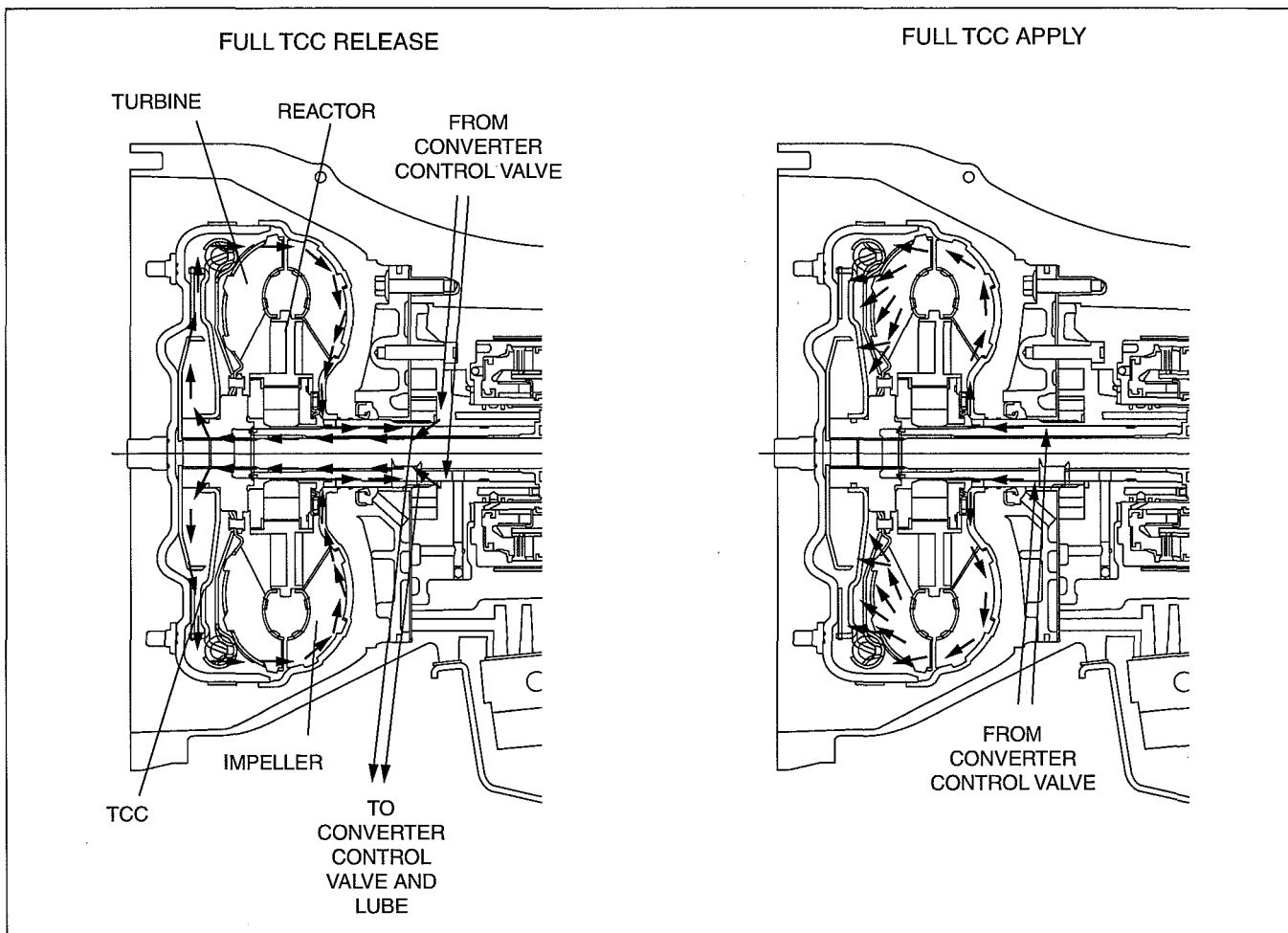


AUTOMATIC TRANSMISSION [5R55S]

TORQUE CONVERTER OUTLINE [5R55S]

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- The torque converter of the 5R55S transmission is a four-element component. It contains an impeller, a turbine and a reactor, plus TCC (torque converter clutch) for increased fuel economy.
- The impeller and cover assembly drives the impeller blades and fluid pump. It contains hydraulic fluid and provides a mating surface for the TCC damper assembly.
- The turbine is driven by fluid from the impeller and transmits power to the turbine shaft and planetary gearsets.
- The reactor redirects fluid flow returned from the turbine to the impeller so that it rotates in the same direction as the impeller. This action assists in torque multiplication.
- The reactor has a one-way clutch to hold it stationary during torque multiplication (at lower vehicle speeds) and allow it to rotate (at higher vehicle speeds).
- TCC has the following features:
 - A piston with damper assembly that connects to the turbine and transmission input shaft
 - A friction paper surface on the piston (or on a ring connected to the piston) that makes contact with the torque converter cover when TCC is applied
 - Fluid that flows through the torque converter and pushes the TCC piston away from the torque converter cover during TCC release
 - Fluid that flows through the torque converter and pushes the TCC piston into contact with the torque converter cover during TCC apply
- TCC operates in three stages:
 - Full release
 - Controlled modulation
 - Full apply
- The TCM controls TCC operation, using the TCC control solenoid.
- TCC control solenoid operation provides the modulation of hydraulic pressure to change the position of the control valves. The control valves change the pressure and direction of fluid flow in the torque converter.
- TCC may be applied in 2nd gear, 3rd gear, 4th gear and 5th gear.



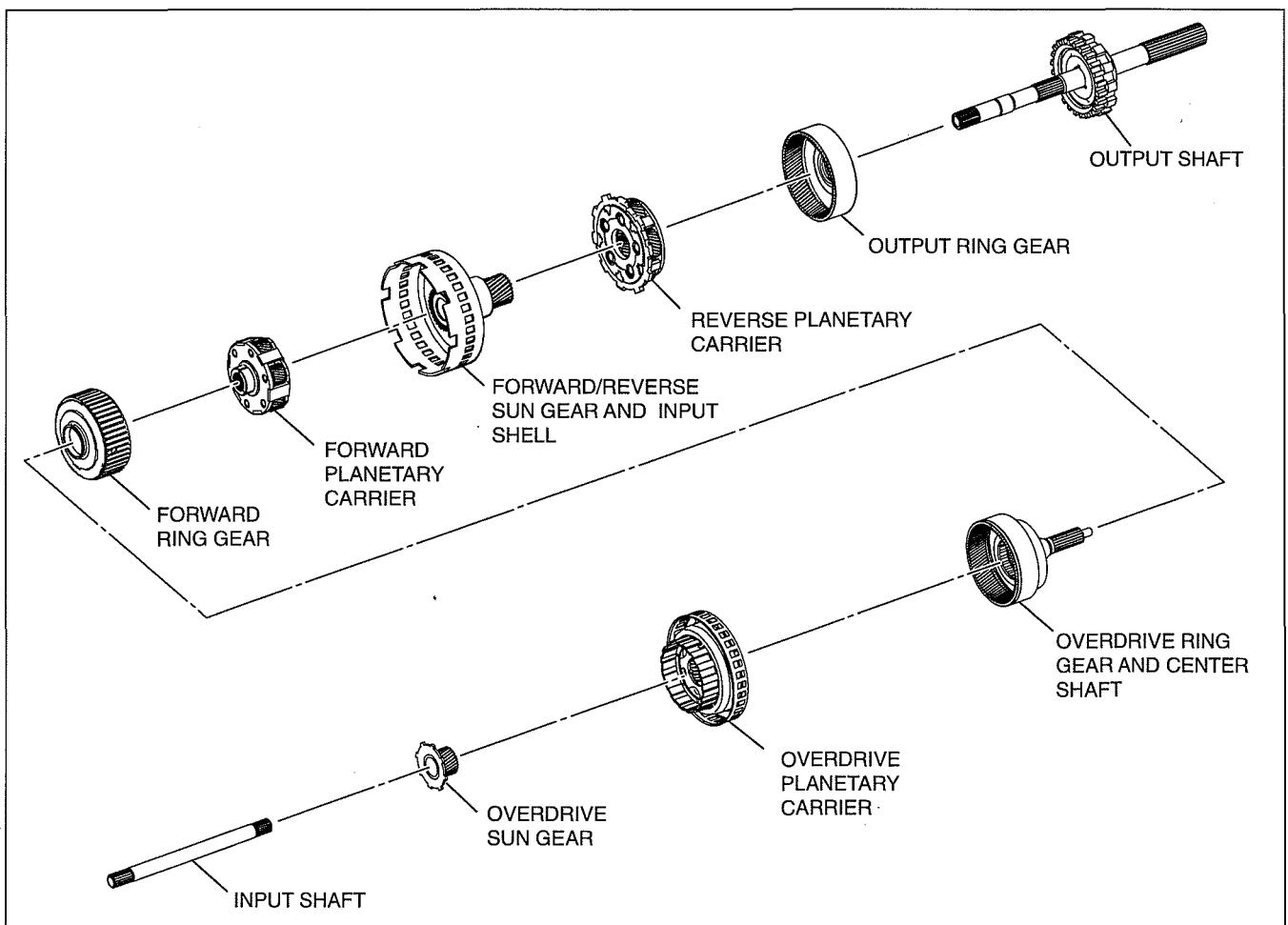
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AUTOMATIC TRANSMISSION [5R55S]

PLANETARY GEARSETS CONSTRUCTION [5R55S]

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- The 5R55S transmission has three planetary gearsets to provide operation in reverse and in five forward speeds.
- The overdrive planetary gearset has the following parts:
 - Overdrive sun gear
 - Overdrive planetary carrier component
 - Overdrive ring gear
- The input shaft rotates the overdrive planetary carrier component as a driving member. The input shaft may also drive the overdrive ring gear directly with a 1.00 to 1 ratio.
- When the transmission case holds the overdrive sun gear stationary, the overdrive planetary carrier drives the overdrive ring gear with an overdrive ratio of either 0.71 to 1.
- The forward and reverse planetary gearsets have the following parts:
 - Forward/reverse sun gear
 - Forward planetary carrier component
 - Forward ring gear
 - Reverse planetary carrier component
 - Output ring gear
- The forward/reverse sun gear is a driving member in reverse. The forward ring gear is a driving member in each forward gear.
- The forward/reverse sun gear and forward ring gear are driving members for a 1.00 to 1 ratio in 4GR and 5GR.
- The forward planetary carrier component and output ring gear are both connected to the output shaft and are driven members in reverse and in all forward speeds.
- The transmission case holds the reverse planetary carrier component stationary in 1GR and 2GR.
- The reverse planetary carrier component is a member that rotates at a different speed than the driving and driven members in 3GR.
- The reverse planetary carrier component rotates at the same speed as the forward/reverse sun gear and forward ring gear as part of the 1.00 to 1 ratio in 4GR and 5GR.



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AUTOMATIC TRANSMISSION [5R55S]

APPLY COMPONENTS CONSTRUCTION [5R55S]

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Clutch/Band Operation

- The apply components of the 5R55S transmission work together to provide a flow of power.
- The coast clutch is applied to provide coast braking in manual 1GR, manual 3GR and manual 4GR. Coast braking occurs without the use of the coast clutch in manual 2GR and in 5GR.

Position/Range		R	D					D (O/D OFF)	3	2	1	Planetary component
Gear position		Reverse	1GR	2GR	3GR	4GR	5GR	Manual 4GR	Manual 3GR	Manual 2GR	Manual 1GR	
Overdrive band				A			A			A		OS
Intermediate band					A				A			FRS
Low/reverse band		A								A	A	RP
Coast clutch								A	A		A	OS
Direct clutch		A				A	A	A				FRS
Forward clutch			A	A	A	A	A	A	A	A	A	FR
Overdrive one-way clutch	Drive	H	H	OR	H	H	OR	H	H	OR	H	OS
	Coast	OR	OR	OR	OR	OR	OR	OR	OR	OR	OR	
Low/ reverse one-way clutch	Drive		H	H	OR	OR	OR	OR	OR	H	H	RP
	Coast		OR	OR	OR	OR	OR	OR	OR	OR	OR	

A : Applied

H : Holding

OR: Overrunning

OS: Overdrive sun gear

RP: Reverse planetary carrier component

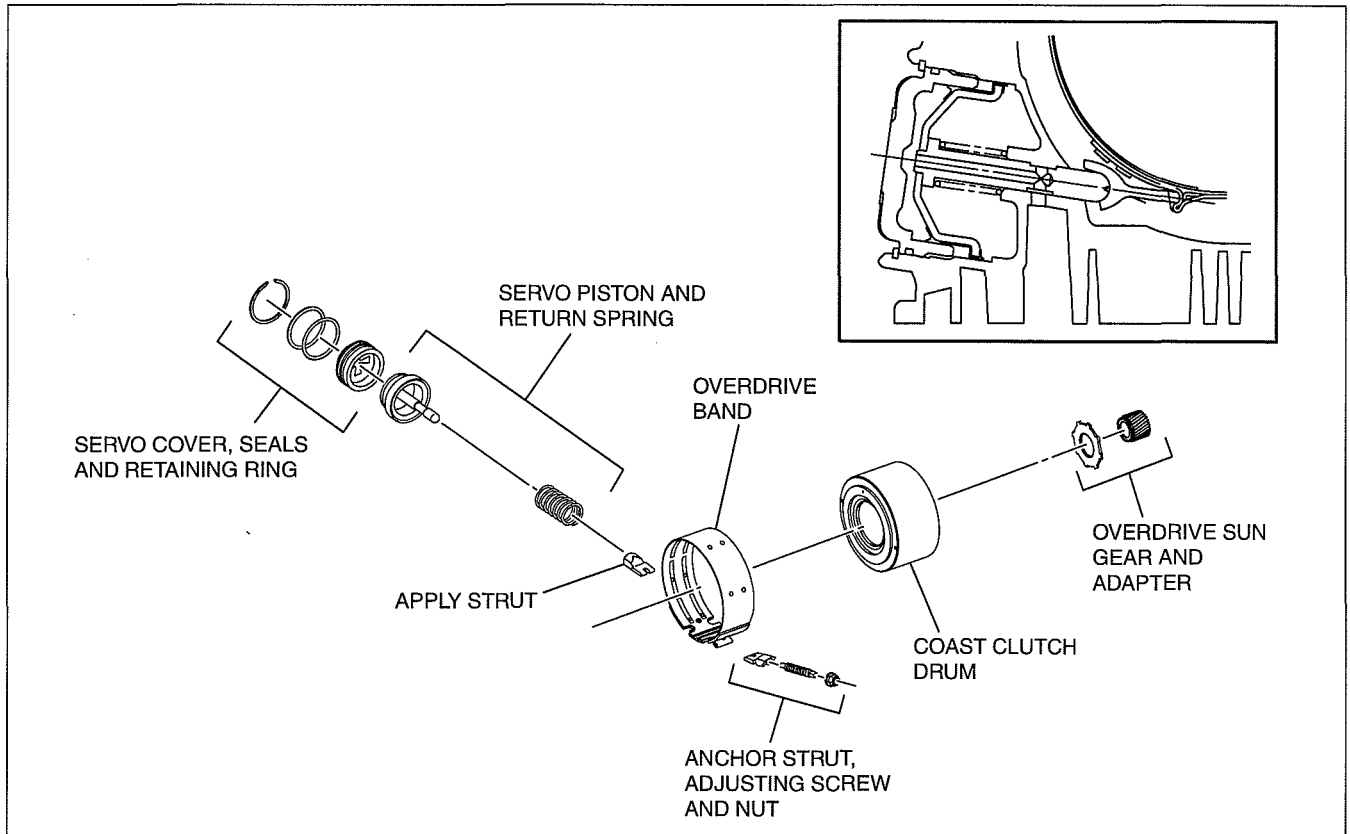
FRS: Forward/reverse sun gear

FR: Forward ring gear

AUTOMATIC TRANSMISSION [5R55S]

Overdrive Band

- The overdrive band connects the overdrive sun gear to the transmission case.
- The overdrive band is applied in 2GR and 5GR, as well as in manual 2GR.
- During 2GR and 5GR operation, hydraulic pressure is applied to the overdrive servo.
 - This pressure causes the piston to move and apply force to the band.
 - This action causes the overdrive band to hold the overdrive drum.
 - This causes the overdrive sun gear to be held stationary through the adapter plate and the overdrive drum.

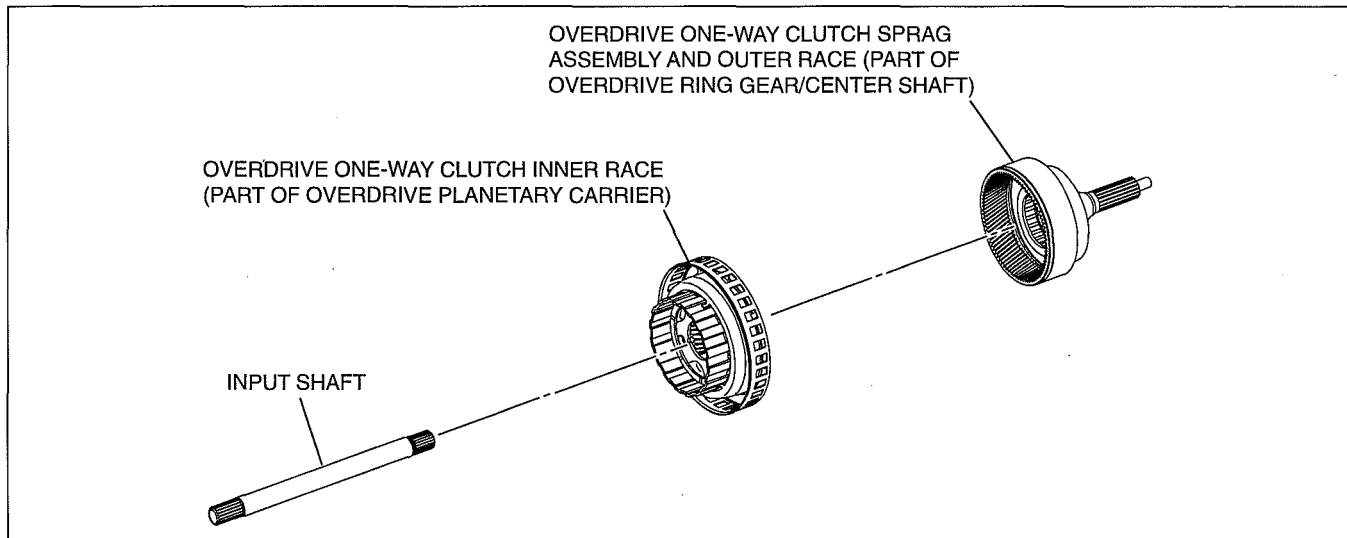


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AUTOMATIC TRANSMISSION [5R55S]

Overdrive One-Way Clutch

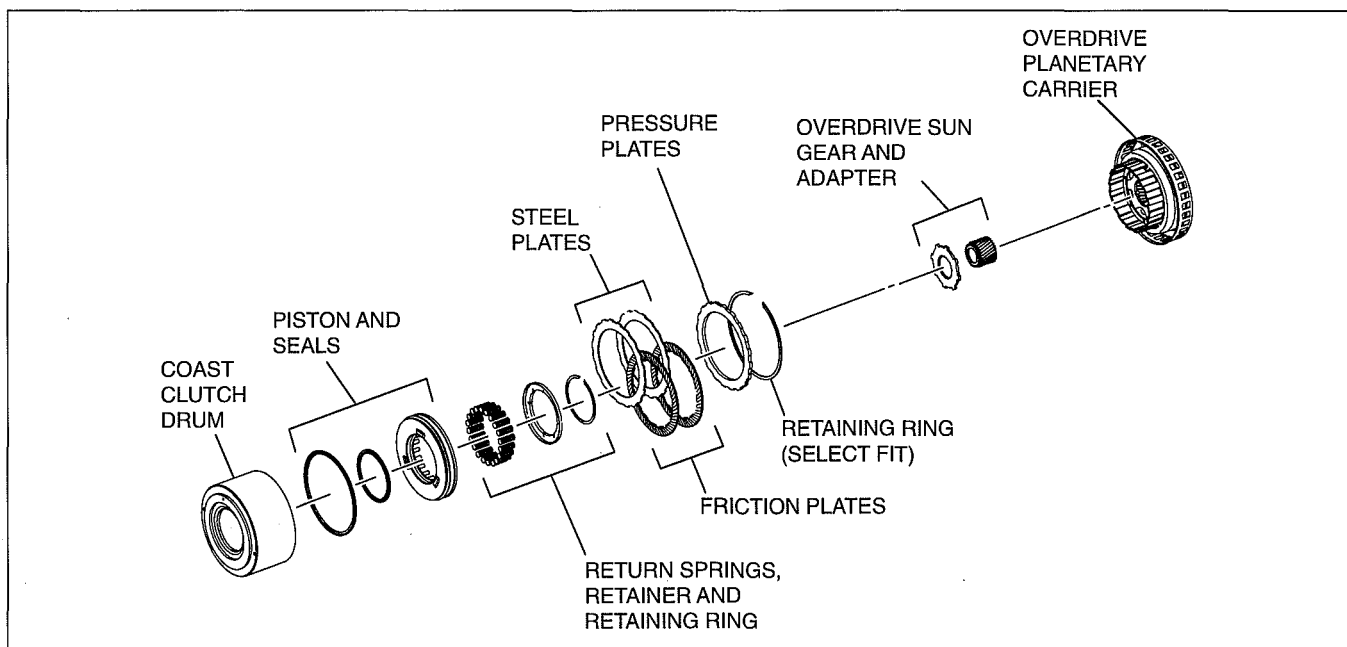
- The overdrive one-way clutch connects the input shaft to the center shaft during drive operation.
- The overdrive one-way clutch transmits torque in Reverse, 1GR, 3GR and 4GR, as well as in manual 3GR.
- The direct one-way clutch is a sprag-type one-way clutch that is pressed into the center shaft.
 - The overdrive one-way clutch is driven by the ring gear of the overdrive planetary carrier.
 - The overdrive one-way clutch holds and drives the outer splines of the center shaft in 1GR, 3GR, 4GR and Reverse gears.
 - The overdrive one-way clutch overruns during all coast operations and at all times in 2GR and 5GR.



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Coast Clutch

- The coast clutch connects the overdrive planetary carrier component to the overdrive sun gear.
- The coast clutch is applied in manual 1GR, manual 3GR, manual 4GR (D range O/D OFF mode) and Reverse positions.
- The coast clutch is a multi-disc clutch made up of steel and friction plates.
 - The coast clutch is applied with hydraulic pressure and disengaged by return springs and the exhaust of the hydraulic pressure.
 - The coast clutch is housed in the overdrive drum.
 - When applied, the coast clutch locks the overdrive sun gear to the overdrive planetary carrier, thus preventing the one-way clutch from overrunning when the vehicle is coasting.
 - This allows the use of engine compression to help slow the vehicle and provide engine braking.

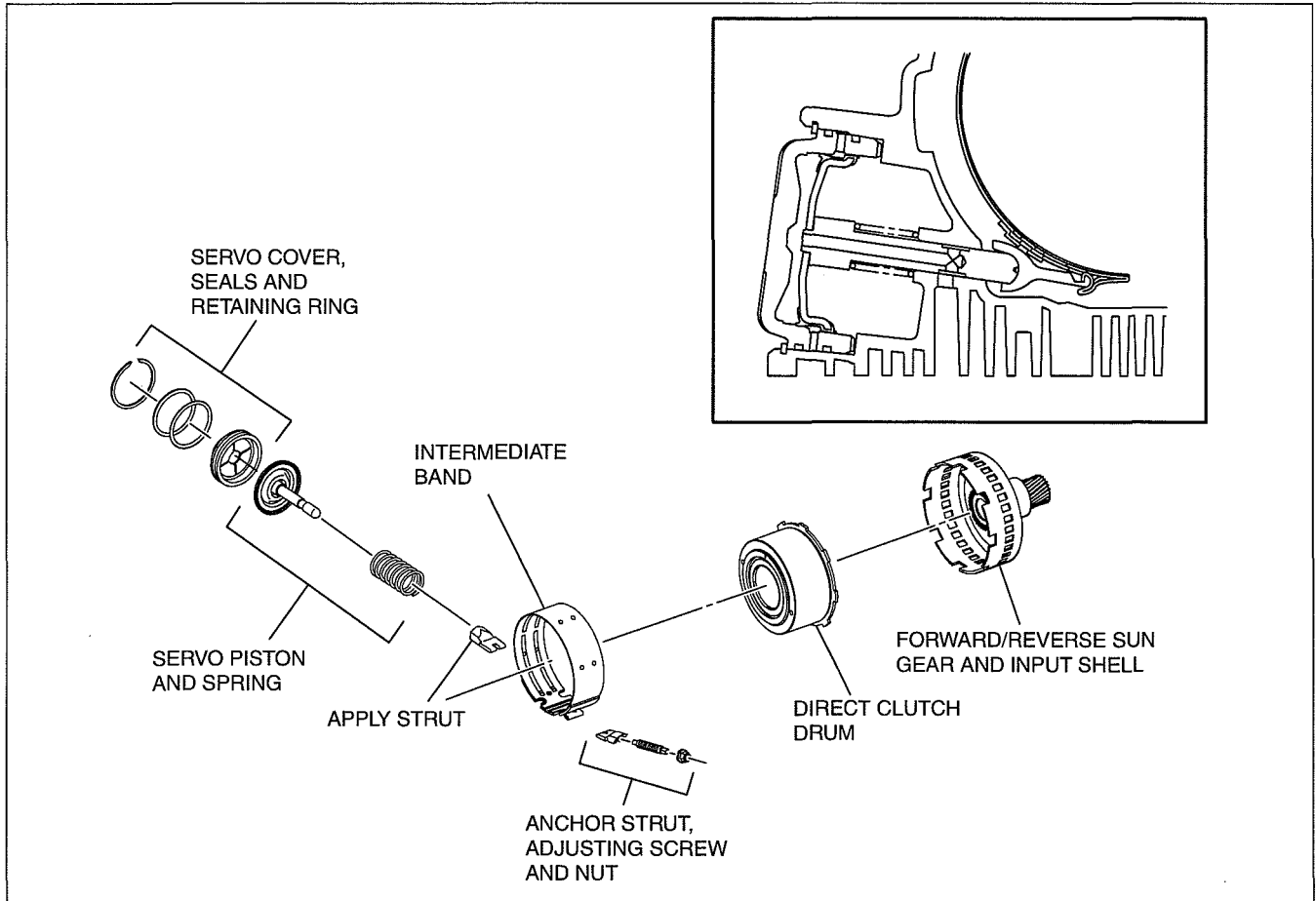


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AUTOMATIC TRANSMISSION [5R55S]

Intermediate Band

- The intermediate band connects the forward/reverse sun gear to the transmission case.
- The intermediate band is applied in 3GR, as well as in manual 3GR.
- During 3GR operation, hydraulic pressure is applied to the intermediate servo.
 - This pressure causes the servo to move and apply force to the intermediate band.
 - This action causes the direct clutch drum to be held.
 - The intermediate band holds the intermediate brake and direct clutch drum to the case in 3GR.
 - This causes the input shell and forward sun gear to be held stationary.

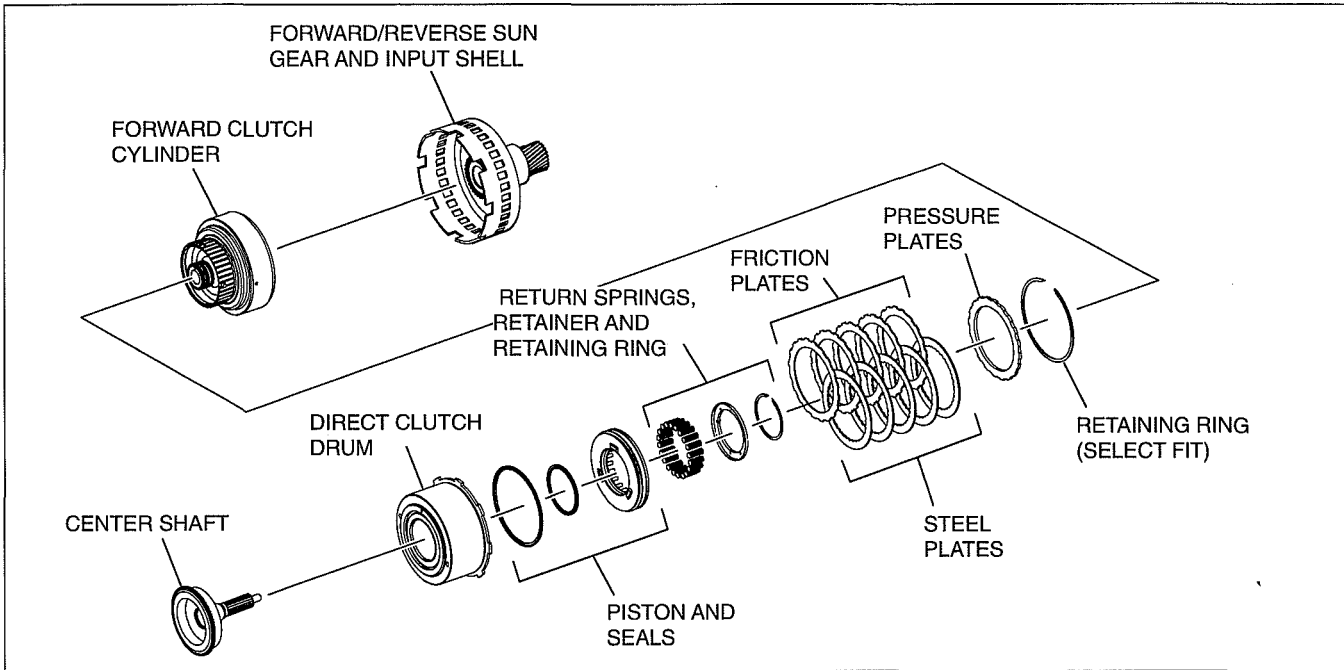


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AUTOMATIC TRANSMISSION [5R55S]

Direct Clutch

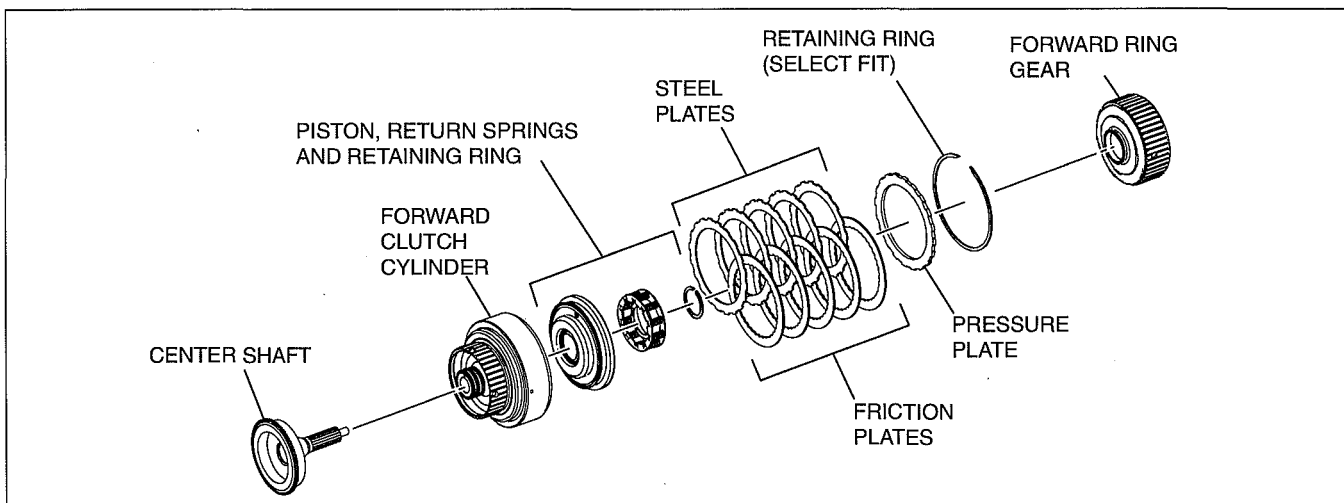
- The direct clutch connects the center shaft to the forward/reverse sun gear.
- The direct clutch is applied in Reverse, 4GR and 5GR, as well as in manual 4GR (D range O/D OFF mode).
- The direct clutch is a multi-disc clutch made up of steel and friction plates.
 - The direct clutch is applied with hydraulic pressure and disengaged by return springs and the exhaust of the hydraulic pressure.
 - It is housed in the direct clutch drum.
 - During 4GR, 5GR and Reverse gear application, the direct clutch is applied transferring torque from the forward clutch cylinder to the direct clutch drum.
 - This action causes the forward sun gear to drive the pinions of the low/reverse planetary carrier.



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Forward Clutch

- The forward clutch connects the center shaft to the forward ring gear.
- The forward clutch is applied in all forward gears.
- The forward clutch is a multi-disc clutch made up of steel and friction plates.
 - The forward clutch is applied with hydraulic pressure and disengaged by return springs and the exhaust of the hydraulic pressure.
 - When applied, the forward clutch provides a direct mechanical coupling between the center shaft and the forward ring gear and hub.

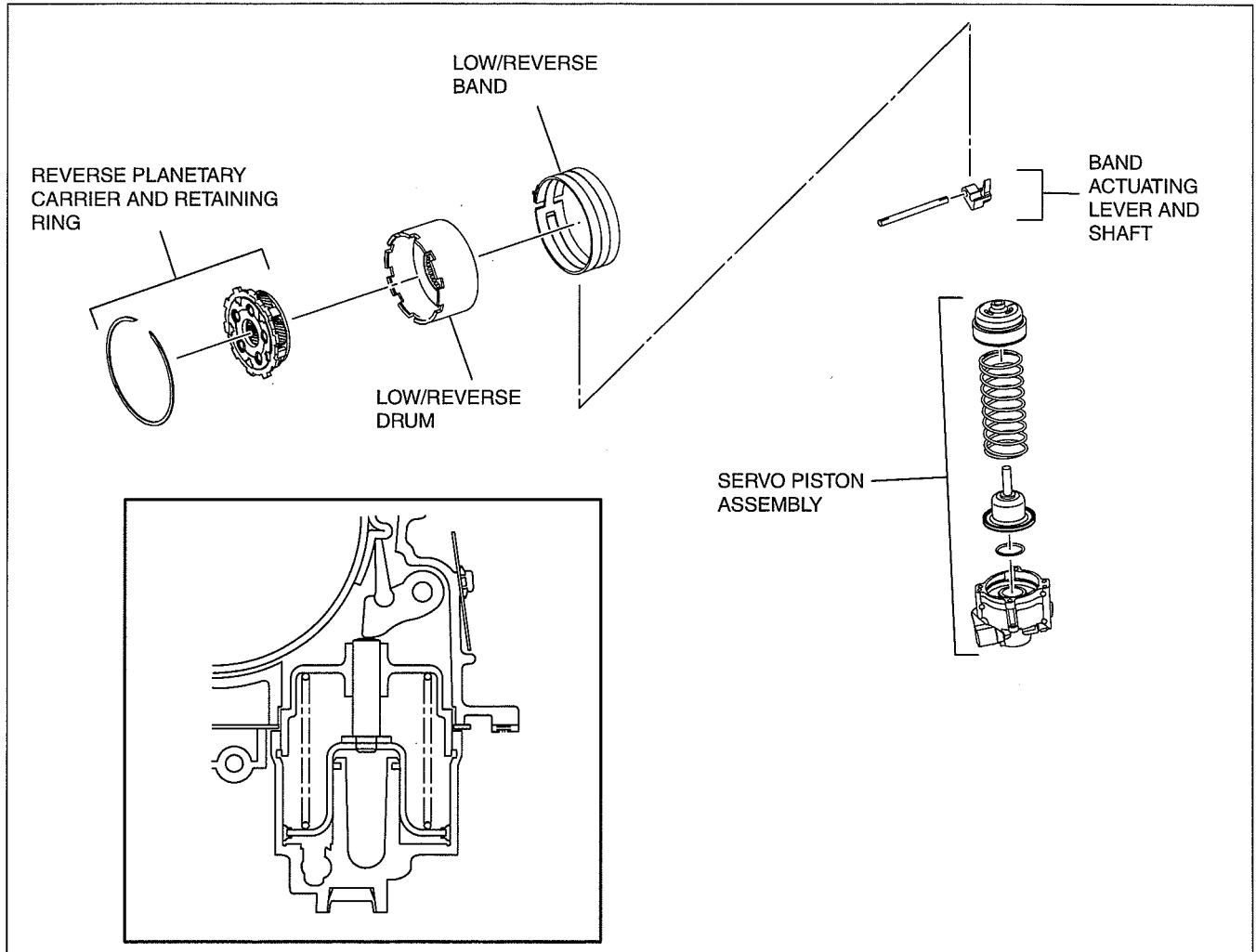


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AUTOMATIC TRANSMISSION [5R55S]

Low/Reverse Band

- The low/reverse band connects the reverse planetary carrier to the transmission case.
- The low/reverse band is applied in Reverse, as well as in manual 1GR and manual 2GR.
- During 2GR operation, 1GR operation and Reverse, hydraulic pressure is applied to the low/reverse servo.
 - This pressure causes the servo to move and apply force to the low/reverse band.
 - This action causes the low/reverse brake drum to be held.
 - This action causes the low/reverse planetary assembly to be held stationary.

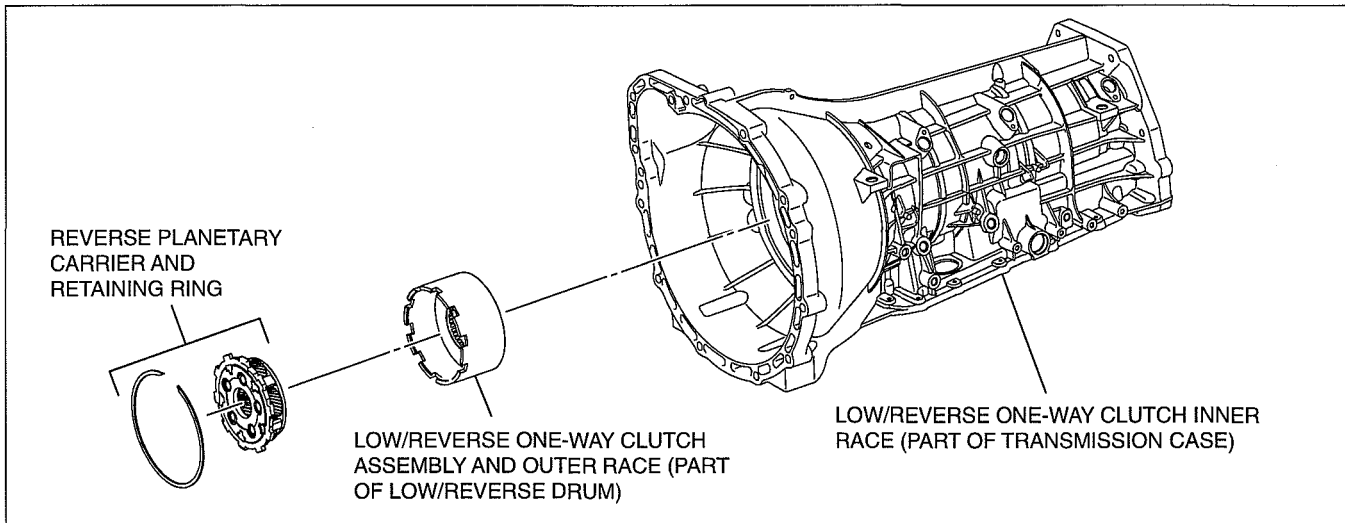


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AUTOMATIC TRANSMISSION [5R55S]

Low/Reverse One-Way Clutch

- The low/reverse one-way clutch connects the reverse planetary carrier to the transmission case during drive operation.
- The low/reverse one-way clutch transmits torque in 1GR and 2GR, as well as in manual 1GR and manual 2GR.
- The low/reverse one-way clutch is a sprag-type one-way clutch.
 - The low/reverse one-way clutch holds the low/reverse drum and low/reverse planetary assembly to the case in 1GR and 2GR.
 - In all other gears the low/reverse one-way clutch overruns.



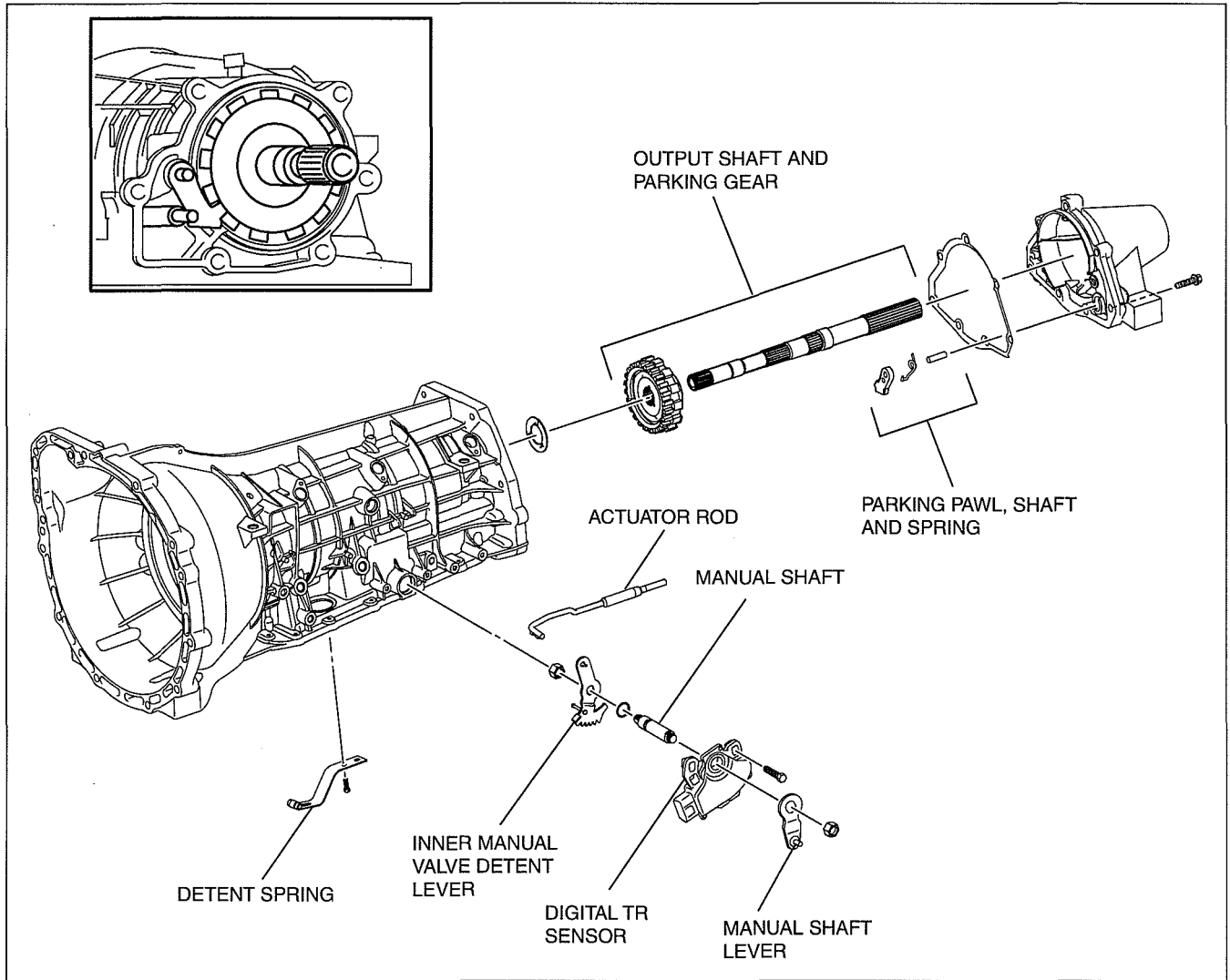
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AUTOMATIC TRANSMISSION [5R55S]

PARKING LOCK CONSTRUCTION [5R55S]

id0513c1321300

- The parking gear is connected to the output shaft. The parking gear has lugs on its outer surface to allow the operation of the parking lock.
- The parking lock prevents the vehicle wheels from rotating by allowing the transmission case to hold the output shaft stationary through the parking gear. When the parking pawl engages the parking gear, it holds the gear stationary.
- When the manual control lever is rotated to the P position, the parking lock works as follows:
 - The inner manual control valve detent lever (connected to the manual control shaft) rotates, pushing a spring-loaded actuator rod.
 - The actuator rod pivots the parking pawl into alignment between the lugs of the parking gear.
 - The parking pawl (connected to the transmission case) holds the parking gear.



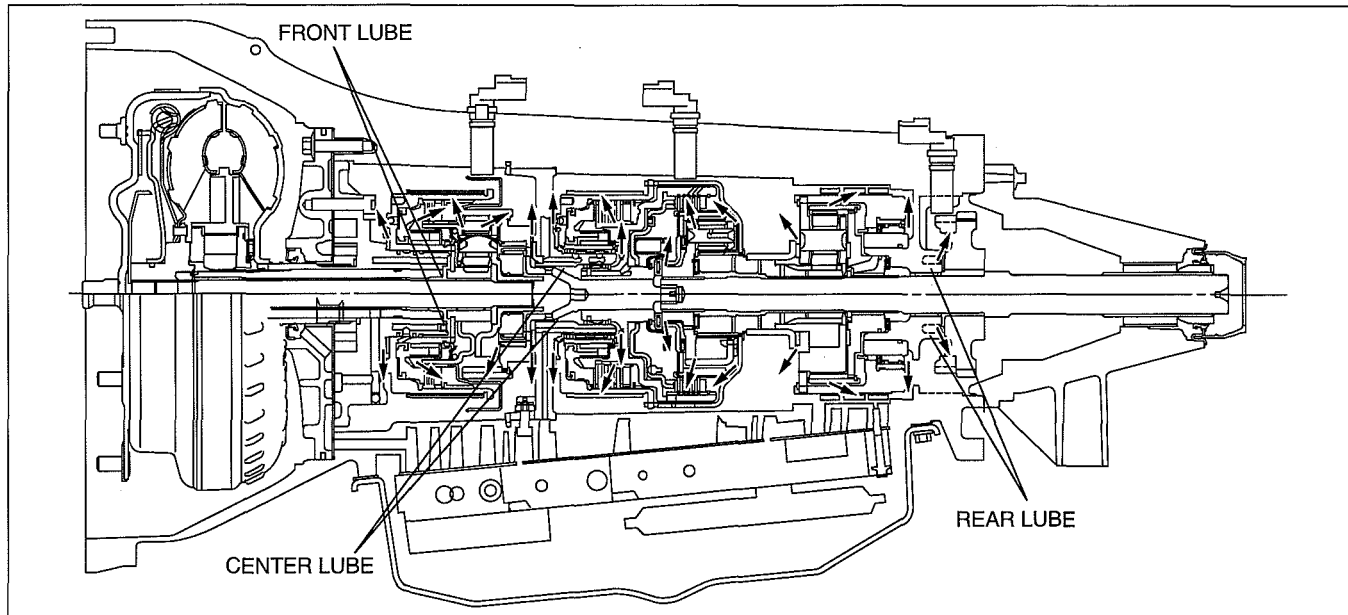
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AUTOMATIC TRANSMISSION [5R55S]

LUBRICATION [5R55S]

id0513c1321500

- The 5R55S transmission provides lubrication for rotating mechanical components through three hydraulic circuits to the front, center and rear areas. Fluid in the torque converter hydraulic circuit provides fluid to lubricate the impeller hub bushing.
- Fluid in this circuit also flows from the transmission case, through a tube, to a cooler located at the front of the vehicle. Fluid from the cooler returns through another tube into the transmission and then through the three lube circuits.
- When the temperature of fluid that exits the torque converter contains less heat, the action of a thermostatically controlled valve in the main control causes the cooler to be bypassed. Fluid flows directly from the torque converter into the lube circuits.
- As transmission fluid temperature increases, the thermostatically controlled valve allows directs the flow of fluid exiting the torque converter to pass through the cooler before it enters the lube circuits.



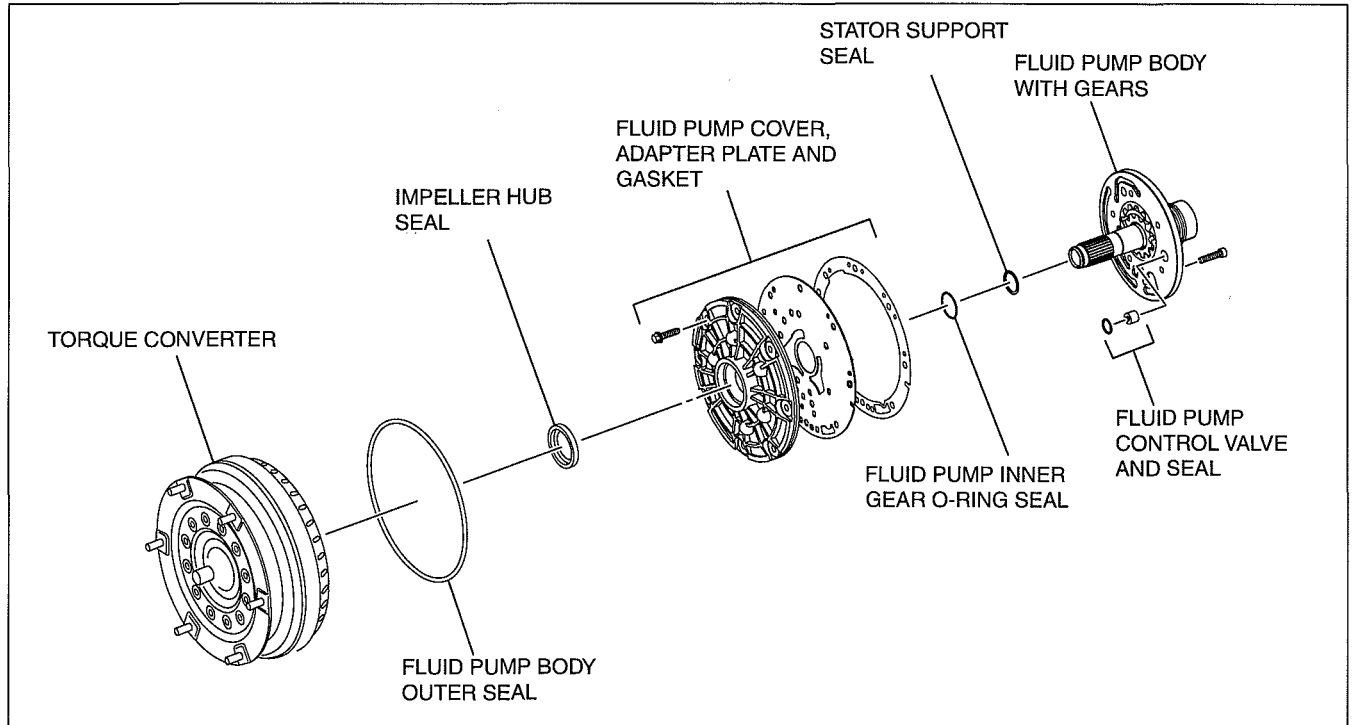
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AUTOMATIC TRANSMISSION [5R55S]

FLUID PUMP CONSTRUCTION [5R55S]

id0513c1323500

- The 5R55S transmission has a gear type pump that supplies fluid under pressure to the hydraulic system.
- The fluid pump provides the fluid pressure necessary to charge the torque converter, main control assembly, transmission cooling system, lubrication system and apply devices.
- The torque converter impeller hub rotates the drive gear of the pump. The drive gear has an O-ring on its impeller hub contact surface.
- The fluid pump has the following parts:
 - Pump cover
 - Adapter plate
 - Pump assembly, with drive and driven gears and an integral stator support



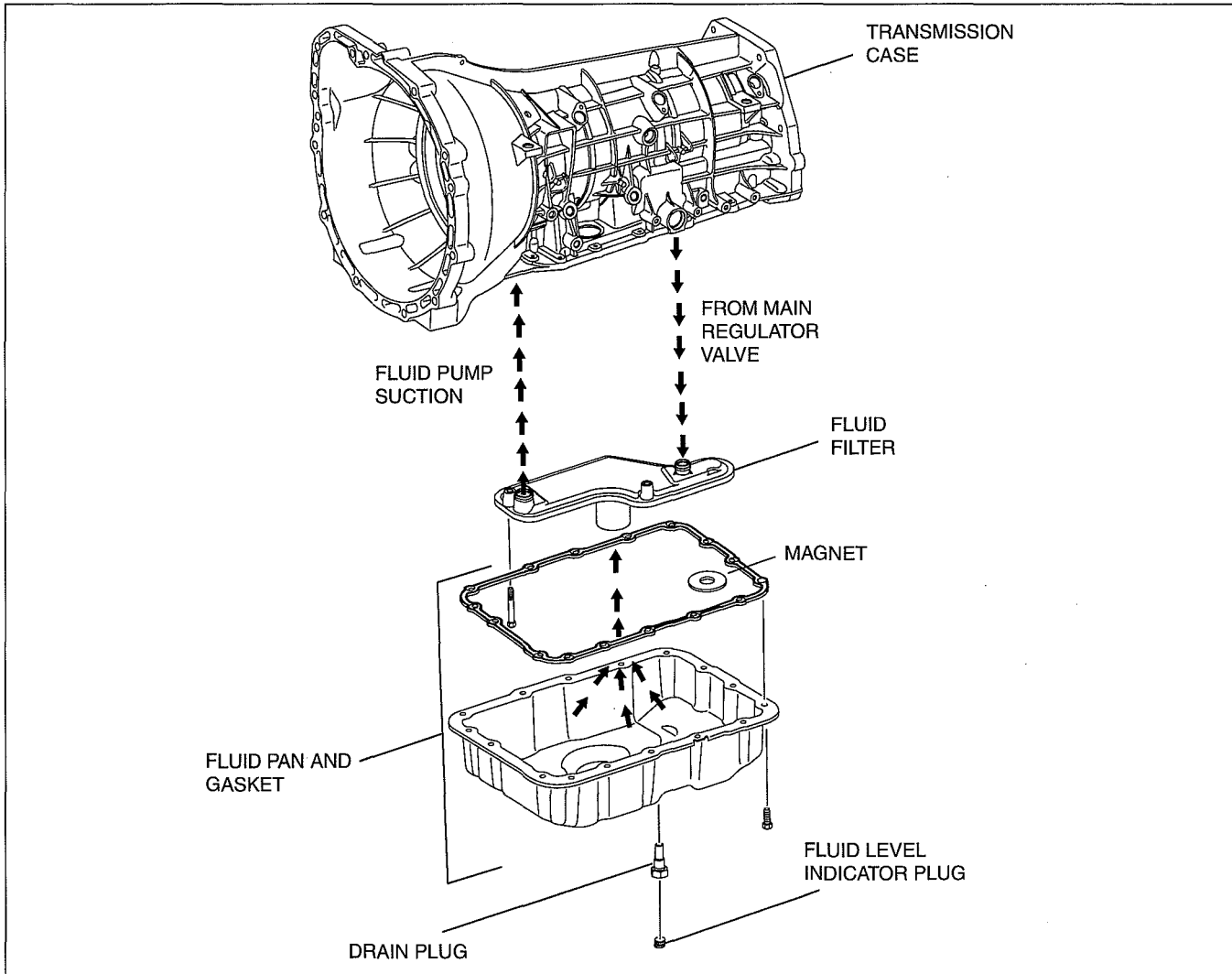
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AUTOMATIC TRANSMISSION [5R55S]

FLUID FILTER CONSTRUCTION [5R55S]

id0513c1321600

- Fluid in a sump area formed by the fluid pan flows from a filter, through passages in the valve body and transmission case, to the suction port of the fluid pump.
- The transmission fluid filter and its accompanying seals are part of the fluid path from the sump (pan) to the fluid pump.
 - The transmission fluid filter has a bypass section which allows fluid vented at the main regulator valve to be recirculated to the fluid pump, without passing through the transmission fluid filter.
- A magnet attached to the inside of the fluid pan collects unwanted magnetic material.



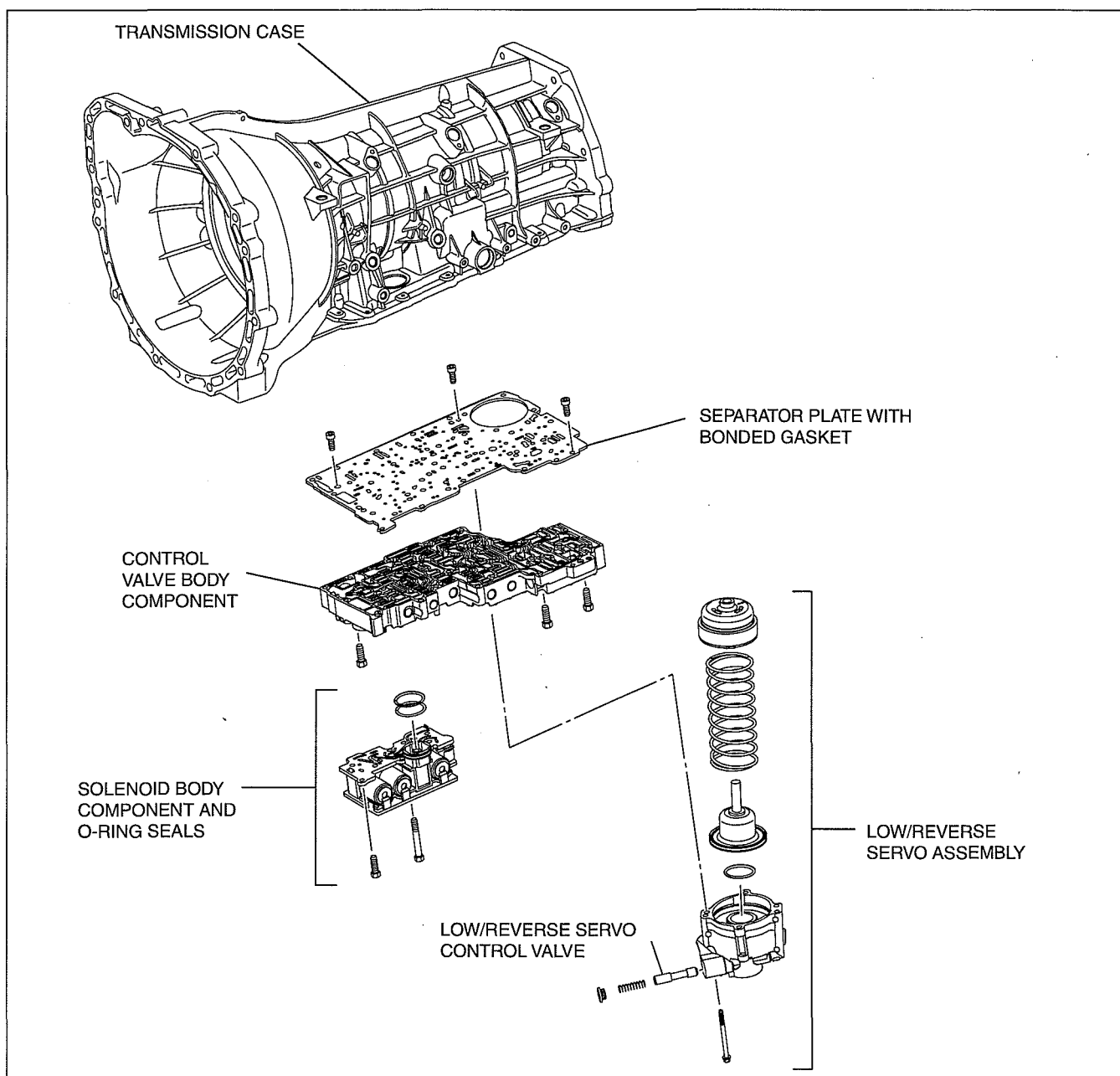
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AUTOMATIC TRANSMISSION [5R55S]

CONTROL VALVE BODY OUTLINE [5R55S]

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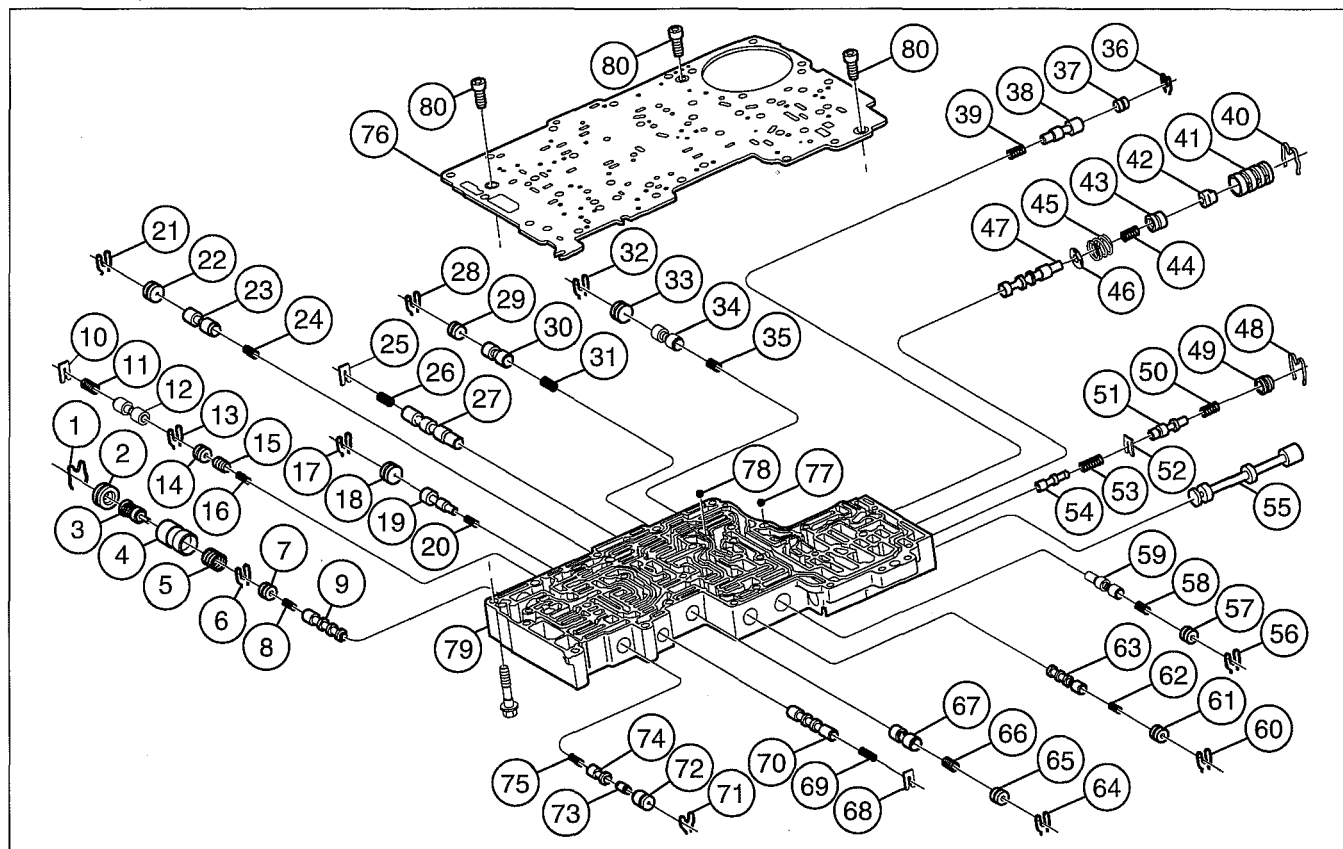
- The control valve body component and related components are part of the pressure side of the hydraulic system.
- The control valve body component consists of the solenoids, the valve body assembly and the separator plate.
- These components combine to convert electrical signals into hydraulic actions.
 - All valves in the control valve body component are anodized aluminum and cannot be sanded, filed or dressed in any other way. If there is any damage to the valves that prevents or restricts their movement, install a new control valve body component.



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AUTOMATIC TRANSMISSION [5R55S]

Control Valve Body Component Exploded View



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1	Retainer clip
2	Retainer plug
3	Thermo valve component
4	Fluid cooler bypass valve
5	Fluid cooler bypass spring
6	Retainer clip
7	Retainer plug
8	Converter clutch control valve spring
9	Converter clutch control valve
10	Plate
11	Coast clutch control spring
12	Coast clutch control valve
13	Retainer clip
14	Retainer plug
15	Converter clutch back pressure valve
16	Converter clutch back pressure spring
17	Retainer clip
18	Retainer plug
19	VFS2 modulator valve
20	VFS2 modulator valve spring
21	Retainer clip
22	Retainer plug
23	Intermediate servo release valve
24	Intermediate servo release valve spring
25	Plate
26	High clutch control spring
27	High clutch control Valve
28	Retainer plug
29	Retainer clip

30	Reverse modulator valve
31	Reverse modulator valve spring
32	Retainer clip
33	Retainer plug
34	Reverse engagement valve
35	Reverse engagement valve spring
36	Retainer clip
37	Retainer plug
38	VFS1 modulator valve
39	VFS1 modulator valve spring
40	Retainer clip
41	Sleeve
42	Booster valve
43	Booster valve
44	Inner spring
45	Outer spring
46	Retainer spring
47	Main regulator valve
48	Retainer clip
49	Retainer plug
50	Converter limit spring
51	Converter limit valve
52	Plate
53	Solenoid regulator valve spring
54	Solenoid regulator valve
55	Manual valve
56	Retainer clip
57	Retainer plug
58	Rear servo control valve spring

AUTOMATIC TRANSMISSION [5R55S]

59	Rear servo control valve
60	Retainer clip
61	Retainer plug
62	RS ISA select valve spring
63	RS ISA select valve
64	Retainer clip
65	Retainer plug
66	Forward engagement control valve spring
67	Forward engagement control valve
68	Plate
69	Overdrive servo control spring

70	Overdrive servo control valve
71	Clip retainer
72	Converter clutch modulator control sleeve
73	Converter clutch modulator control valve
74	Converter clutch modulator valve
75	Converter clutch modulator control spring
76	Separator plate
77	Lubrication check ball
78	Shuttle valve ball
79	Lower control valve body
80	Bolt

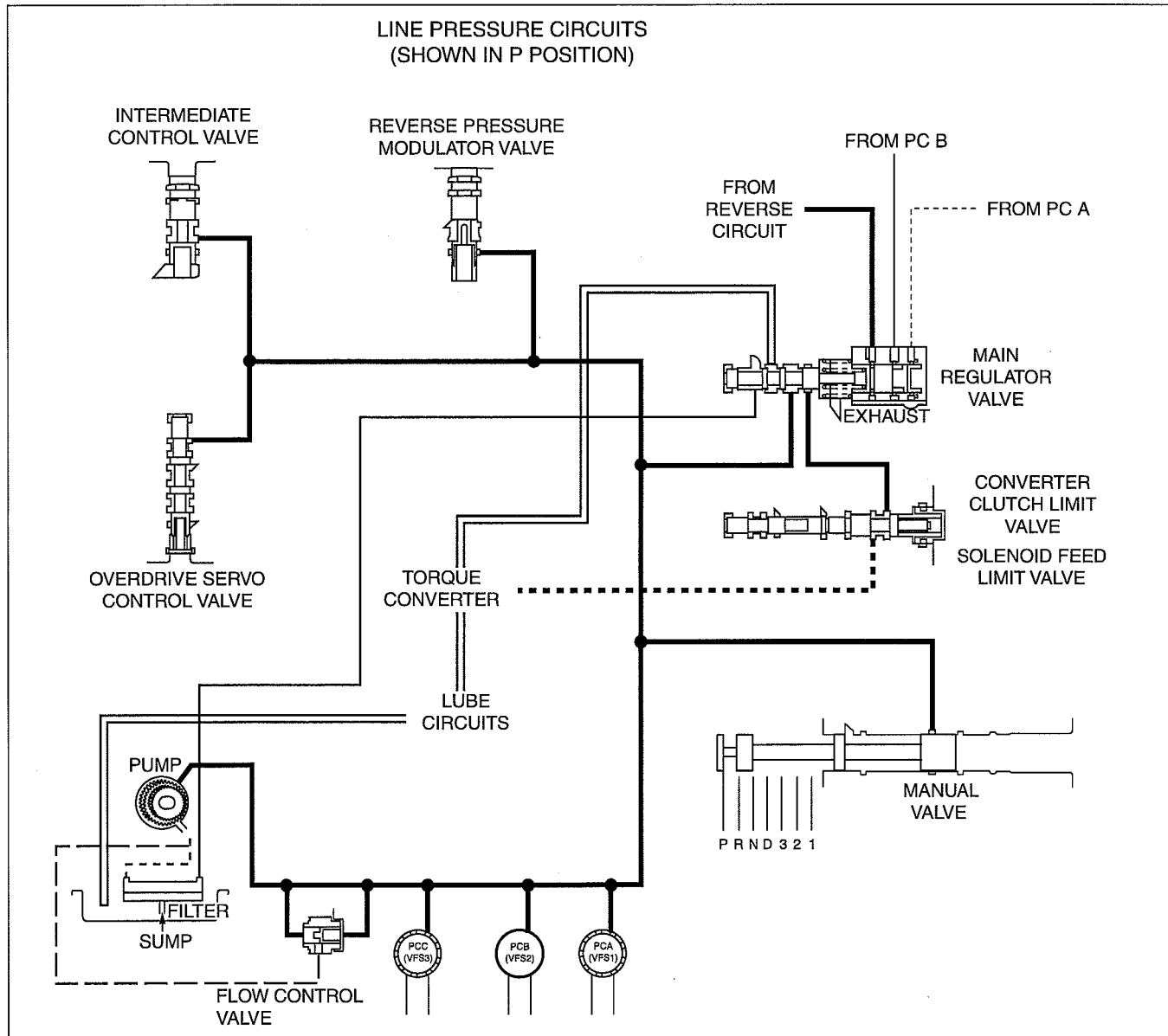
AUTOMATIC TRANSMISSION [5R55S]

PRESSURE CONTROL HYDRAULIC CIRCUITS CONSTRUCTION [5R55S]

id0513c1321800

Line Pressure Hydraulic Circuit

- Fluid under pressure from the pump flows to several hydraulic controls in the valve body, including the main regulator valve. The main regulator valve varies line pressure as part of controlling shift feel and efficient apply component operation.
- The TCM controls the position of the main regulator valve with the pressure control solenoid A and B.
- When pressure control solenoid A and B pressures are lower, the main regulator valve position provides lower line pressure. As the fluid pressures from pressure control solenoid A and B increase, the main regulator valve provides higher line pressure.
- When the transmission shifter is in the R position, fluid under pressure from the manual valve to the apply components also affects the position of the main regulator valve. This results in a pressure boost.

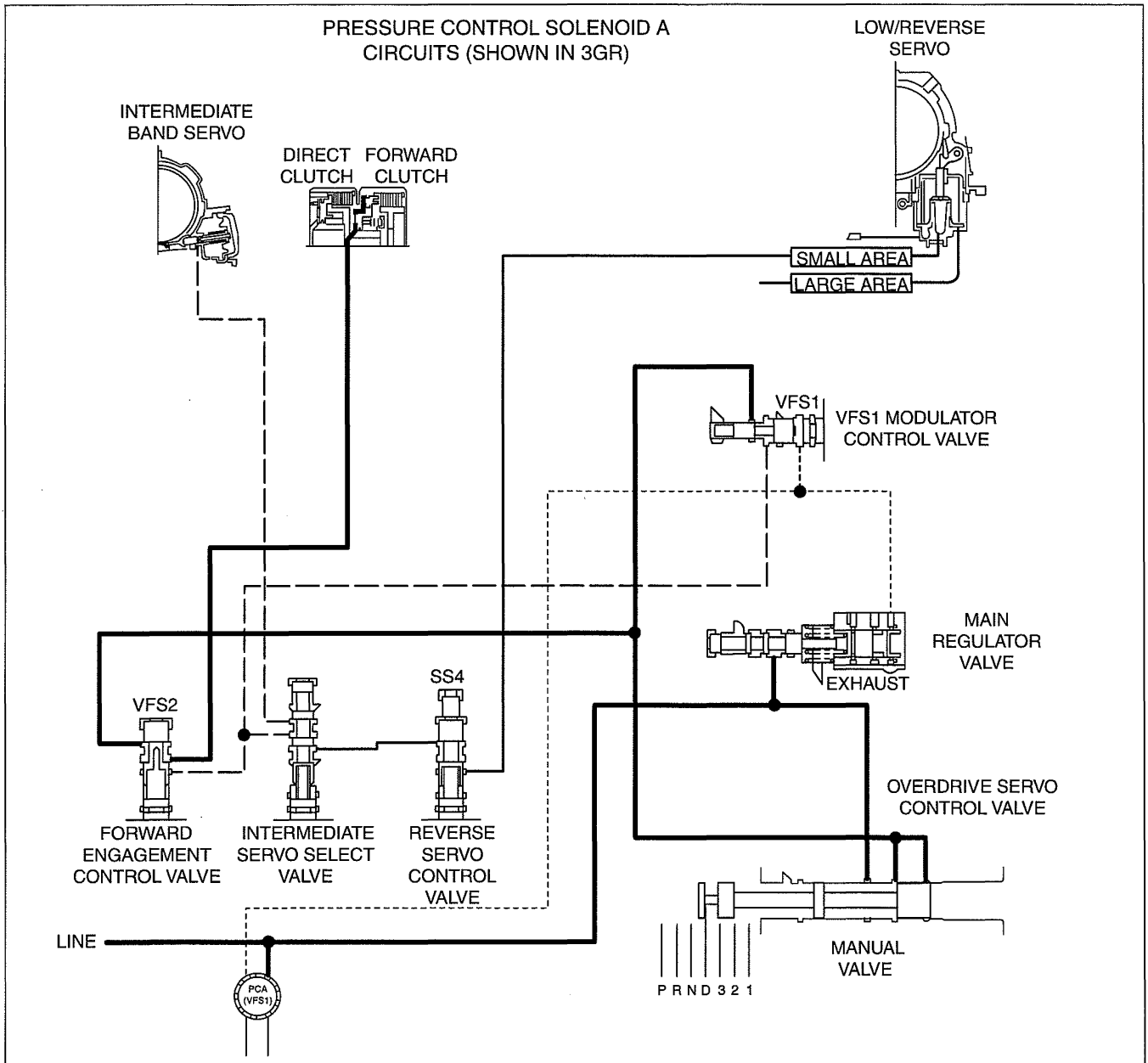


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AUTOMATIC TRANSMISSION [5R55S]

Pressure Control Solenoid A Hydraulic Circuit

- In the 5R55S transmission, pressure control solenoid A affects the operation of the main regulator valve and the VFS1 modulator control valve.
- VFS1 pressure indirectly affects the feel of the forward clutch engagement. VFS1 pressure is directly applied to the intermediate servo. It is also directly applied to the low/reverse servo in manual 1GR and manual 2GR.

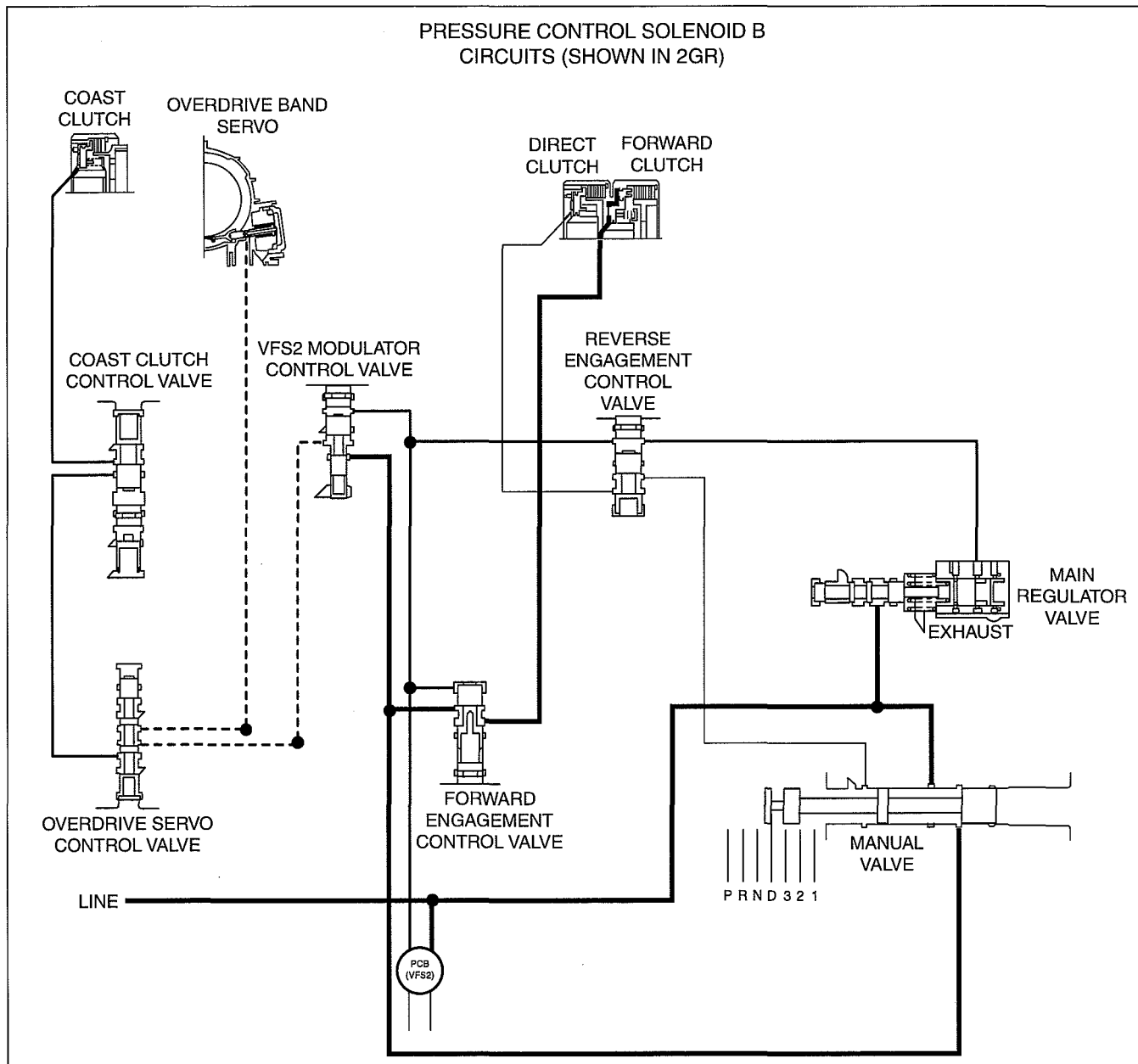


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AUTOMATIC TRANSMISSION [5R55S]

Pressure Control B Solenoid B Hydraulic Circuit

- In the 5R55S transmission, pressure control solenoid B affects the operation of the main regulator valve, the VFS2 modulator control valve, the forward engagement control valve and the reverse engagement control valve.
- VFS2 pressure indirectly affects the feel of the forward clutch engagement, as well as the direct clutch engagement in reverse. VFS2 pressure is directly applied to the overdrive servo or the coast clutch.

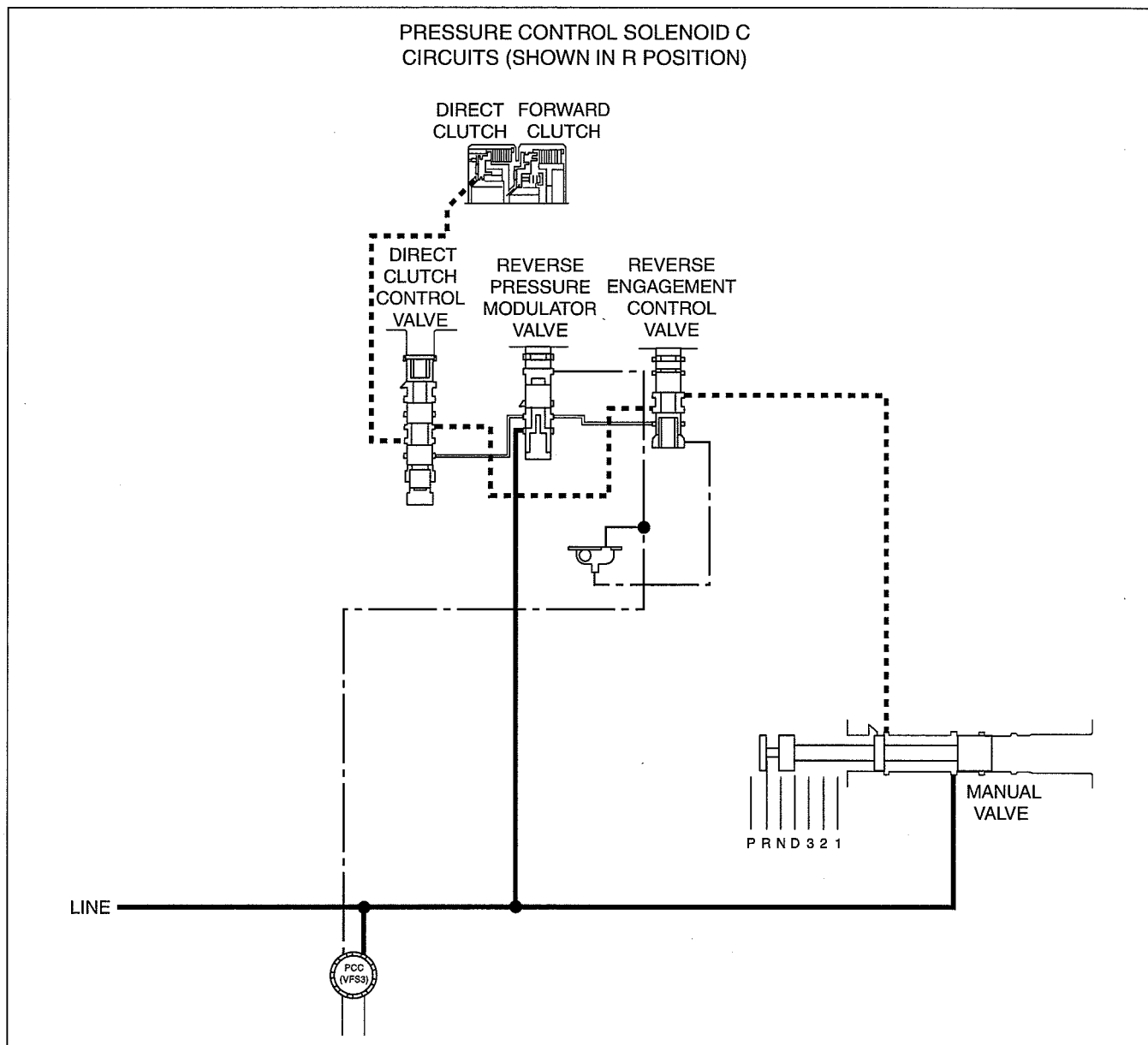


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AUTOMATIC TRANSMISSION [5R55S]

Pressure Control Solenoid C Hydraulic Circuit

- In the 5R55S transmission, pressure control solenoid C solenoid affects the operation of the reverse pressure modulator valve and the reverse engagement control valve.
- Pressure control solenoid C pressure indirectly affects the feel of direct clutch engagement in reverse. Pressure control solenoid C pressure is directly applied to the direct clutch in 4GR and 5GR.



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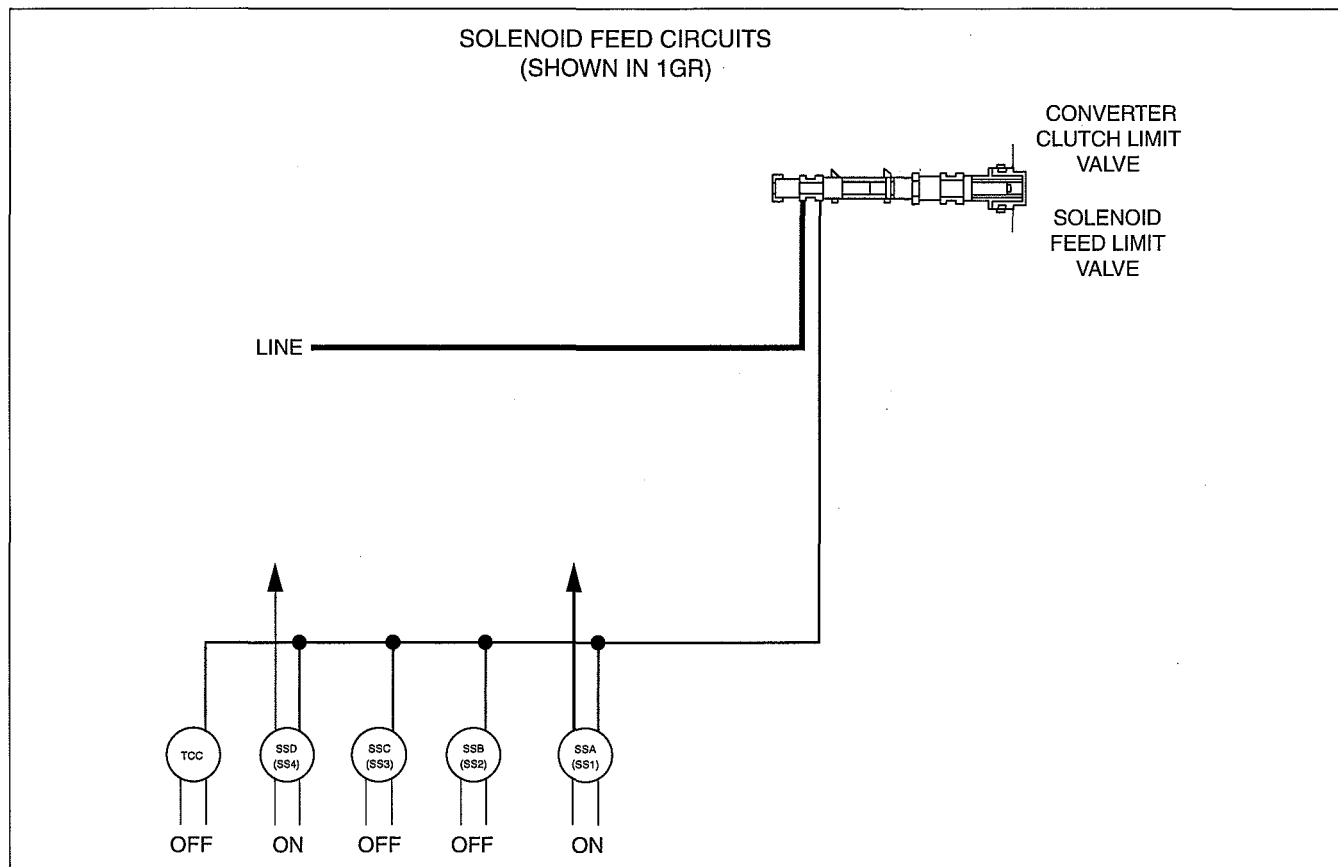
AUTOMATIC TRANSMISSION [5R55S]

SHIFT SOLENOID HYDRAULIC CIRCUITS CONSTRUCTION [5R55S]

id0513c1321900

Solenoid Feed Hydraulic Circuit

- In the 5R55S transmission, fluid under line pressure moves to the solenoid feed limit valve. The solenoid feed limit valve provides fluid under a steady regulated pressure to the shift solenoids and TCC control solenoid.
- The TCM operates the shift solenoids with on/off control. When the TCM turns off each solenoid, fluid flow is blocked. When the TCM turns on each solenoid, full fluid flow occurs.
- The TCM operates the TCC control solenoid with varying Pulse-Width Modulation (PWM) control. This results in variable fluid flow and pressure from the TCC control solenoid to the hydraulic valve it controls.

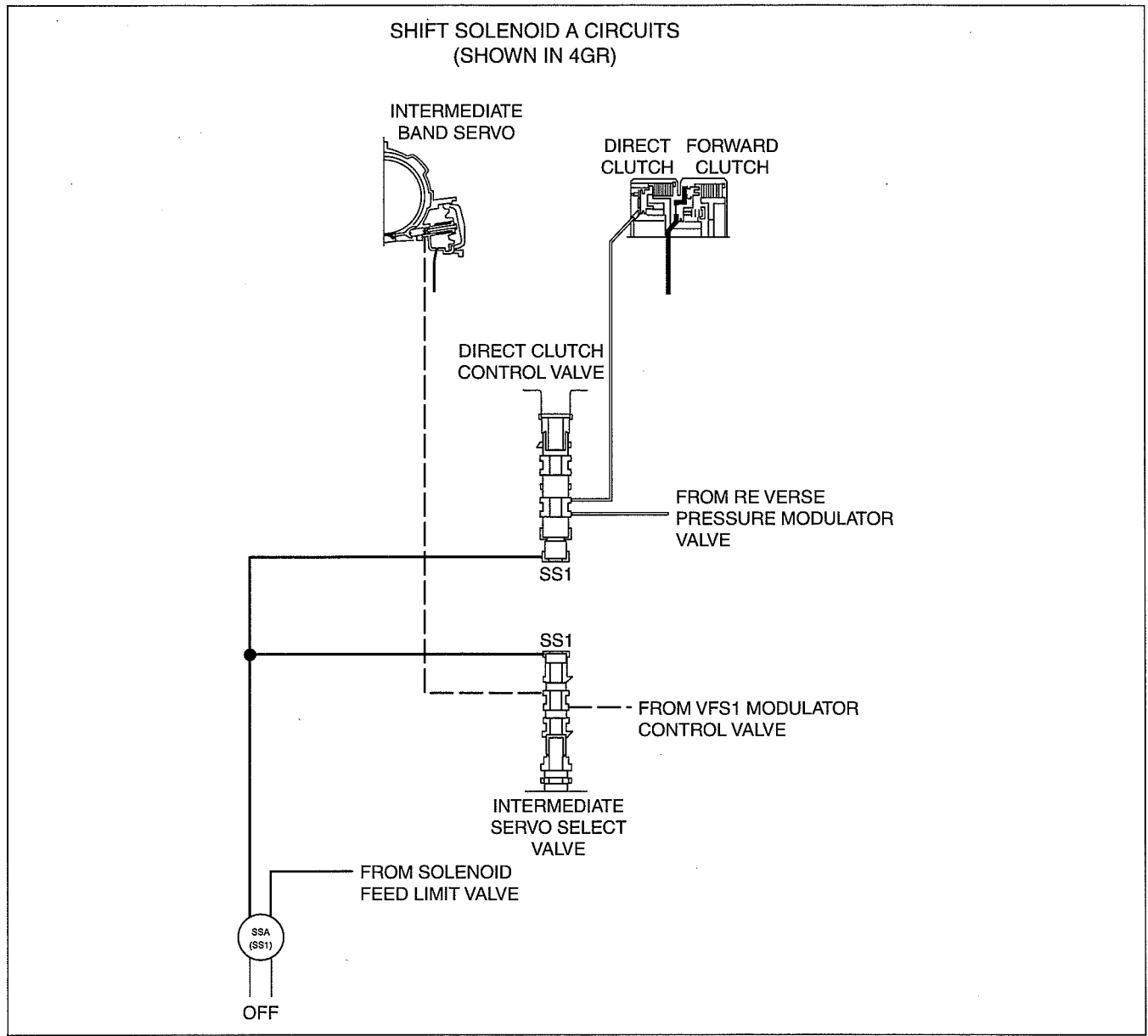


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AUTOMATIC TRANSMISSION [5R55S]

Shift Solenoid A Hydraulic Circuit

- In the 5R55S transmission, shift solenoid A affects the operation of the direct clutch. When shift solenoid A is on in forward speeds, the direct clutch is released. When shift solenoid A is turned off, the direct clutch is applied. shift solenoid A is also related to the operation of the intermediate band.
- When the transmission shifter is moved to the R position under conditions that may cause transmission damage, the TCM turns off shift solenoid A to inhibit reverse operation. This affects various hydraulic control valves in order to keep the direct clutch released.

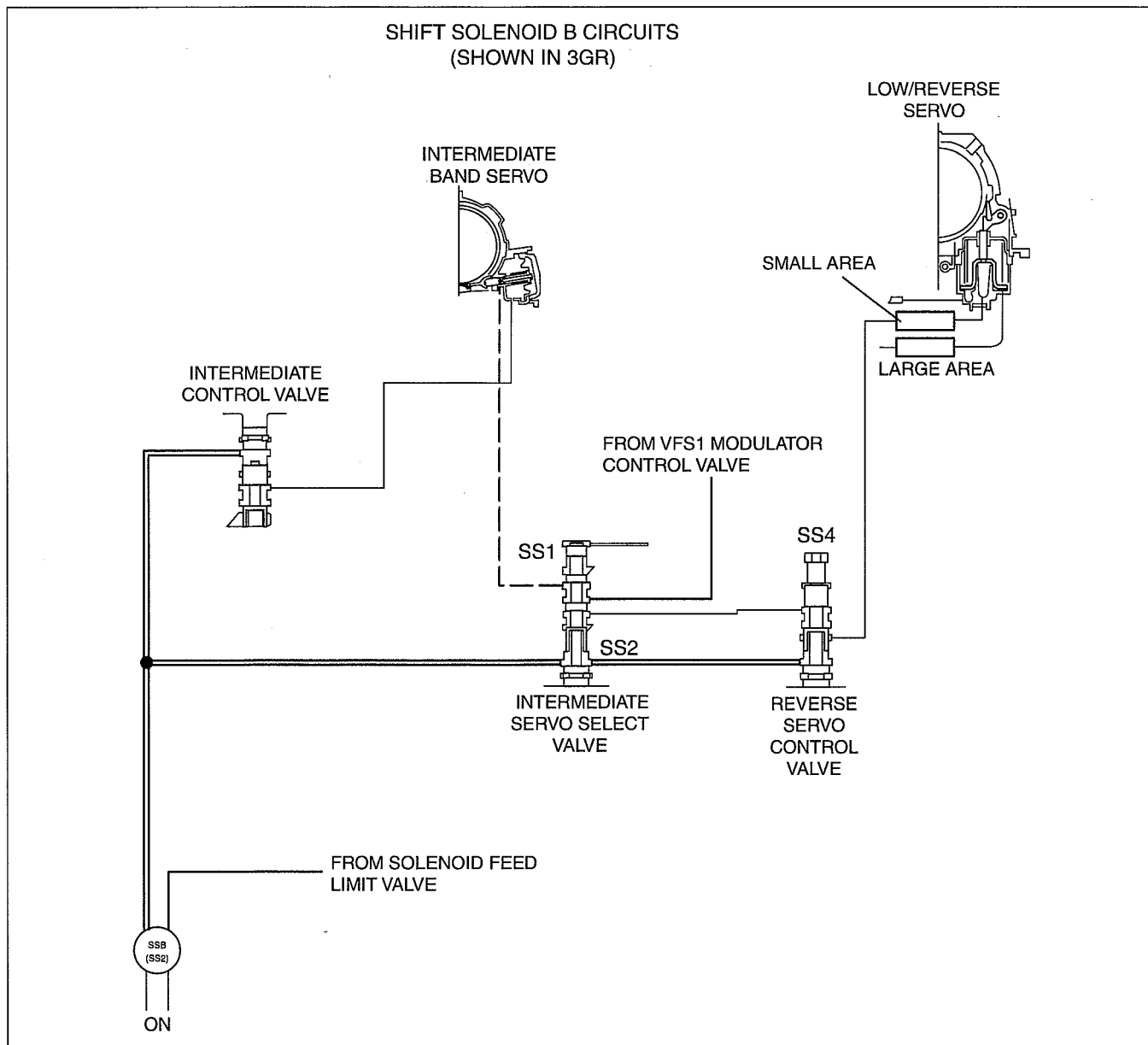


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AUTOMATIC TRANSMISSION [5R55S]

Shift Solenoid B Hydraulic Circuit

- In the 5R55S transmission, shift solenoid B affects the operation of the intermediate band. When shift solenoid B is on, the intermediate band is applied. When shift solenoid B is turned off, the intermediate band is released. shift solenoid B is also related to the operation of the low/reverse band in manual 1st gear and manual 2nd gear.
- The TCM does not use shift solenoid B to inhibit reverse operation in 5R55S transmission.

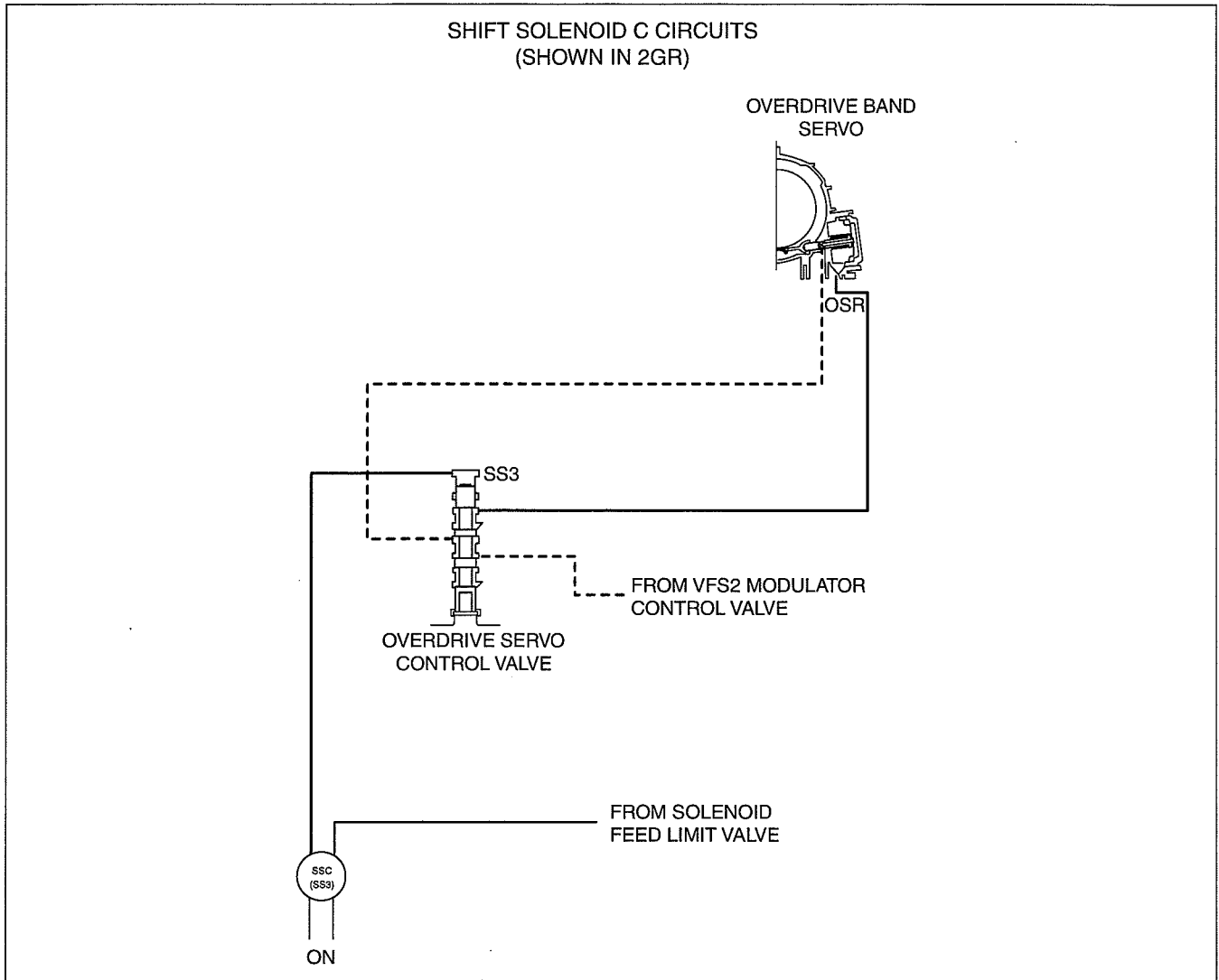


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AUTOMATIC TRANSMISSION [5R55S]

Shift Solenoid C Hydraulic Circuit

- In the 5R55S transmission, shift solenoid C affects the operation of the overdrive band. When shift solenoid C is on, the overdrive band is applied. When shift solenoid C is turned off, the overdrive band is released.

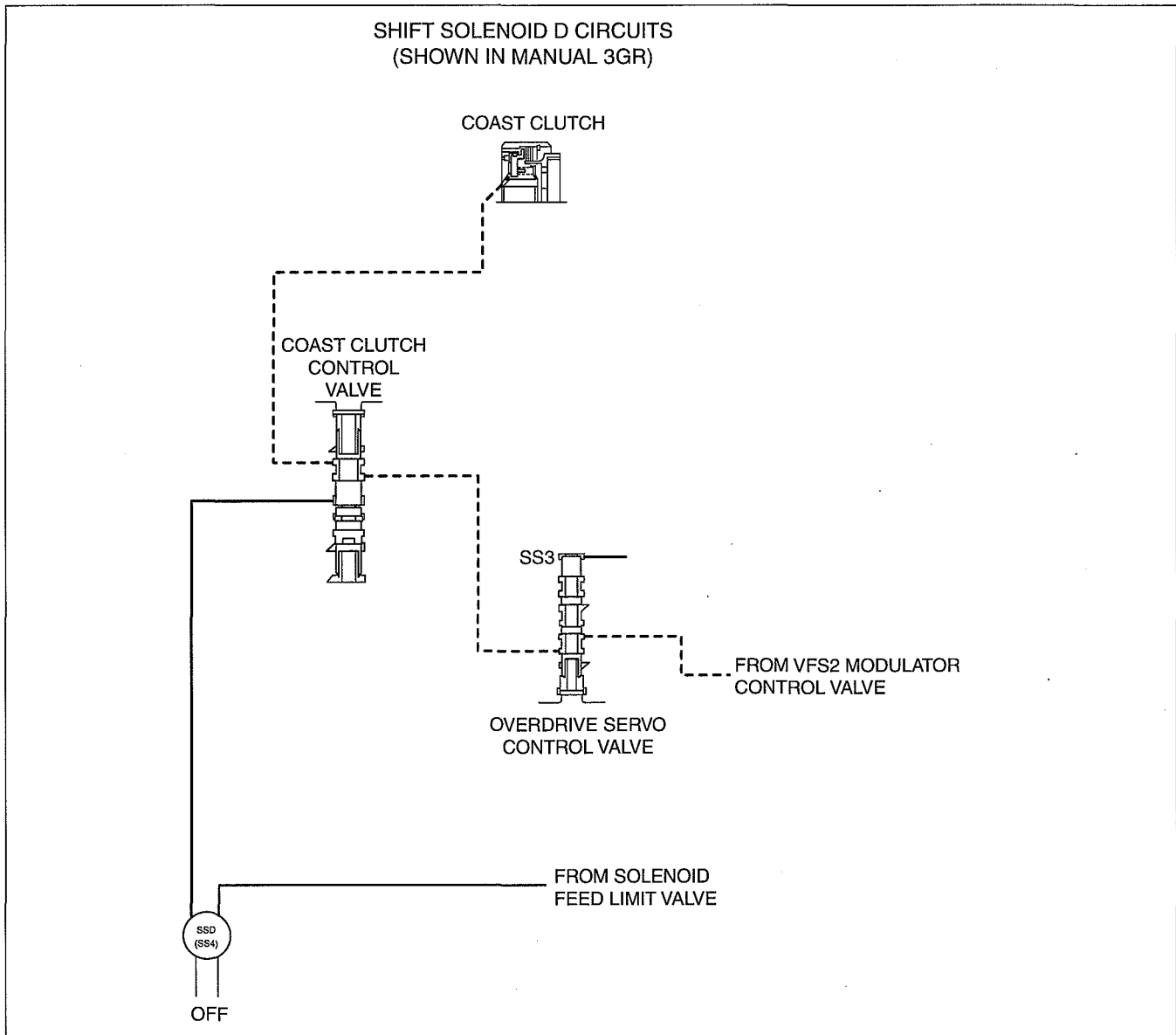


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AUTOMATIC TRANSMISSION [5R55S]

Shift Solenoid D Hydraulic Circuit

- In the 5R55S transmission, shift solenoid D affects the operation of the coast clutch. When shift solenoid D is on, the coast clutch is released. When shift solenoid D is turned off, the coast clutch is applied.

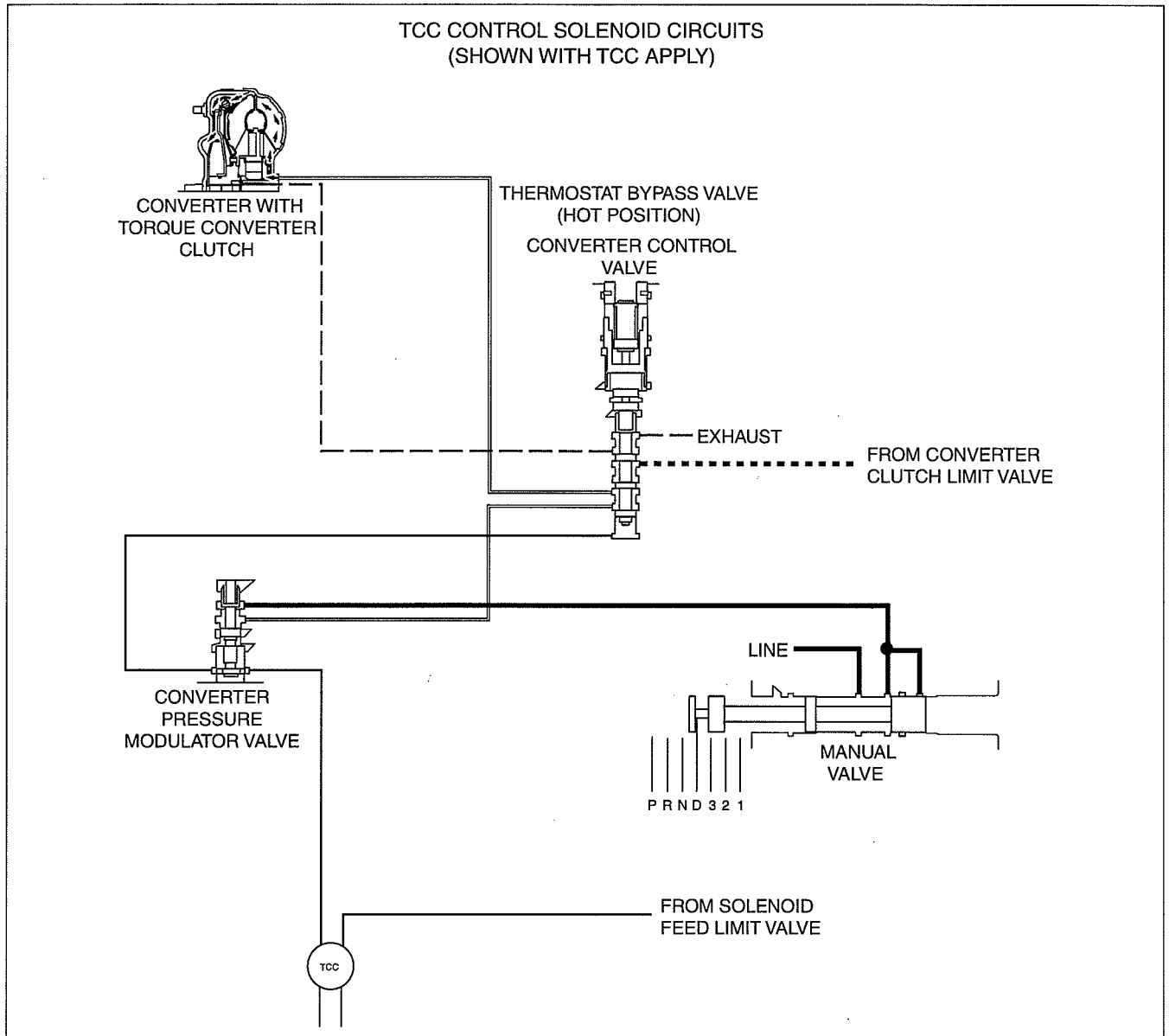


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AUTOMATIC TRANSMISSION [5R55S]

Torque Converter Clutch (TCC) Control Solenoid Hydraulic Circuit

- In the 5R55S transmission, the TCC control solenoid affects the operation of the torque converter clutch in forward speeds.
- TCC pressure affects the positions of the converter pressure modulator valve and the converter control valve.



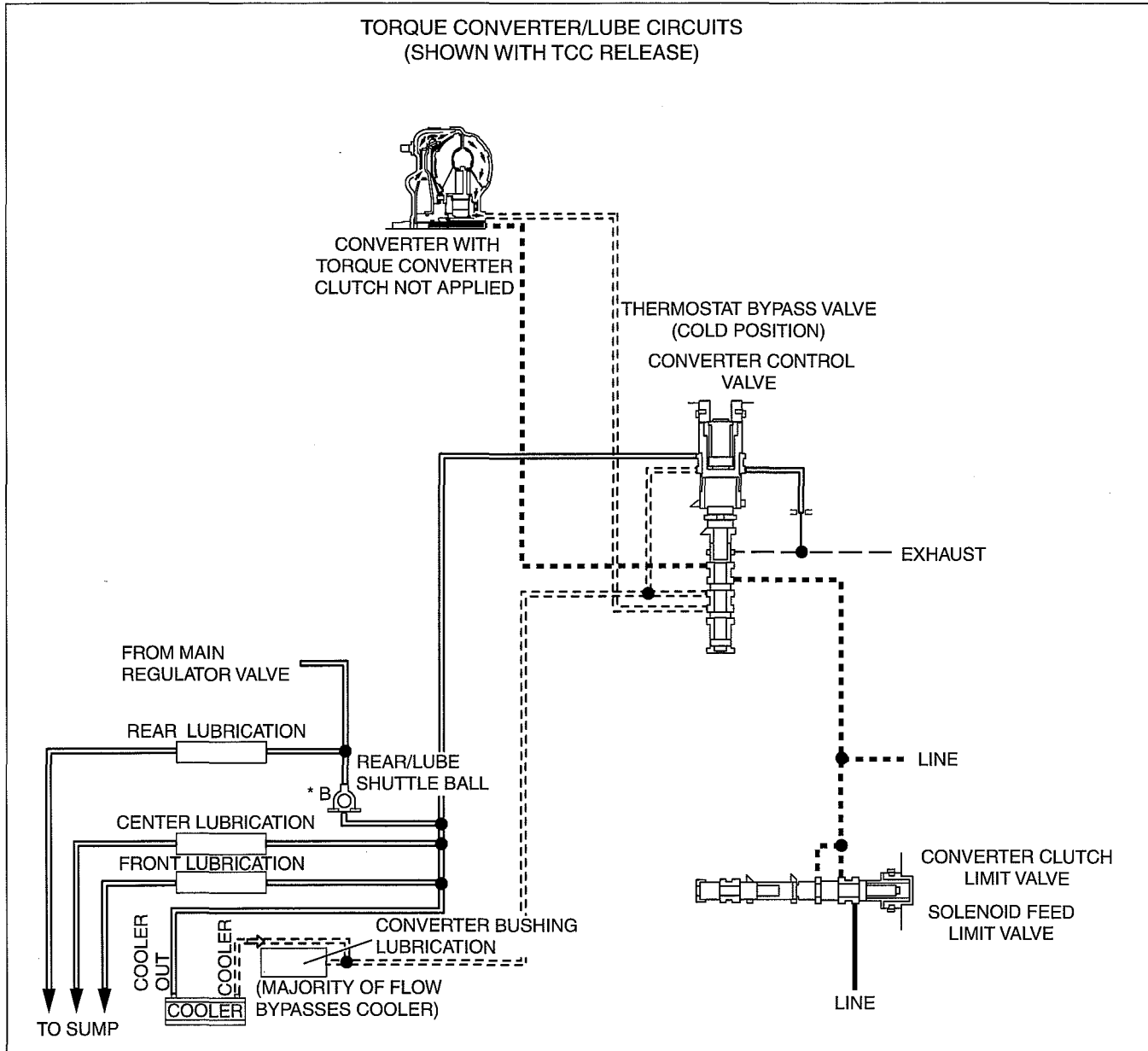
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AUTOMATIC TRANSMISSION [5R55S]

TORQUE CONVERTER/LUBE HYDRAULIC CIRCUITS CONSTRUCTION [5R55S]

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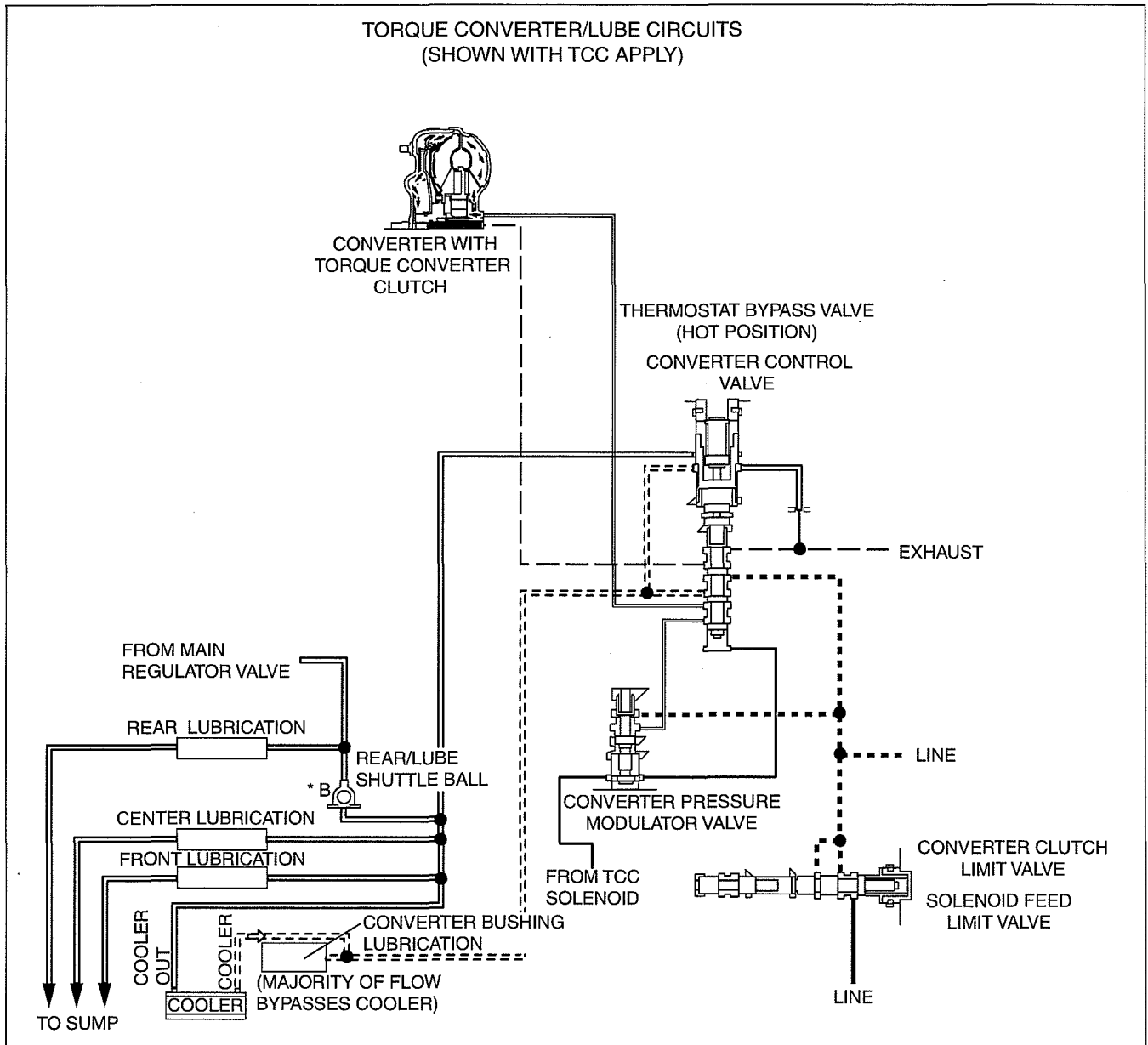
- In the 5R55S transmission, fluid for torque converter operation flows as follows during TCC release:
 - From the LINE circuit, through the converter clutch limit valve, through the converter control valve, to the torque converter
 - From the torque converter, through the converter control valve, into the lube circuits
- The thermostat bypass valve allows fluid from the torque converter to bypass the fluid cooler and enter into the lube circuits when fluid temperature is cold.



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AUTOMATIC TRANSMISSION [5R55S]

- When TCC is applied, fluid for torque converter operation flows as follows:
 - From the LINE circuit, through the converter clutch limit valve, through the converter control valve, into the lube circuits
 - From the LINE circuit, through the converter pressure modulator valve, through the converter control valve, to the torque converter



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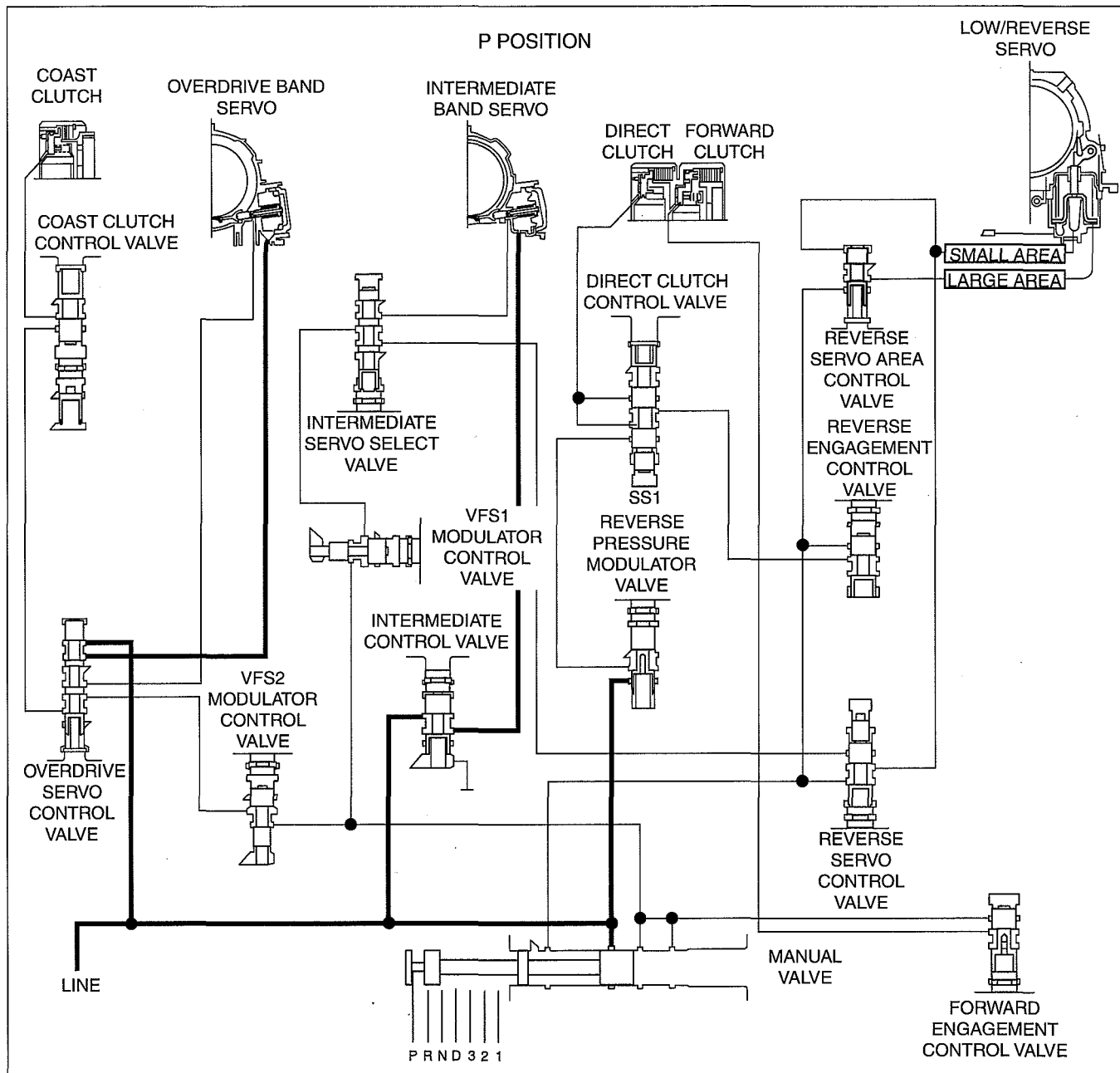
AUTOMATIC TRANSMISSION [5R55S]

APPLY COMPONENT HYDRAULIC CIRCUITS CONSTRUCTION [5R55S]

id0513c1322100

P and N Positions (Park and Neutral)

- With the transmission in the P or N position, fluid under pressure is present at the following components:
 - Overdrive servo-release
 - Intermediate servo-release

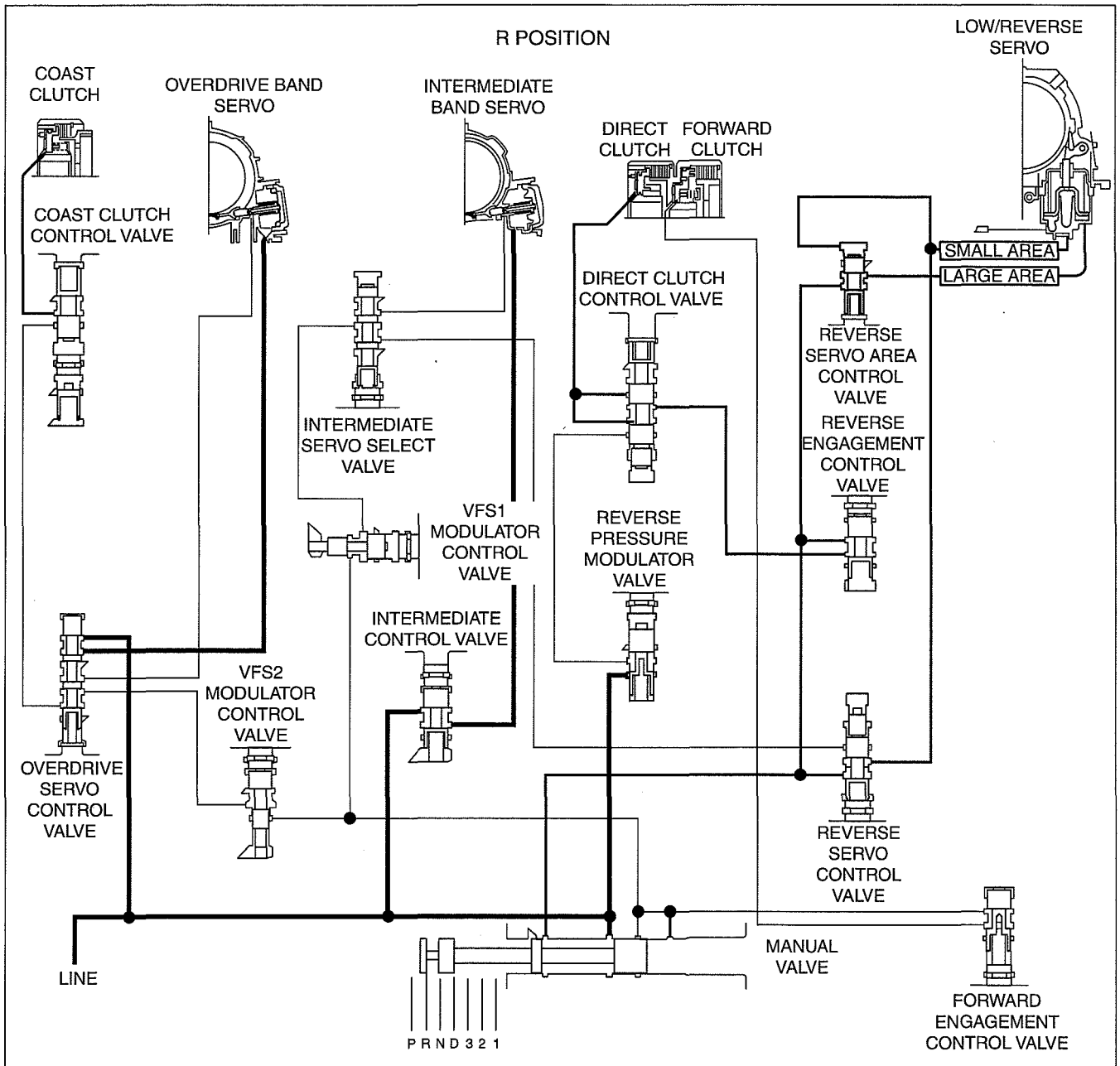


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AUTOMATIC TRANSMISSION [5R55S]

R Position (Reverse)

- With the transmission in the R position, fluid under pressure is present at the following components:
 - Overdrive servo-release
 - Intermediate servo-release
 - Direct clutch
 - Low/reverse servo-apply

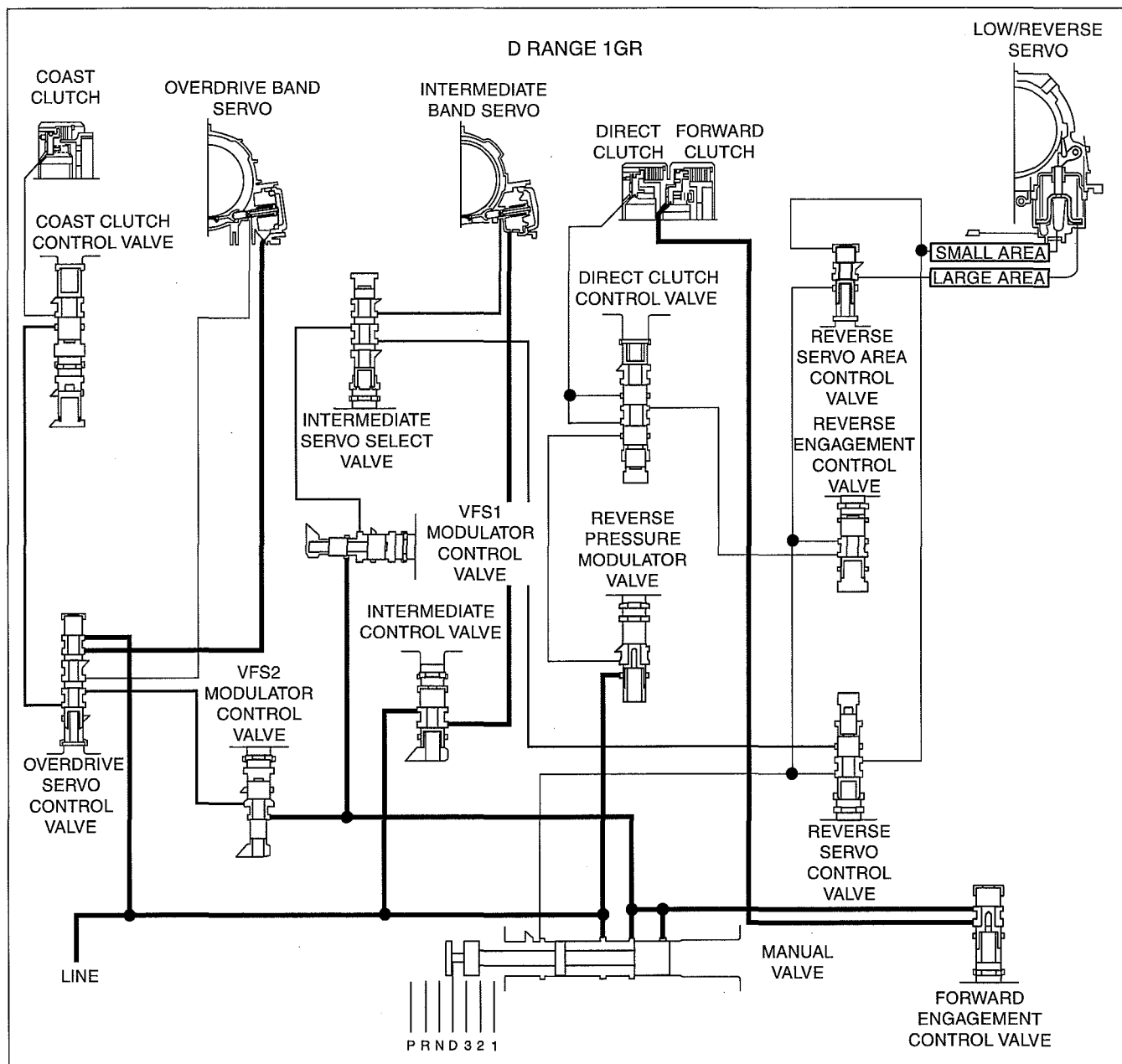


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AUTOMATIC TRANSMISSION [5R55S]

D Range, 1GR

- With the transmission in the D range, fluid under pressure is present at the following components in 1GR:
 - Overdrive servo-release
 - Intermediate servo-release
 - Forward clutch

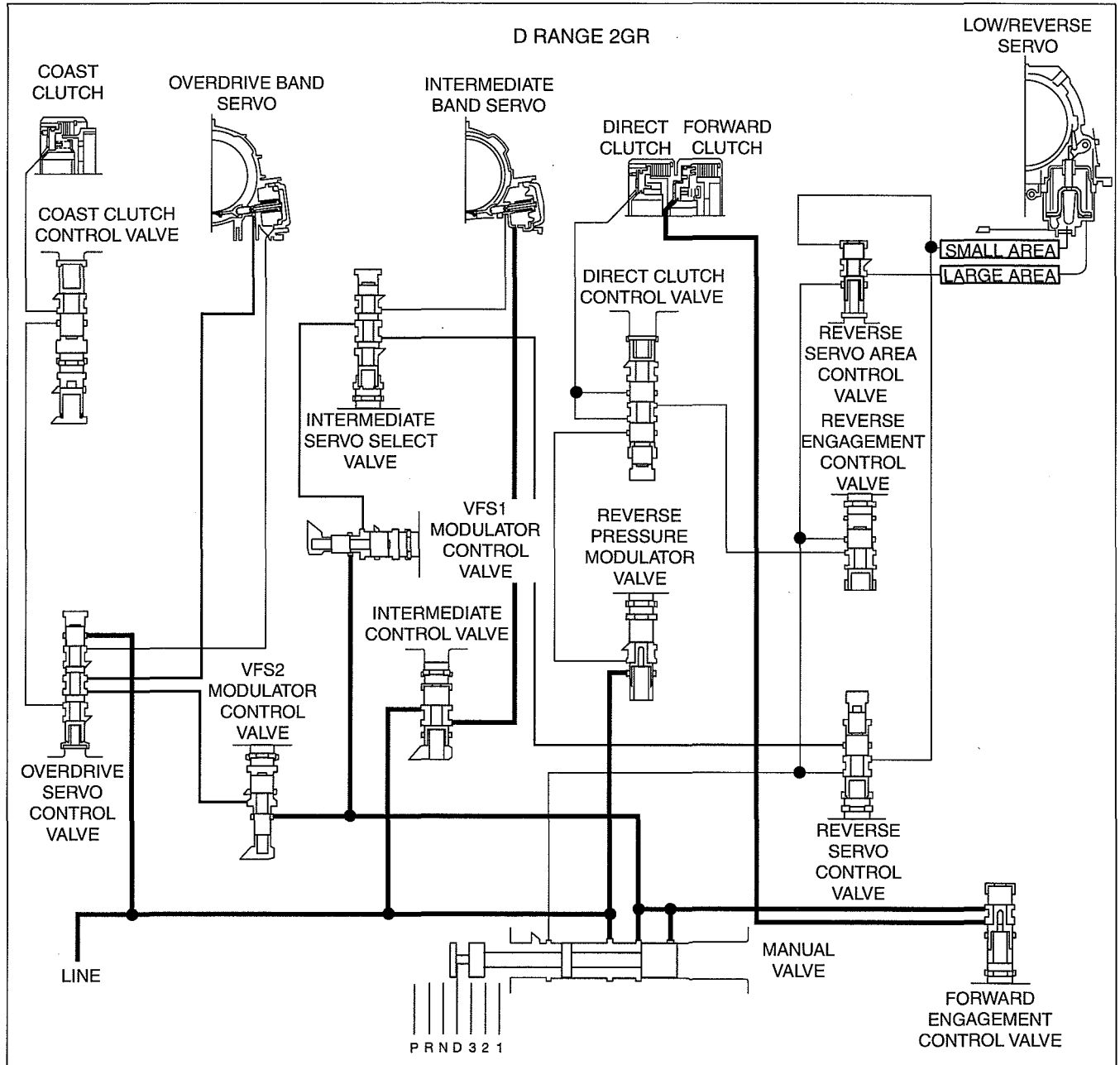


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AUTOMATIC TRANSMISSION [5R55S]

D Range, 2GR

- With the transmission in the D range, fluid under pressure is present at the following components in 2GR:
 - Overdrive servo-apply
 - Intermediate servo-release
 - Forward clutch

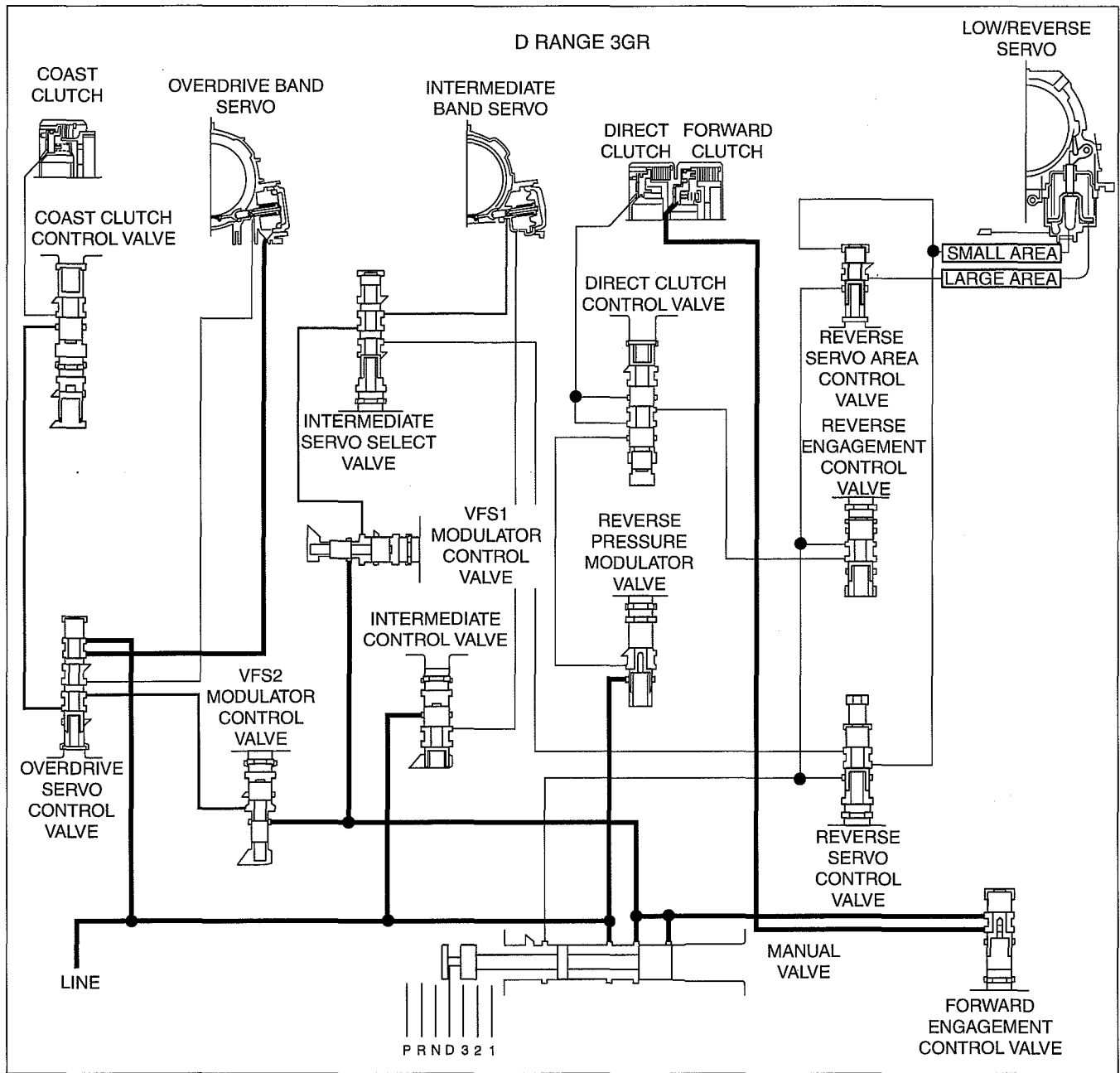


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AUTOMATIC TRANSMISSION [5R55S]

D range, 3GR

- With the transmission in the D range, fluid under pressure is present at the following components in 3GR:
 - Overdrive servo-release
 - Intermediate servo-apply
 - Forward clutch

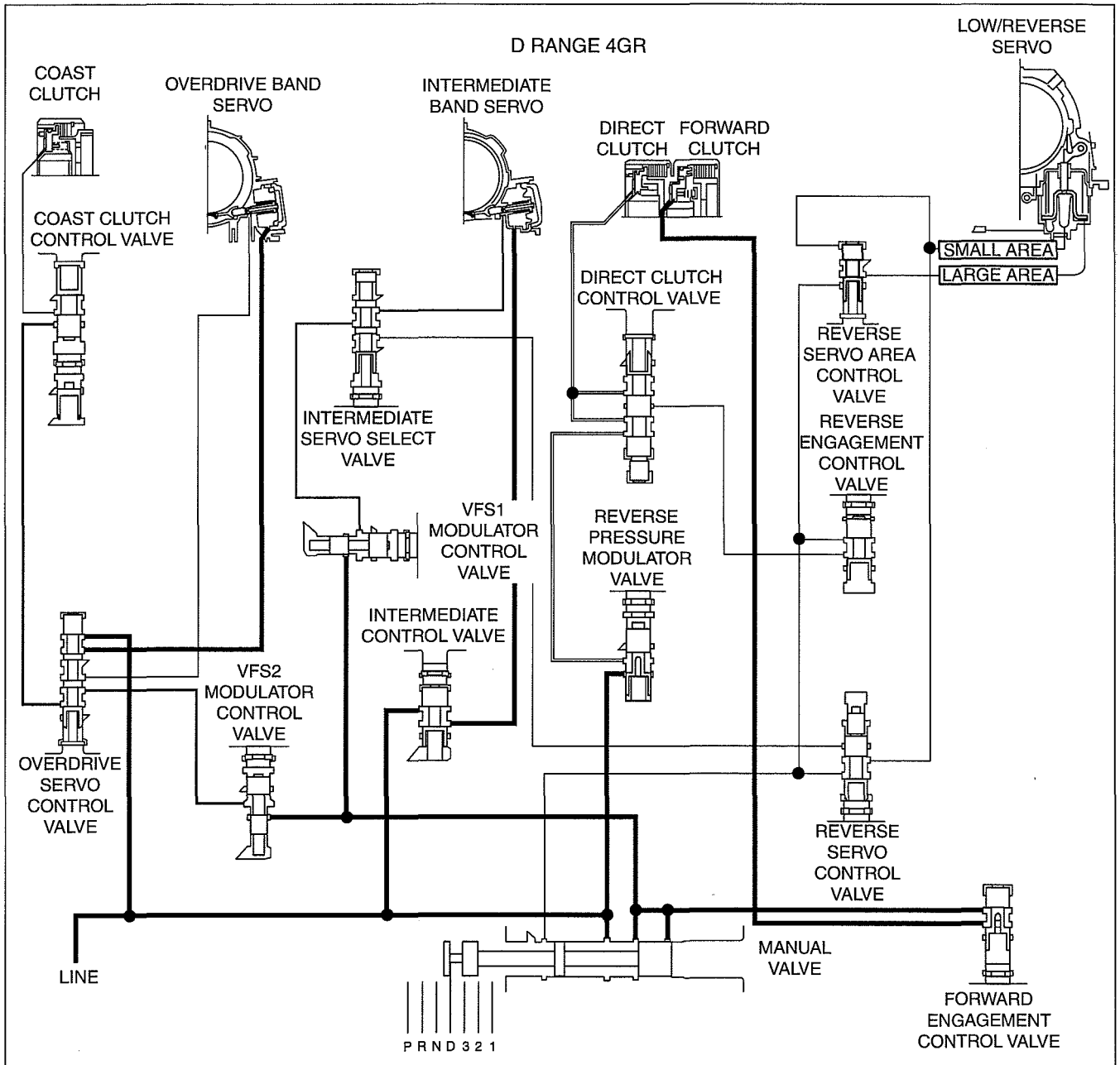


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AUTOMATIC TRANSMISSION [5R55S]

D range, 4GR

- With the transmission in the D range, fluid under pressure is present at the following components in 4GR:
 - Overdrive servo-release
 - Intermediate servo-release
 - Forward clutch
 - Direct clutch

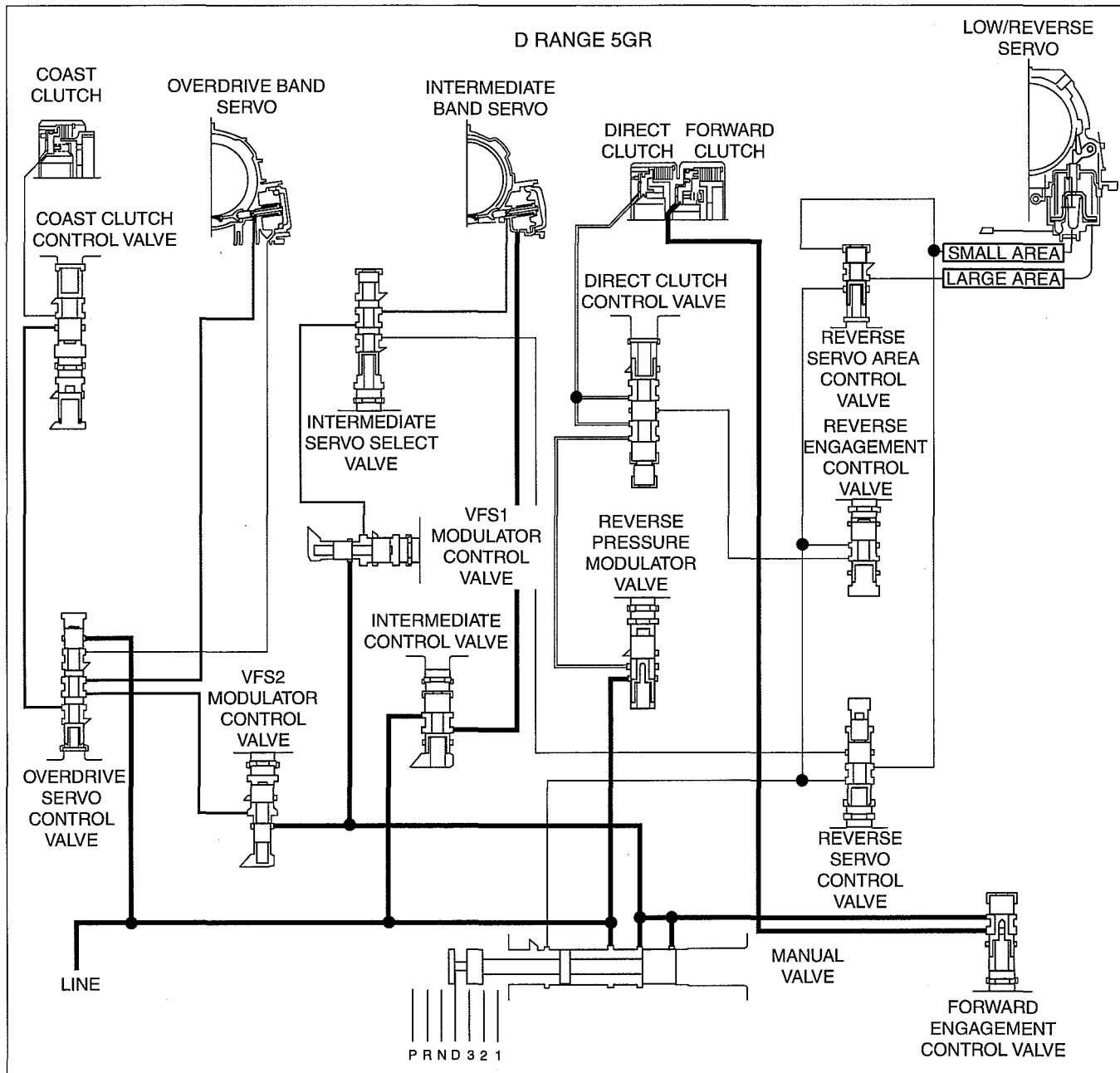


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AUTOMATIC TRANSMISSION [5R55S]

D range, 5GR

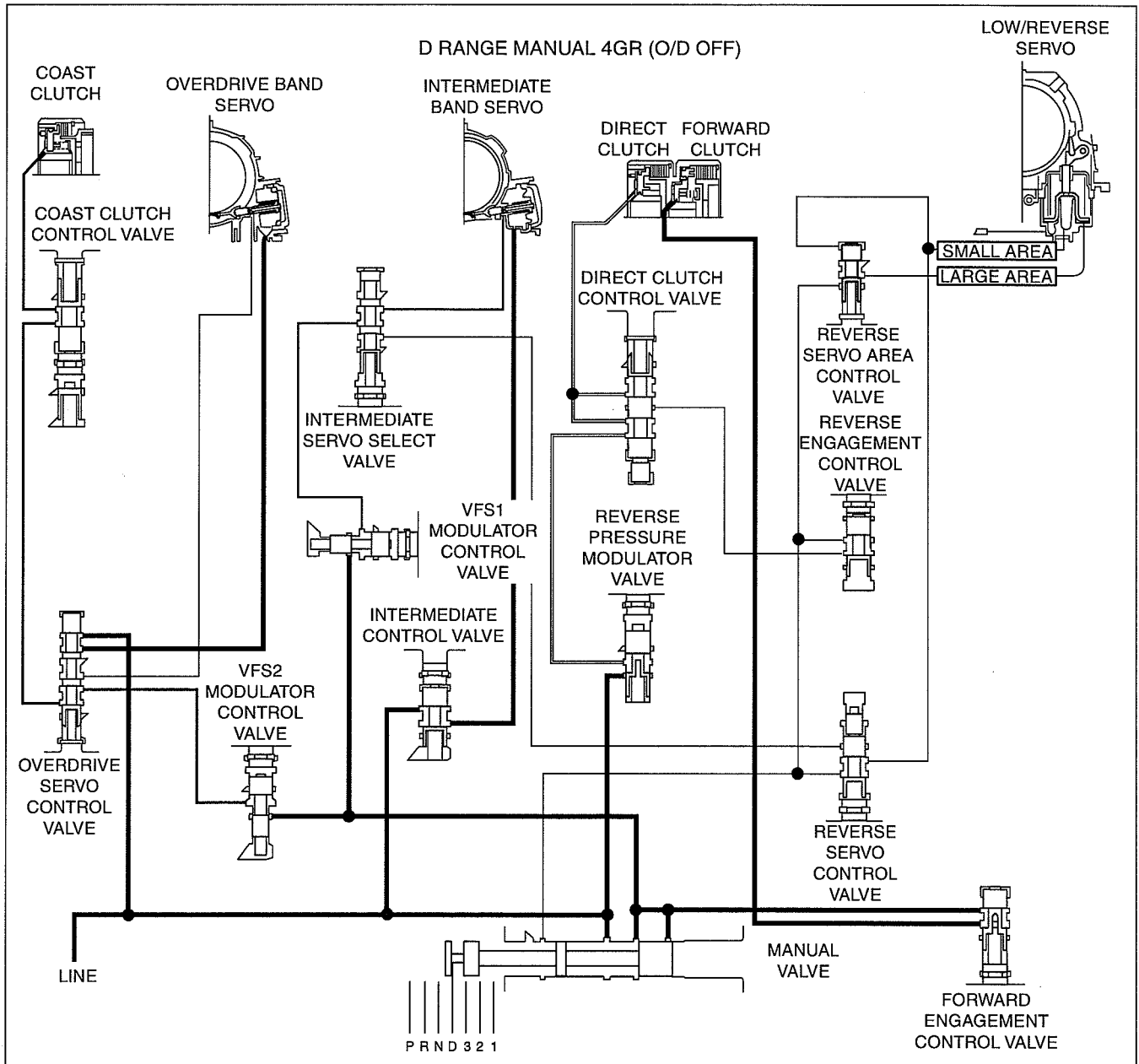
- With the transmission in the D range, fluid under pressure is present at the following components in 5GR:
 - Overdrive servo-apply
 - Intermediate servo-release
 - Forward clutch
 - Direct clutch



AUTOMATIC TRANSMISSION [5R55S]

D range O/D OFF Mode, 4GR (Manual 4GR)

- With the transmission in the D range (O/D OFF mode), fluid under pressure is present at the following components in manual 4GR:
 - Overdrive servo-release
 - Intermediate servo-release
 - Forward clutch
 - Direct clutch
 - Coast clutch

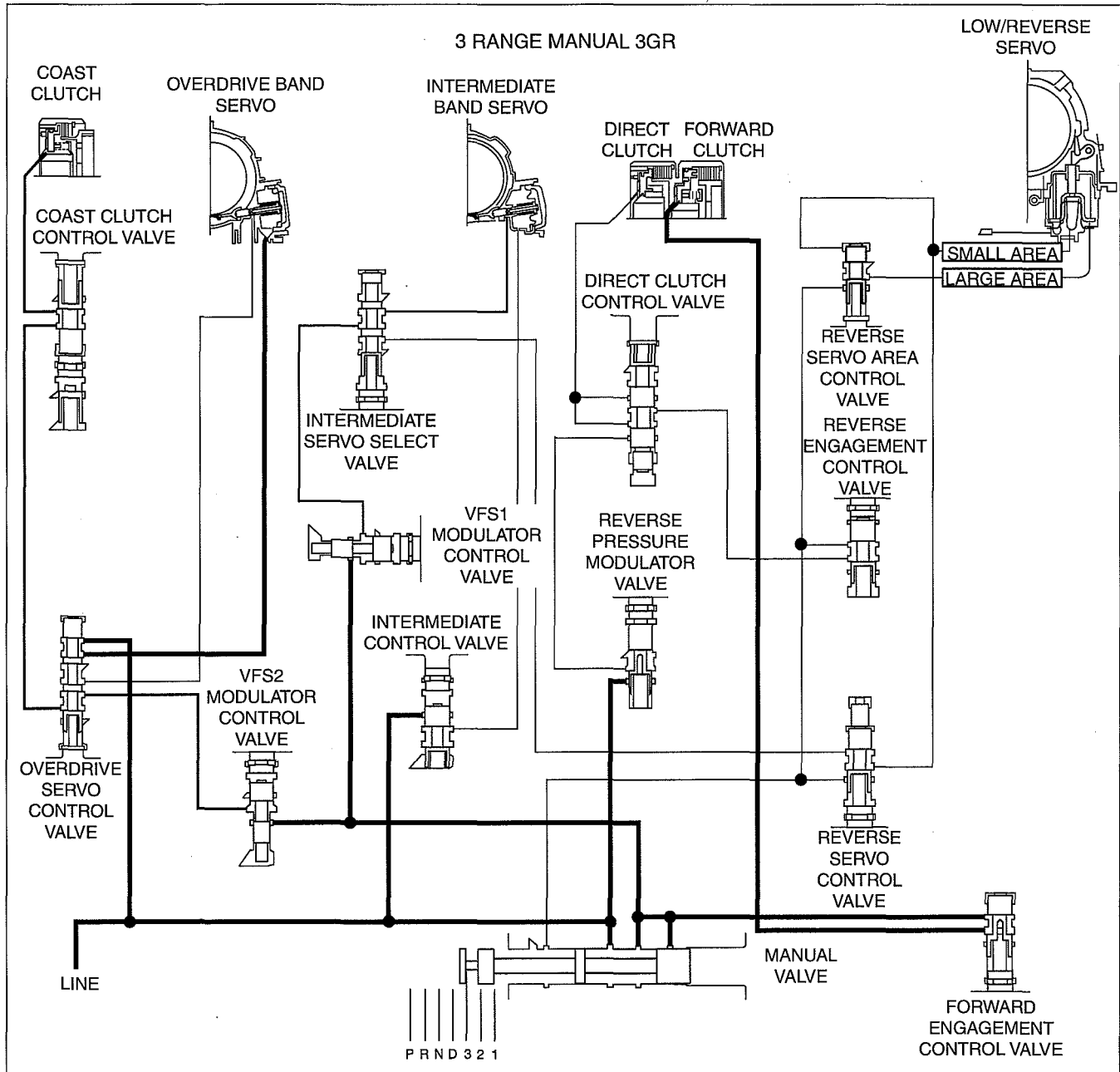


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AUTOMATIC TRANSMISSION [5R55S]

3 Range (Manual 3GR)

- With the transmission in the 3 range, fluid under pressure is present at the following components in manual 3GR:
 - Overdrive servo-release
 - Intermediate servo-apply
 - Forward clutch
 - Coast clutch

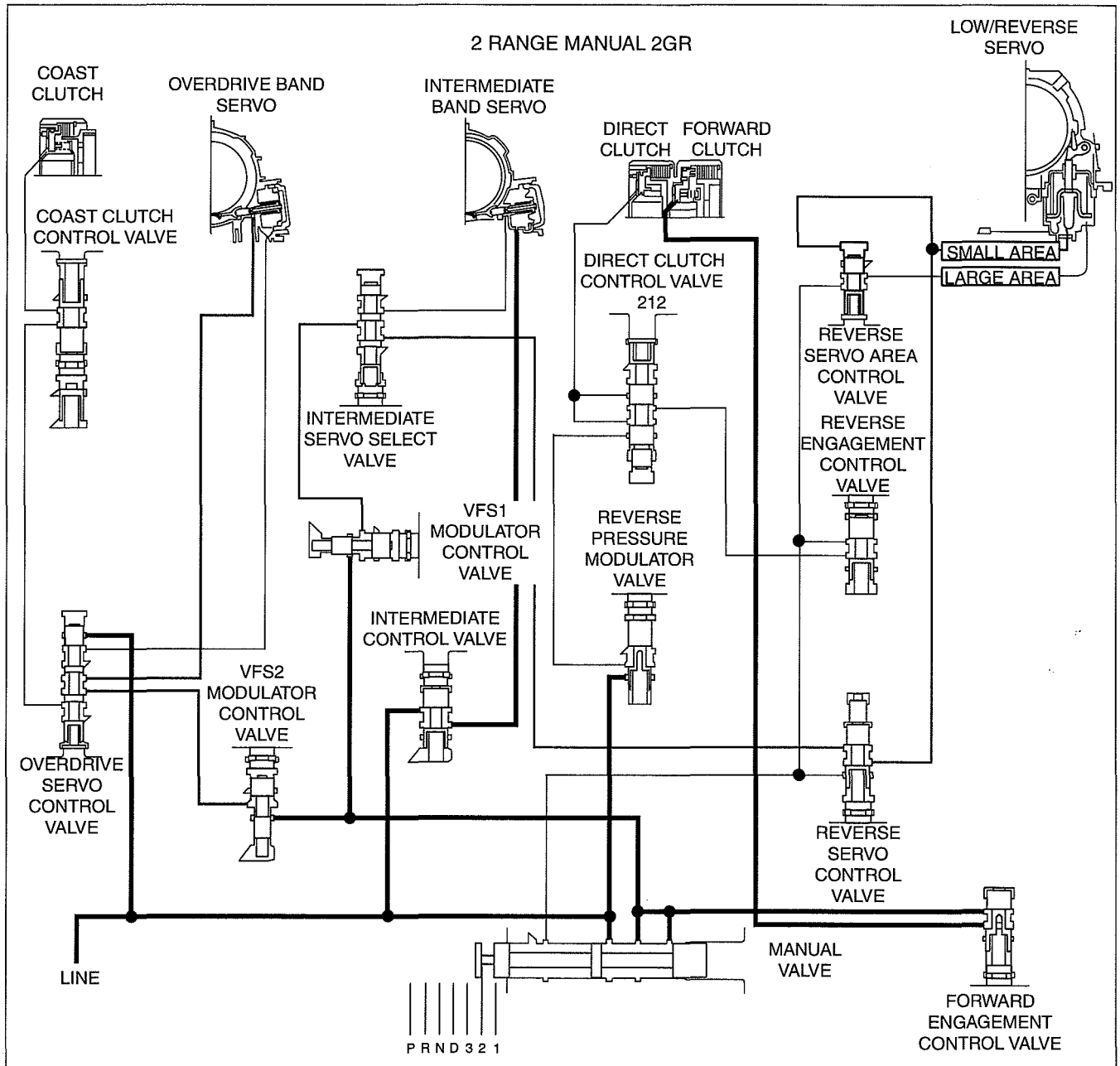


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AUTOMATIC TRANSMISSION [5R55S]

2 Range (Manual 2GR)

- With the transmission in the 2 range, fluid under pressure is present at the following components in manual 2GR:
 - Overdrive servo-apply
 - Intermediate servo-release
 - Forward clutch
 - Low/reverse servo-apply

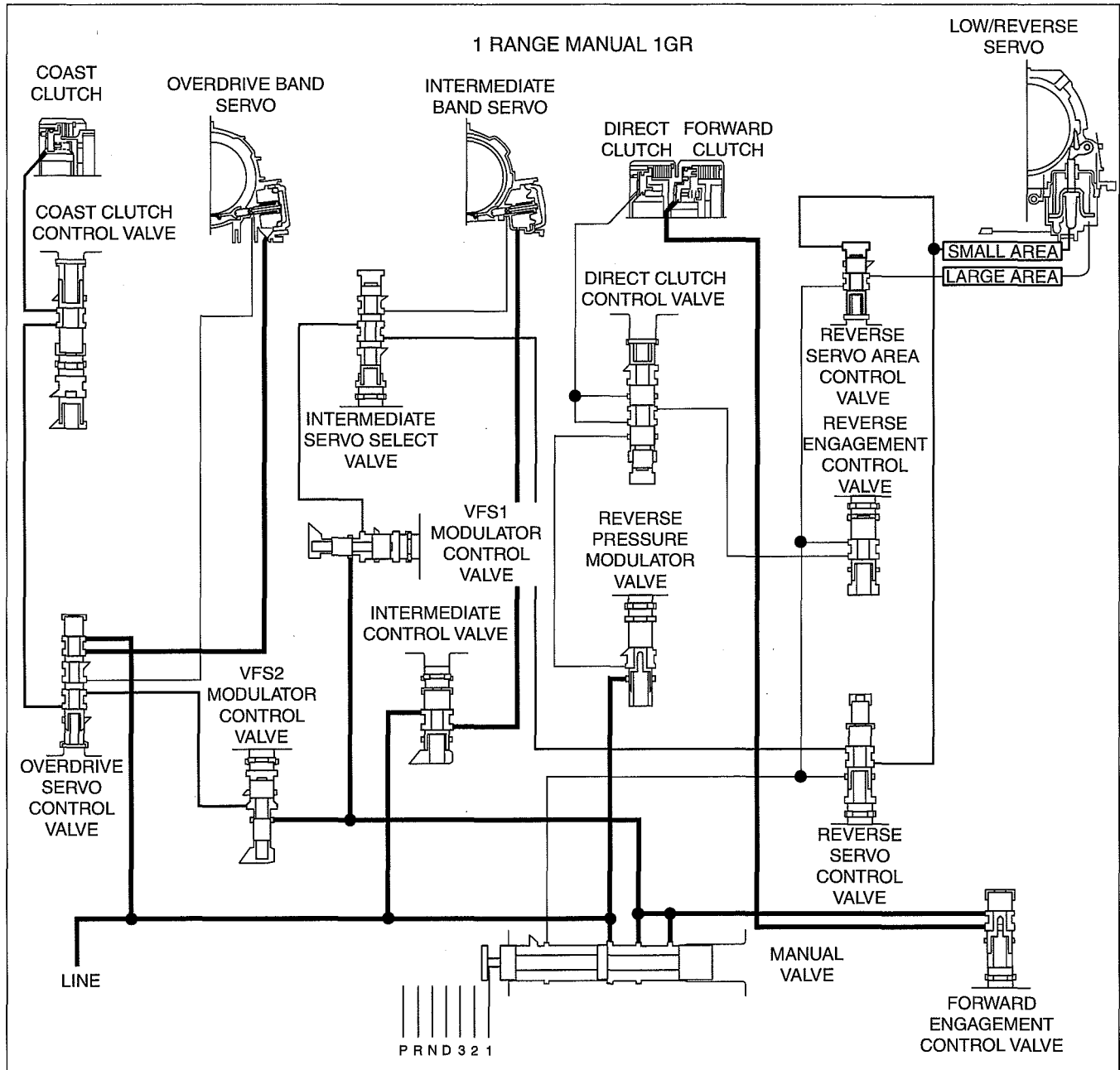


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AUTOMATIC TRANSMISSION [5R55S]

1 Range (Manual 1GR)

- With the transmission in the 1 range, fluid under pressure is present at the following components in manual 1GR:
 - Overdrive servo-release
 - Intermediate servo-release
 - Forward clutch
 - Coast clutch
 - Low/reverse servo-apply



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AUTOMATIC TRANSMISSION [5R55S]

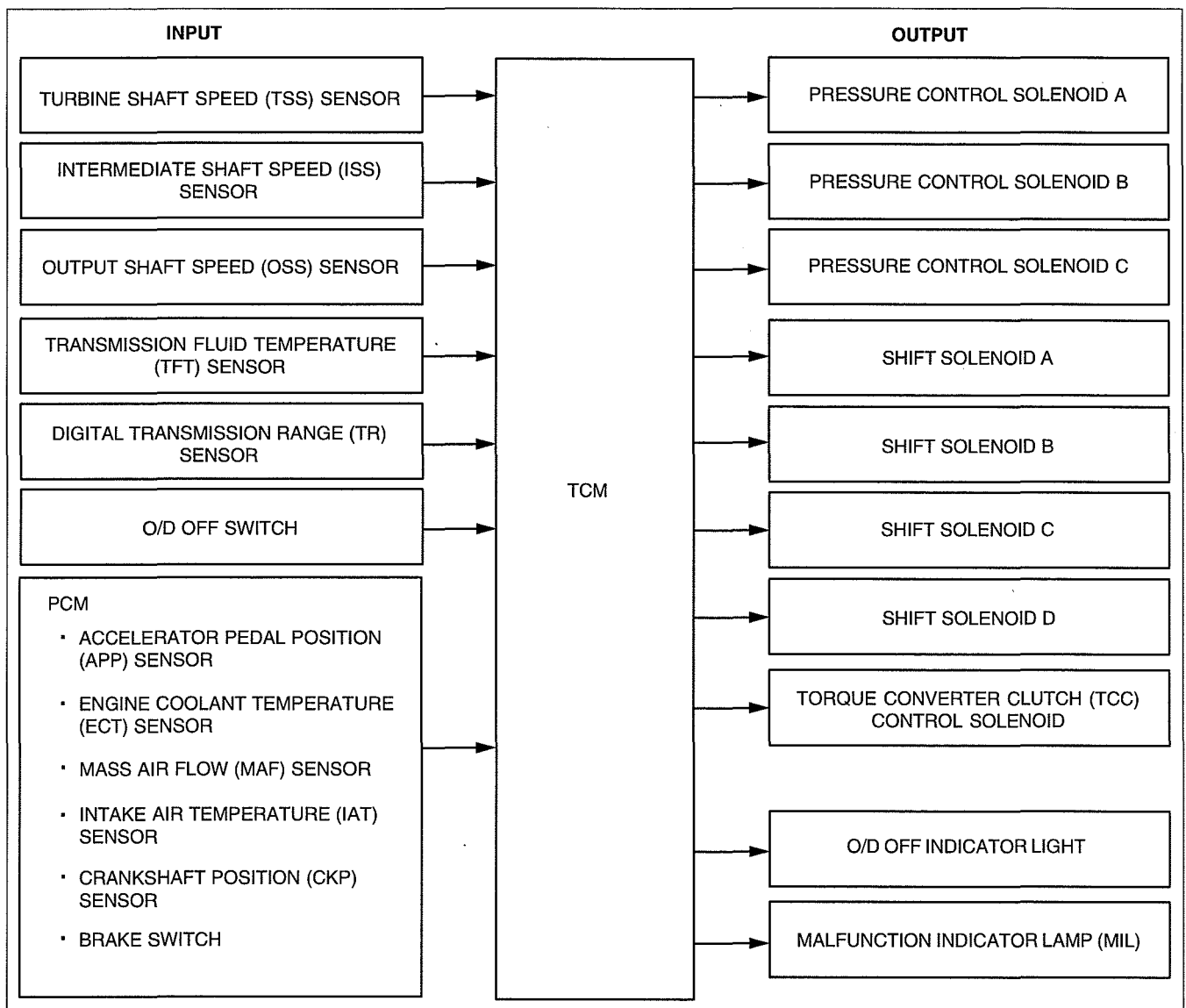
ELECTRONIC CONTROL SYSTEM OUTLINE [5R55S]

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- The TCM and its input/output network control the following transmission operations:
 - Shift timing
 - Line pressure (shift feel)
 - Torque converter clutch
- The transmission control strategy combined with the engine control provides optimum powertrain operation under all conditions.
- When determining the best operating strategy for transmission operation, the TCM uses input information from the PCM via the CAN bus. In addition, the TCM receives input signals from certain transmission-related sensors and switches.
- The TCM also uses these signals when determining transmission operating strategy.
- Using all of these input signals, the TCM can determine when the time and conditions are right for a shift or when to apply or release the torque converter clutch.
- It will also determine the pressure needed to optimize shift feel.
- To accomplish this the PCM uses 3 pressure controls, 1 torque converter clutch and 4 shift solenoids to control transmission operation.

ELECTRONIC CONTROL SYSTEM BLOCK DIAGRAM [5R55S]

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AUTOMATIC TRANSMISSION [5R55S]

ELECTRONIC CONTROL SYSTEM STRUCTURE [5R55S]

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TCM

- The operation of the transmission is controlled by the TCM.
- Many input sensors provide information to the TCM. The TCM then controls the actuators which determine transmission operation.

Air Conditioning (A/C) Clutch

- An electromagnetic clutch is energized when the clutch cycling pressure switch closes.
- The switch is located on the suction accumulator/drier.
- The closing of the switch completes the circuit to the clutch and draws it into engagement with the compressor driveshaft.
- When the A/C is engaged, operating pressures are adjusted to compensate for additional load on the engine.

Brake Switch

- The brake switch tells the PCM when the brakes are applied.
- The PCM sends this message to TCM via CAN bus.
- The torque converter clutch disengages when the brakes are applied.
- The brake switch closes when the brakes are applied and opens when they are released.

Engine Coolant Temperature (ECT) Sensor

- The ECT sensor detects temperature of engine coolant and supplies the information to the PCM.
- The PCM sends this message to TCM via CAN bus.
- The ECT sensor is used to control torque converter clutch (TCC) operation.

Crankshaft Position (CKP) Sensor

- The CKP sensor detects pulse wheel rotational signals as crank angle signals.
- The PCM sends this message to TCM via CAN bus.
- The PCM uses engine speed signal in the transmission strategy, wide-open throttle (WOT) shift control, torque converter clutch control and operating pressures.

Mass Air Flow (MAF) Sensor

- The MAF sensor measures the mass of air flowing into the engine.
- The PCM sends this message to TCM via CAN bus.
- The MAF sensor output signal is used by the PCM to calculate injector pulse width.
- For transmission strategies, the MAF sensor is used to regulate electronic pressure control, shift and torque converter clutch scheduling.

O/D OFF Switch

- The O/D OFF switch is a momentary contact switch that allows the driver to cancel operation of 5GR.
- The O/D OFF switch is located on the left side of the selector lever.
- When the driver initially presses the O/D OFF switch a signal is sent to the TCM.
- The TCM uses the shift solenoids to disengage/disable 5GR operation and activate the coast clutch. At the same time, the TCM illuminates the O/D OFF Indicator Light to notify the driver that 5GR is canceled.
- When the O/D OFF switch is pressed again, 5GR operation is enabled, the coast clutch is released and the O/D OFF indicator light is turned off.
- Whenever the ignition is cycled (vehicle shut off, then started again), the O/D OFF switch is turned off and 5GR will be enabled, even if the O/D OFF switch had been on when the ignition was shut off.

O/D OFF Indicator Light

- The O/D OFF indicator light is located in the instrument panel and is labeled O/D OFF. It is illuminated in conjunction with the O/D OFF switch.

Accelerator Pedal Position (APP) Sensor

- The APP sensor is mounted on the accelerator pedal. The APP sensor detects the position of the accelerator pedal and inputs this information as a voltage to the PCM.
- The PCM sends this message to TCM via CAN bus.
- The TCM uses the APP sensor information to aid in determining shift scheduling, presser control and torque converter control.

Digital Transmission Range (TR) Sensor

- The digital TR sensor is located on the outside of the transmission at the manual lever.
- The digital TR sensor completes the start circuit in PARK, NEUTRAL and the back-up lamp circuit in REVERSE.
- The digital TR sensor also opens and closes a set of 4 switches that are monitored by the TCM to determine the position of the manual lever (P, R, N, D, 3, 2, 1).

Turbine Shaft Speed (TSS) Sensor

- The TSS sensor is a magnetic pickup that sends the TCM torque converter turbine speed information.
- The TSS sensor is mounted externally on the case.
- The TCM uses TSS information to help determine appropriate operating pressures and TCC operation.

Output Shaft Speed (OSS) Sensor

- The OSS sensor is a magnetic pickup, located at the park gear trigger wheel assembly, that sends a signal to the TCM to indicate transmission output shaft speed.
- The OSS sensor is mounted externally on the case.
- The OSS is used for torque converter clutch control, shift scheduling and to determine pressure control.

Intermediate Shaft Speed (ISS) Sensor

- The ISS sensor is a magnetic pickup that sends planetary sun gear speed information to the TCM.
- It is mounted externally on the center of the case.
- The TCM uses the intermediate shaft speed sensor information to aid in determining pressure requirements.

Pressure Control Solenoid A, B and C

- The pressure control solenoids are a variable-force style (VFS) solenoid. The VFS-type solenoid is an electro-hydraulic actuator combining a solenoid and a regulating valve.
- The line pressure tap is used to verify output pressure from pressure control solenoid A or B by turning either one off while verifying the output from the other solenoid.
- The second pressure tap is used to verify the output from the pressure control solenoid C.
- There are 3 pressure control solenoids located in the solenoid body assembly used to control line pressure, band and clutch application pressure within the transmission.
- The TCM varies the current to the pressure control solenoid.
- The TCM has an adaptive learn strategy to electronically control the transmission which will automatically adjust the shift feel. When the battery has been disconnected or a new battery installed, certain transmission operating parameters may be lost. The TCM must relearn these parameters.
- During this learning process you may experience slightly firm shifts, delayed or early shifts. This operation is considered normal and will not affect the function of the transmission. Normal operation will return once these parameters are stored by the TCM.

Torque Converter Clutch (TCC) Control Solenoid

- The TCC control solenoid is a pulse width modulating type solenoid which is used to control the apply and release of the TCC.

Shift Solenoid A, B, C and D

- Four ON/OFF shift solenoids allow the TCM to control shift scheduling.
 - The solenoids are 3-way, normally open style.
 - The shift solenoids shift solenoid A, B, C and D provide gear selection of 1GR through 5GR and REVERSE gears by directing pressure control pressures to the appropriate elements.
- Coast braking and manual gears are also controlled by the shift solenoids.

Transmission Fluid Temperature (TFT) Sensor

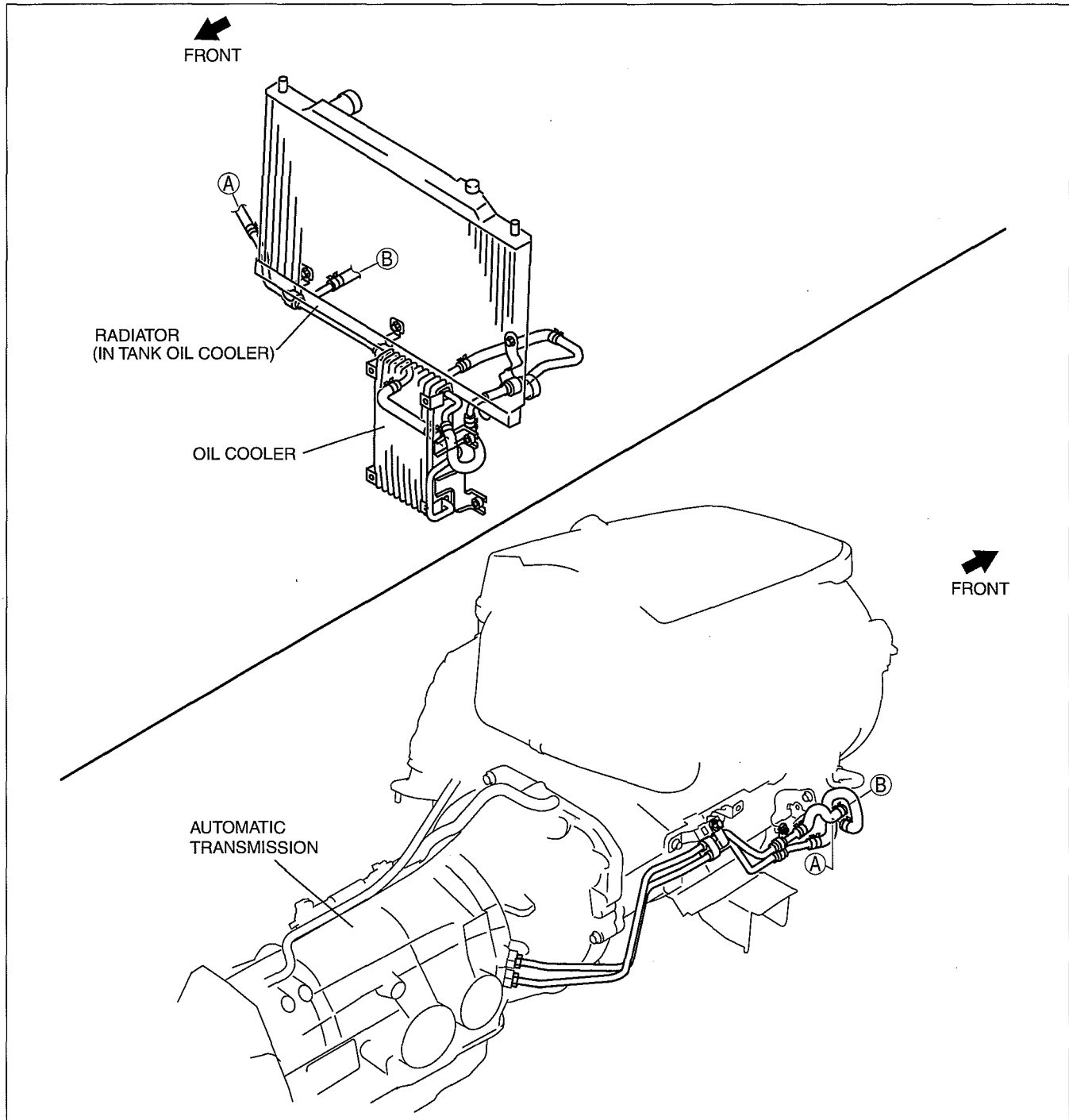
- The TFT sensor is a thermistor-type sensor that varies a reference voltage signal. The resistance in the TFT varies with temperature.
- The TCM monitors the voltage signal across the TFT and uses this information to determine the transmission fluid temperature.
- The TFT is located on the solenoid body.
- The TCM uses the TFT signal to help determine shift scheduling, torque converter clutch operation and pressure control requirements.
- It sends a voltage signal to the TCM. The voltage signal varies with transmission fluid temperature. The TCM uses this signal to determine whether a cold start shift schedule is necessary. The shift schedule is compensated when the transmission fluid temperature is cold. The TCM also inhibits TCC operation at low transmission fluid temperatures and uses it to determine PC solenoid operations.

AUTOMATIC TRANSMISSION [5R55S]

OIL COOLER CONSTRUCTION [5R55S]

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- A water-cooling type AT oil cooler has been adopted and is installed in the radiator. The oil cooler cools the heated ATF in the AT.
- An air-cooling type AT oil cooler has also adopted.



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05-14 AUTOMATIC TRANSMISSION SHIFT MECHANISM

AUTOMATIC TRANSMISSION
SHIFT MECHANISM OUTLINE..... 05-14-1

AUTOMATIC TRANSMISSION
SHIFT MECHANISM
STRUCTURAL VIEW05-14-1

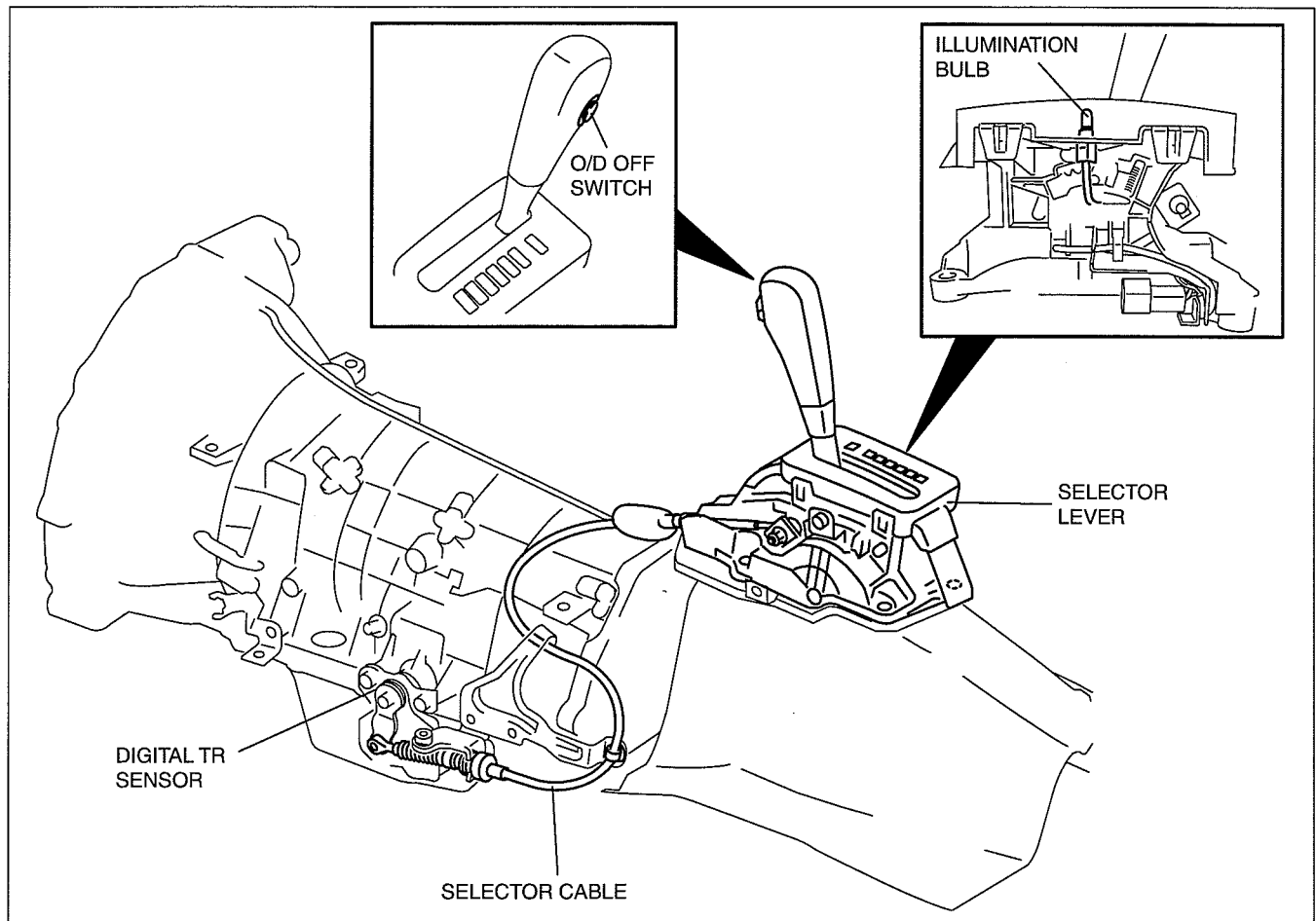
AUTOMATIC TRANSMISSION SHIFT MECHANISM OUTLINE

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- The cable type shift mechanism has been adopted.
- The floor-shift type selector lever has been adopted.

AUTOMATIC TRANSMISSION SHIFT MECHANISM STRUCTURAL VIEW

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